

# Revisit for Module and Representation Theory

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## 1 Concepts

1. faithful, injective, projective, flat, Noetherian, semisimple, Artinian, Soc, essential epi idem, Jacobson radical, projective cover, DVR.

Composition series, Wedderburn's Theorem, idempotents, Schur's lemma, the cartan matrix, Krull-Schmidt, Socle Series, Radical Series, Top, Nakayama lemma, Idempotent Lifting Jordan-Hölder, Loewy

## 2 Module

**Theorem 2.1** (Universal property of direct sums). *Let  $A$  be a unital associative ring, and let  $\{M_i\}_{i \in I}$  be a family of left  $A$ -modules. The direct sum*

$$\bigoplus_{i \in I} M_i$$

*together with the canonical inclusions*

$$\iota_i : M_i \longrightarrow \bigoplus_{i \in I} M_i$$

*satisfies the following universal property: for every left  $A$ -module  $N$  and every family of homomorphisms  $f_i : M_i \rightarrow N$ , there exists a unique homomorphism*

$$f : \bigoplus_{i \in I} M_i \longrightarrow N$$

*such that  $f \circ \iota_i = f_i$  for all  $i \in I$ . Equivalently,*

$$\mathrm{Hom}_A\left(\bigoplus_{i \in I} M_i, N\right) \cong \prod_{i \in I} \mathrm{Hom}_A(M_i, N).$$

**Theorem 2.2** (Universal property of direct products). *Let  $A$  be a unital associative ring, and let  $\{M_i\}_{i \in I}$  be a family of left  $A$ -modules. The direct product*

$$\prod_{i \in I} M_i$$

*together with the canonical projections*

$$\pi_i : \prod_{i \in I} M_i \longrightarrow M_i$$

*satisfies the following universal property: for every left  $A$ -module  $N$  and every family of homomorphisms  $g_i : N \rightarrow M_i$ , there exists a unique homomorphism*

$$g : N \longrightarrow \prod_{i \in I} M_i$$

*such that  $\pi_i \circ g = g_i$  for all  $i \in I$ . Equivalently,*

$$\operatorname{Hom}_A \left( N, \prod_{i \in I} M_i \right) \cong \prod_{i \in I} \operatorname{Hom}_A(N, M_i).$$

**Theorem 2.3** (Tensor–Hom adjunction). *Let  $A$  and  $B$  be unital associative rings.*

(i) *Let  ${}_B X_A$  be a  $(B, A)$ -bimodule. Then the functor*

$$X \otimes_A - : A\text{-Mod} \longrightarrow B\text{-Mod}$$

*is left adjoint to*

$$\operatorname{Hom}_B(X, -) : B\text{-Mod} \longrightarrow A\text{-Mod}.$$

*Equivalently, for every left  $A$ -module  $M$  and every left  $B$ -module  $N$ , there is a natural isomorphism*

$$\operatorname{Hom}_B(X \otimes_A M, N) \cong \operatorname{Hom}_A(M, \operatorname{Hom}_B(X, N)).$$

(ii) *Let  ${}_A X_B$  be an  $(A, B)$ -bimodule. Then the functor*

$$- \otimes_A X : \operatorname{Mod}\text{-}A \longrightarrow \operatorname{Mod}\text{-}B$$

*is left adjoint to*

$$\operatorname{Hom}_B(X, -) : \operatorname{Mod}\text{-}B \longrightarrow \operatorname{Mod}\text{-}A.$$

*Equivalently, for every right  $A$ -module  $M$  and every right  $B$ -module  $N$ , there is a natural isomorphism*

$$\operatorname{Hom}_B(M \otimes_A X, N) \cong \operatorname{Hom}_A(M, \operatorname{Hom}_B(X, N)).$$

**Theorem 2.4.** *The following three conditions on  $V_A$  are equivalent.*

- (1)  *$V$  is a Noetherian module.*
- (2) *(Ascending chain condition.) For every infinite chain of  $A$ -submodules of  $V$ ,*

$$V_1 \subset V_2 \subset \cdots \subset V_n \subset \cdots, \tag{2.1}$$

*there is a number  $m$  such that  $V_m = V_{m+1} = \cdots$ .*

- (3) *Every  $A$ -submodule of  $V$  is finitely generated over  $A$ .*

*Remark 2.5* (Finiteness in Category Theory). **(1) Finite Presentation via Colimits:**

In classical module categories, “finiteness” is characterized by filtered colimits. A module  $M$  is finitely presented if and only if the functor  $\text{Hom}(M, -)$  commutes with filtered colimits:

$$\varinjlim \text{Hom}(M, N_i) \xrightarrow{\sim} \text{Hom}(M, \varinjlim N_i)$$

Intuitively, any morphism from  $M$  to an infinite filtered system must factor through a finite stage.

**(2) Finiteness via Functor Preservation in  $\infty$ -Categories:**

In higher category theory, objects lack “elements.” Instead, an object  $X$  is defined as “finite” based on how its associated functors interact with limits and colimits.

- **Compact (Finitely Presented) Objects:**

The functor  $\text{Map}(X, -)$  commutes with *filtered colimits*.

*Example:* Finite CW complexes in the  $\infty$ -category of Spaces.

- **Dualizable Objects:**

The functor  $X \otimes -$  commutes with *all limits* (e.g., infinite products). This happens because having a dual allows the left adjoint  $X \otimes -$  to act like the right adjoint  $\underline{\text{Map}}(X^\vee, -)$ .

*Example:* Finite spectra in the stable  $\infty$ -category of Spectra.

- **Perfect Complexes:**

The ultimate intersection of both finiteness conditions.  $\mathbb{R}\text{Map}(X, -)$  preserves filtered colimits, and  $X \otimes^{\mathbb{L}} -$  preserves limits.

*Example:* Perfect complexes in the derived  $\infty$ -category  $\mathcal{D}(R)$  or  $\text{QCoh}(X)$  in derived algebraic geometry.

**Theorem 2.6.** *The following three conditions on a module  $V_A$  are equivalent:*

(1)  *$V$  is an Artinian module.*

(2) *(Descending chain condition.) For every infinite chain of  $A$ -submodules of  $V$ ,*

$$V_1 \supset V_2 \supset \cdots \supset V_n \supset \cdots,$$

*there is a number  $m$  such that  $V_m = V_{m+1} = \cdots$ .*

(3) *(Minimal condition.) Every non-empty set of  $A$ -submodules of  $V$  contains a minimal element with respect to inclusion.*

**Theorem 2.7.** *Let  $W$  be an  $A$ -submodule of  $V$ . Then  $V$  is Noetherian (respectively, Artinian) if and only if both  $W$  and  $V/W$  are Noetherian (respectively, Artinian).*

**Definition 2.8.** A nonzero  $A$ -module  $V$  is said to be **irreducible** (or **simple**) if it has no proper  $A$ -submodule other than 0.

A chain of  $A$ -submodules of  $V$

$$V = V_0 \supset V_1 \supset \cdots \supset V_n = 0$$

is said to be a **composition series** of  $V$  if each  $V_i/V_{i+1}$  is irreducible. And the simple modules

$$V/V_1, V_1/V_2, \dots, V_{n-1}/V_n = V_{n-1}$$

are called the **composition factors** of the series.

**Theorem 2.9.** *A non-zero module has a composition series if and only if it is both Noetherian and Artinian.*

*Proof.* Use [Theorem 2.7](#) and induction on the length of chain.  $\square$

**Theorem 2.10** (Jordan–Hölder). *If  $V$  has a composition series, any two composition factors obtained from the two series of  $V$  are the same up to order of occurrence and isomorphism.*

**Theorem 2.11** (Equivalent descriptions of the Jacobson radical). *Let  $A$  be a unital associative ring. For  $a \in A$ , the following conditions are equivalent:*

- (i)  $a \in J(A)$ .
- (ii)  $a$  annihilates every simple left  $A$ -module, i.e.  $aS = 0$  for every simple left  $A$ -module  $S$ .
- (iii)  $a$  annihilates every simple right  $A$ -module, i.e.  $Ta = 0$  for every simple right  $A$ -module  $T$ .
- (iv)  $a$  belongs to every maximal left ideal of  $A$ .
- (v)  $a$  belongs to every maximal right ideal of  $A$ .
- (vi)  $1 - xa \in A^\times$  for every  $x \in A$ .
- (vii)  $1 - ax \in A^\times$  for every  $x \in A$ .

Equivalently,

$$J(A) = \bigcap_{S \text{ simple}} \text{Ann}_A(S) = \bigcap_{\mathfrak{m} \text{ maximal left}} \mathfrak{m} = \bigcap_{\mathfrak{n} \text{ maximal right}} \mathfrak{n}.$$

Moreover,  $J(A)$  is the largest two-sided ideal  $I \subset A$  such that  $1 - a \in A^\times$  for every  $a \in I$ .

**Theorem 2.12.** *Let  $A$  be a unital associative ring. If  $I \subset A$  is a nil left(right, double-sided) ideal, then  $I \subset J(A)$ .*

*Proof.* Use the (vi) of [Theorem 2.11](#).  $\square$

**Theorem 2.13** (Jacobson radical of an Artinian ring). *Let  $A$  be a unital associative ring. If  $A$  is left Artinian or right Artinian, then  $J(A)$  is the largest nilpotent two-sided ideal of  $A$ .*

**Theorem 2.14.** *Let  $A$  be a unital associative ring. If  $S$  is a simple left  $A$ -module, then there exists a maximal left ideal  $\mathfrak{m} \subset A$  such that*

$$S \cong A/\mathfrak{m}$$

*as left  $A$ -modules.*

*Remark 2.15.* Let  $A$  be a unital associative ring, and let  $\mathfrak{m} \subset A$  be a maximal left ideal. Then  $A/\mathfrak{m}$  is a simple left  $A$ -module, and

$$\text{Ann}_A(A/\mathfrak{m}) = \{a \in A \mid aA \subset \mathfrak{m}\}.$$

This is the largest two-sided ideal contained in  $\mathfrak{m}$ . In general it need not equal  $\mathfrak{m}$ , since  $\mathfrak{m}$  is only a left ideal. Thus  $\mathfrak{m}$  is the annihilator of the element  $1 + \mathfrak{m}$ , whereas  $\text{Ann}_A(A/\mathfrak{m})$  is the annihilator of the whole module.

**Theorem 2.16** (Azumaya–Nakayama). *Let  $A$  be a unital associative ring.*

(i) Let  $V$  be a left  $A$ -module and let  $W$  be an  $A$ -submodule of  $V$  such that  $V/W$  is finitely generated over  $A$ . Then

$$V = W + J(A)V \implies V = W.$$

(ii) If  $V$  is a finitely generated left  $A$ -module, then

$$J(A)V = V \implies V = 0.$$

**Theorem 2.17.** Let  $A$  be a unital associative ring, and let  $M$  be a left  $A$ -module of finite length. Then

$$M \text{ is semisimple} \iff \text{rad}(M) = 0.$$

**Theorem 2.18.** Let  $A$  be a unital associative ring.

(i) If  $A$  is semisimple, then  $J(A) = 0$ .

(ii) If  $A$  is left Artinian or right Artinian, then

$$A \text{ is semisimple} \iff J(A) = 0.$$

**Definition 2.19.** Let  $A$  be a unital associative ring, and let  $M$  be a left  $A$ -module. The radical of  $M$  is defined by

$$\text{rad}(M) = \bigcap_{N \subset M \text{ maximal}} N,$$

where the intersection is taken over all maximal submodules of  $M$ .

**Corollary 2.20.** Let  $A$  be a unital associative ring, and let  $M$  be a left  $A$ -module. Then

$$J(A)M \subset \text{rad}(M).$$

**Theorem 2.21.** Let  $A$  be a unital associative ring, and let  $M$  be a left  $A$ -module. If  $M$  is semisimple, then

$$\text{rad}(M) = 0.$$

**Theorem 2.22.** Let  $A$  be a left Artinian ring, and let  $M$  be a finitely generated left  $A$ -module. Then

$$\text{rad}(M) = J(A)M.$$

*Proof.* Use [Theorem 2.21](#). □

**Theorem 2.23.** Let  $A$  be a unital associative ring, and let  $M$  be a left  $A$ -module. A decomposition

$$M = M_1 \oplus \cdots \oplus M_n$$

is equivalent to a decomposition of  $\text{id}_M$  in  $\text{End}_A(M)$  as

$$\text{id}_M = p_1 + \cdots + p_n,$$

where  $p_i^2 = p_i$  and  $p_i p_j = 0$  for  $i \neq j$ . Under this correspondence,

$$M_i = \text{Im}(p_i).$$

In particular,  $M$  is indecomposable if and only if  $\text{id}_M$  is a primitive idempotent in  $\text{End}_A(M)$ .

**Theorem 2.24.** *Let  $A$  be a unital associative ring. Then*

$$\text{End}_A(A_A) \cong A \quad \text{End}_A({}_A A) \cong A^{op}.$$

**Theorem 2.25.** *Let  $A$  be a unital associative ring, and let  $e \in A$  be an idempotent.*

(i) *If  $e = e_1 + \cdots + e_n$  is a decomposition into pairwise orthogonal idempotents, then*

$$eA = e_1A \oplus \cdots \oplus e_nA$$

*as right  $A$ -modules, and*

$$Ae = Ae_1 \oplus \cdots \oplus Ae_n$$

*as left  $A$ -modules.*

(ii) *Conversely, every direct sum decomposition*

$$eA = I_1 \oplus \cdots \oplus I_n$$

*into right ideals is of the form  $I_i = e_iA$ , where  $e = e_1 + \cdots + e_n$  is a decomposition into pairwise orthogonal idempotents. Similarly, every direct sum decomposition*

$$Ae = L_1 \oplus \cdots \oplus L_n$$

*into left ideals is of the form  $L_i = Ae_i$ , where  $e = e_1 + \cdots + e_n$  is a decomposition into pairwise orthogonal idempotents.*

**Theorem 2.26.** *Let  $A$  be a unital associative ring, let  $M$  be a left  $A$ -module, and set  $E = \text{End}_A(M)$ . For a nonzero idempotent  $p \in E$ , write  $pM = \text{Im}(p)$ . Then*

$$p \text{ is primitive in } E \iff pM \text{ is indecomposable as a left } A\text{-module.}$$

*Moreover,*

$$\text{End}_A(pM) \cong pEp.$$

**Theorem 2.27.** *Let  $A$  be a unital associative ring, and let  $V$  be a left  $A$ -module. Suppose that  $V$  has a composition series. Then*

$$V \text{ is indecomposable} \iff \text{End}_A(V) \text{ is a local ring.}$$

**Theorem 2.28** (Krull–Schmidt–Azumaya). *Let  $A$  be a unital associative ring, and let  $V$  be a left  $A$ -module. Suppose that*

$$V = V_1 \oplus \cdots \oplus V_n$$

*is a finite direct sum decomposition into indecomposable submodules, and that  $\text{End}_A(V_i)$  is a local ring for every  $i$ .*

*Then this decomposition is unique up to isomorphism and permutation. That is, if*

$$V = W_1 \oplus \cdots \oplus W_m$$

*is another finite direct sum decomposition into indecomposable submodules, then  $m = n$ , and there exists a permutation  $\sigma \in S_n$  such that*

$$V_i \cong W_{\sigma(i)}$$

*for all  $1 \leq i \leq n$ .*

**Lemma 2.29.** *Let  $A$  be a unital associative ring, and let  $M$  be an Artinian left  $A$ -module. Set  $E = \text{End}_A(M)$ . Then every nonzero idempotent  $p \in E$  admits a finite decomposition*

$$p = p_1 + \cdots + p_n$$

*where  $p_1, \dots, p_n \in E$  are pairwise orthogonal primitive idempotents. Equivalently,*

$$\text{Im}(p) = \text{Im}(p_1) \oplus \cdots \oplus \text{Im}(p_n)$$

*is a finite direct sum decomposition into indecomposable left  $A$ -modules.*

*Proof.* Since  $p$  is idempotent,  $\text{Im}(p)$  is a direct summand of  $M$ . Hence  $\text{Im}(p)$  is Artinian. Repeatedly decompose  $\text{Im}(p)$  whenever it is decomposable. This process terminates, because otherwise one obtains an infinite strictly descending chain of submodules of  $M$ . Thus

$$\text{Im}(p) = M_1 \oplus \cdots \oplus M_n$$

with each  $M_i$  indecomposable.

Let  $p_i$  be the projection from  $M$  onto  $M_i$ , with kernel equal to the sum of the other summands and  $\ker(p)$ . Then the  $p_i$  are pairwise orthogonal idempotents in  $E$ , and

$$p = p_1 + \cdots + p_n.$$

Since  $\text{Im}(p_i) = M_i$  is indecomposable,  $p_i$  is primitive. □

**Corollary 2.30.** *Let  $A$  be a left Artinian ring, and let  $M$  be a finitely generated left  $A$ -module. Then  $M$  admits a finite direct sum decomposition*

$$M = M_1 \oplus \cdots \oplus M_n$$

*where each  $M_i$  is indecomposable. Moreover, this decomposition is unique up to isomorphism and permutation.*

*Proof.* Use [Lemma 2.29](#) and [Theorem 2.28](#). □

**Theorem 2.31.** *Let  $A$  be a unital associative ring, and let  $P$  be a left  $A$ -module. The following conditions are equivalent:*

- (i)  $P$  is projective.
- (ii) For every epimorphism  $M \twoheadrightarrow N$  and every homomorphism  $P \rightarrow N$ , there exists a lift  $P \rightarrow M$ .
- (iii) The functor

$$\text{Hom}_A(P, -) : A\text{-Mod} \longrightarrow \text{Ab}$$

*is exact.*

- (iv)  $P$  is a direct summand of a free left  $A$ -module.
- (v) Every epimorphism  $M \twoheadrightarrow P$  splits.

*If  $P$  is finitely generated, then these conditions are also equivalent to  $P$  being a direct summand of  $A^n$  for some  $n \geq 1$ .*

**Theorem 2.32.** *Let  $A$  be a unital ring, and let  $I$  be a left  $A$ -module. The following conditions are equivalent:*



(i)  $I$  is injective, i.e., for every submodule  $M \subset N$  and every homomorphism  $f : M \rightarrow I$ , there exists an extension  $g : N \rightarrow I$  such that  $g|_M = f$ .

(ii) The functor

$$\mathrm{Hom}_A(-, I) : (A\text{-Mod})^{\mathrm{op}} \longrightarrow \mathrm{Ab}$$

is exact.

(iii) (Baer criterion) For every left ideal  $J \subset A$  and homomorphism  $f : J \rightarrow I$ , there exists an extension  $g : A \rightarrow I$  with  $g|_J = f$ .

**Theorem 2.33.** Let  $A$  be a unital associative ring, and let  $F$  be a left  $A$ -module. The following conditions are equivalent:

(i)  $F$  is flat, i.e. the functor

$$- \otimes_A F : \mathrm{Mod}\text{-}A \longrightarrow \mathrm{Ab}$$

is exact. Equivalently, for every monomorphism  $M' \hookrightarrow M$  of right  $A$ -modules, the induced map

$$M' \otimes_A F \longrightarrow M \otimes_A F$$

is a monomorphism.

(ii) For every finitely generated right ideal  $I \subset A$ , the natural map

$$I \otimes_A F \longrightarrow A \otimes_A F$$

is injective.

(iii)  $F$  is a filtered colimit of finitely generated free left  $A$ -modules.

**Theorem 2.34** (Wedderburn–Artin). Let  $A$  be a simple Artinian ring. Then there exist a division ring  $D$  and a positive integer  $n$  such that

$$A \cong M_n(D),$$

the ring of  $n \times n$  matrices over  $D$ .

Moreover, as a left  $A$ -module, the left regular module  ${}_A A$  decomposes as

$${}_A A \cong S^{\oplus n},$$

where  $S = Ae_{11}$  is a minimal left ideal of  $A$ , simple and projective, corresponding to the first column of the matrix block. Each of the  $n$  summands is isomorphic to  $S$ , so the left regular module is a direct sum of mutually isomorphic minimal left ideals.

**Theorem 2.35** (Idempotent lifting modulo a nilpotent ideal). Let  $A$  be a unital associative ring, and let  $I \subset A$  be a two-sided nilpotent ideal. Let  $\pi : A \rightarrow A/I$  be the quotient map.

(i) Every idempotent of  $A/I$  lifts to an idempotent of  $A$ . That is, if  $\bar{e} \in A/I$  satisfies  $\bar{e}^2 = \bar{e}$ , then there exists an idempotent  $e \in A$  such that  $\pi(e) = \bar{e}$ .

(ii) More generally, if

$$\bar{e} = \bar{e}_1 + \cdots + \bar{e}_n$$

is a decomposition into pairwise orthogonal idempotents in  $A/I$ , and  $e \in A$  is an idempotent lift of  $\bar{e}$ , then there exist pairwise orthogonal idempotents  $e_1, \dots, e_n \in A$  such that

$$e = e_1 + \cdots + e_n, \quad \pi(e_i) = \bar{e}_i$$

for all  $1 \leq i \leq n$ .

**Theorem 2.36.** *Let  $A$  be a unital associative ring, and let  $I \subset A$  be a two-sided nilpotent ideal. Set  $\bar{A} = A/I$ , and let  $\pi : A \rightarrow \bar{A}$  be the quotient map. If  $e \in A$  is an idempotent and  $\bar{e} = \pi(e)$ , then*

$$e \text{ is primitive in } A \iff \bar{e} \text{ is primitive in } \bar{A}.$$

**Lemma 2.37.** *Let  $A$  be a unital associative ring, and let  $f : M \rightarrow N$  be an epimorphism of left  $A$ -modules. If  $M$  is finitely generated and*

$$\ker f \subset J(A)M,$$

*then  $f$  is essential. That is, if  $L \subset M$  is a submodule such that  $f(L) = N$ , then  $L = M$ .*

**Theorem 2.38.** *Let  $A$  be a semiperfect ring, let  $J = J(A)$ , and set  $\bar{A} = A/J$ .*

*(i) Every finitely generated left  $A$ -module  $M$  has a projective cover*

$$p : P \twoheadrightarrow M,$$

*where  $P$  is a finitely generated projective left  $A$ -module. Moreover, the induced map*

$$P/JP \longrightarrow M/JM$$

*is an isomorphism.*

*(ii) Let  $f : P \twoheadrightarrow M$  be an epimorphism of left  $A$ -modules, and suppose that  $P$  is finitely generated and projective. Then*

$$f \text{ is essential} \iff \ker f \subset JP.$$

*(iii) If  $p : P \twoheadrightarrow M$  is the projective cover in (i), then the induced map*

$$\bar{p} : P/JP \longrightarrow M/JM$$

*is a projective cover in  $\bar{A}\text{-Mod}$ .*

### 3 Representation Theory

**Theorem 3.1** (Characters are class functions). *Let  $G$  be a finite group, and let  $\chi$  be a complex character of  $G$ . If  $g, h \in G$  are conjugate, then*

$$\chi(g) = \chi(h).$$

*Equivalently, every character of  $G$  is constant on conjugacy classes.*

**Theorem 3.2** (Number of irreducible characters). *Let  $G$  be a finite group, and let  $\text{Irr}(G)$  be the set of irreducible complex characters of  $G$ . Then*

$$|\text{Irr}(G)| = |\text{Cl}(G)|,$$

*where  $\text{Cl}(G)$  denotes the set of conjugacy classes of  $G$ .*

**Theorem 3.3** (Linear characters and the abelianization). *Let  $G$  be a finite group, and let  $G' = [G, G]$  be its commutator subgroup. Then the one-dimensional complex characters of  $G$  are exactly the characters inflated from the abelianization  $G/G'$ . Equivalently,*

$$\text{Irr}_1(G) \cong \text{Hom}(G/G', \mathbb{C}^\times).$$

*In particular,*

$$|\text{Irr}_1(G)| = |G/G'|.$$

**Theorem 3.4** (Degree formula for irreducible characters). *Let  $G$  be a finite group, and let  $\text{Irr}(G)$  be the set of irreducible complex characters of  $G$ . Then*

$$|G| = \sum_{\chi \in \text{Irr}(G)} \chi(1)^2.$$

Moreover, for every  $\chi \in \text{Irr}(G)$ ,

$$\chi(1) \mid |G|.$$

**Theorem 3.5** (Row orthogonality of irreducible characters). *Let  $G$  be a finite group, and let  $\text{Irr}(G)$  be the set of irreducible complex characters of  $G$ . For class functions  $\chi, \psi$ , define*

$$\langle \chi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \chi(g) \overline{\psi(g)}.$$

Then for  $\chi, \psi \in \text{Irr}(G)$ ,

$$\langle \chi, \psi \rangle = \delta_{\chi, \psi}.$$

Equivalently, if  $C_1, \dots, C_r$  are the conjugacy classes of  $G$ , then

$$\frac{1}{|G|} \sum_{\ell=1}^r |C_\ell| \chi(C_\ell) \overline{\psi(C_\ell)} = \delta_{\chi, \psi}.$$

**Theorem 3.6** (Column orthogonality of irreducible characters). *Let  $G$  be a finite group, and let  $\text{Irr}(G)$  be the set of irreducible complex characters of  $G$ . If  $g, h \in G$ , then*

$$\sum_{\chi \in \text{Irr}(G)} \chi(g) \overline{\chi(h)} = \begin{cases} |C_G(g)|, & g \text{ is conjugate to } h, \\ 0, & g \text{ is not conjugate to } h. \end{cases}$$

In particular,

$$\sum_{\chi \in \text{Irr}(G)} |\chi(g)|^2 = |C_G(g)|.$$

**Theorem 3.7** (Basic facts on ordinary characters of finite groups). *Let  $G$  be a finite group, and let  $\text{Irr}(G)$  denote the set of irreducible complex characters of  $G$ . Then the following facts hold.*

(i) *Every character is a class function. That is, if  $g, h \in G$  are conjugate, then*

$$\chi(g) = \chi(h).$$

(ii) *The irreducible characters form an orthonormal basis of the space of complex class functions on  $G$ , with respect to the inner product*

$$\langle \chi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \chi(g) \overline{\psi(g)}.$$

In particular,

$$\langle \chi, \psi \rangle = \delta_{\chi, \psi}$$

for  $\chi, \psi \in \text{Irr}(G)$ .

(iii) *The number of irreducible characters equals the number of conjugacy classes of  $G$ :*

$$|\text{Irr}(G)| = |\text{Cl}(G)|.$$

- (iv) The one-dimensional characters of  $G$  are exactly the characters inflated from the abelianization  $G/G'$ , where  $G' = [G, G]$ . Equivalently,

$$\text{Irr}_1(G) \cong \text{Hom}(G/G', \mathbb{C}^\times).$$

In particular,

$$|\text{Irr}_1(G)| = |G/G'|.$$

- (v) The regular character  $\rho_{\text{reg}}$  satisfies

$$\rho_{\text{reg}}(1) = |G|, \quad \rho_{\text{reg}}(g) = 0 \quad (g \neq 1),$$

and decomposes as

$$\rho_{\text{reg}} = \sum_{\chi \in \text{Irr}(G)} \chi(1)\chi.$$

- (vi) The degrees of the irreducible characters satisfy

$$|G| = \sum_{\chi \in \text{Irr}(G)} \chi(1)^2.$$

Moreover, for every  $\chi \in \text{Irr}(G)$ ,

$$\chi(1) \mid |G|.$$

- (vii) The row orthogonality relations are

$$\frac{1}{|G|} \sum_{C \in \text{Cl}(G)} |C| \chi(C) \overline{\psi(C)} = \delta_{\chi, \psi}.$$

- (viii) The column orthogonality relations are as follows. If  $g, h \in G$ , then

$$\sum_{\chi \in \text{Irr}(G)} \chi(g) \overline{\chi(h)} = \begin{cases} |C_G(g)|, & g \text{ is conjugate to } h, \\ 0, & g \text{ is not conjugate to } h. \end{cases}$$

In particular,

$$\sum_{\chi \in \text{Irr}(G)} |\chi(g)|^2 = |C_G(g)|.$$

- (ix) If  $z \in Z(G)$  and  $\chi \in \text{Irr}(G)$ , then  $z$  acts by a scalar on any representation affording  $\chi$ . Hence

$$|\chi(z)| = \chi(1).$$

Consequently,

$$\chi(1)^2 \leq |G : Z(G)|.$$

- (x) If  $N \triangleleft G$ , then inflation gives an injection

$$\text{Irr}(G/N) \hookrightarrow \text{Irr}(G).$$

Its image consists exactly of those irreducible characters  $\chi$  of  $G$  such that

$$N \subset \ker \chi.$$

(xi) For every character  $\chi$ , its kernel is

$$\ker \chi = \{g \in G \mid \chi(g) = \chi(1)\}.$$

If  $\chi$  is irreducible, this is the kernel of any representation affording  $\chi$ , and hence is a normal subgroup of  $G$ .

(xii) If  $H \leq G$ , then induction and restriction satisfy Frobenius reciprocity:

$$\langle \text{Ind}_H^G \varphi, \chi \rangle_G = \langle \varphi, \text{Res}_H^G \chi \rangle_H$$

for every character  $\varphi$  of  $H$  and every character  $\chi$  of  $G$ .

(xiii) If  $G$  acts on a finite set  $X$ , then the associated permutation character is

$$\pi_X(g) = |\text{Fix}_X(g)|.$$

(xiv) If  $\chi$  and  $\psi$  are characters of  $G$ , then  $\chi + \psi$  is the character of a direct sum and  $\chi\psi$  is the character of a tensor product. The character of the dual representation is

$$\chi^\vee(g) = \chi(g^{-1}) = \overline{\chi(g)}.$$

(xv) For every character  $\chi$  and every  $g \in G$ , the value  $\chi(g)$  is an algebraic integer. In particular, if  $\chi(g) \in \mathbb{Q}$ , then  $\chi(g) \in \mathbb{Z}$ .

(xvi) If  $A \triangleleft G$  is abelian, then for every  $\chi \in \text{Irr}(G)$ ,

$$\chi(1) \mid |G : A|.$$

## 4 Week 2

### Problem 1

Let  $R$  be a ring, and let  $M$  be an  $R$ -module. Suppose that  $S$  is a ring and  $\theta : S \rightarrow R$  is a ring homomorphism satisfying  $\theta(1_S) = 1_R$ . Let  $S$  act on  $M$  by  $sx = \theta(s)x$ , where  $x \in M, s \in S$ . Prove that  $M$  is an  $S$ -module.

**Solution:** Define  $s \cdot_S x = \theta(s) \cdot_R x$ . We verify the axioms:

- $s \cdot_S (x + y) = \theta(s) \cdot_R (x + y) = \theta(s) \cdot_R x + \theta(s) \cdot_R y = s \cdot_S x + s \cdot_S y$ .
- $(s_1 + s_2) \cdot_S x = \theta(s_1 + s_2) \cdot_R x = (\theta(s_1) + \theta(s_2)) \cdot_R x = s_1 \cdot_S x + s_2 \cdot_S x$ .
- $(s_1 s_2) \cdot_S x = \theta(s_1 s_2) \cdot_R x = \theta(s_1) \cdot_R (\theta(s_2) \cdot_R x) = s_1 \cdot_S (s_2 \cdot_S x)$ .
- $1_S \cdot_S x = \theta(1_S) \cdot_R x = 1_R \cdot_R x = x$ .

Thus  $M$  is an  $S$ -module. ■

### Problem 2

Let  $R$  be a ring, and let  $M$  and  $N$  be two right  $R$ -modules. Let  $\text{Hom}_{\mathbb{Z}}(M, N)$  be the set of group homomorphisms from  $M$  to  $N$  (as abelian groups). The action of  $R$  on  $\text{Hom}_{\mathbb{Z}}(M, N)$  is given as follows: for any  $a \in R, \varphi \in \text{Hom}_{\mathbb{Z}}(M, N)$ , define

$$(\varphi a)(x) = \varphi(xa), \quad \forall x \in M.$$

Prove that  $\text{Hom}_{\mathbb{Z}}(M, N)$  is a left  $R$ -module.

**Solution:** For any  $a, b \in R$ ,  $\varphi, \psi \in \text{Hom}_{\mathbb{Z}}(M, N)$ , and  $x \in M$ :

- $(a \cdot (\varphi + \psi))(x) = (\varphi + \psi)(x \cdot a) = \varphi(x \cdot a) + \psi(x \cdot a) = (a \cdot \varphi)(x) + (a \cdot \psi)(x).$
- $((a + b) \cdot \varphi)(x) = \varphi(x \cdot (a + b)) = \varphi(x \cdot a + x \cdot b) = \varphi(x \cdot a) + \varphi(x \cdot b) = (a \cdot \varphi)(x) + (b \cdot \varphi)(x).$
- $((a \cdot b) \cdot \varphi)(x) = \varphi(x \cdot (a \cdot b)) = \varphi((x \cdot a) \cdot b) = (b \cdot \varphi)(x \cdot a) = (a \cdot (b \cdot \varphi))(x).$
- $(1_R \cdot \varphi)(x) = \varphi(x \cdot 1_R) = \varphi(x)$ , so  $1_R \cdot \varphi = \varphi$ .

Thus,  $\text{Hom}_{\mathbb{Z}}(M, N)$  is a left  $R$ -module. ■

### Problem 3

Let  $R$  be a ring, and let  $M$  be an  $R$ -module. Denote  $\text{Ann}_R M = \{r \in R \mid rm = 0, \forall m \in M\}$ . We call  $M$  a **faithful**  $R$ -module if  $\text{Ann}_R M = \{0\}$ . Define the action of the quotient ring  $R/\text{Ann}_R M$  on  $M$  by  $\bar{r}x = rx$ , where  $x \in M, r \in R, \bar{r} = r + \text{Ann}_R M$ . Prove that  $M$  is a faithful  $R/\text{Ann}_R M$ -module.

**Solution:** Let  $I = \text{Ann}_R M$  and  $S = R/I$ .

1.  **$M$  is an  $S$ -module:** The action is well-defined because if  $\bar{r} = \bar{r}'$ , then  $r - r' \in I$ , so  $(r - r') \cdot_R x = 0$ , implying  $r \cdot_R x = r' \cdot_R x$ . The module axioms follow directly from the  $R$ -module structure.
2.  **$M$  is faithful over  $S$ :** Suppose  $\bar{r} \in \text{Ann}_S M$ . Then for all  $x \in M$ ,

$$\bar{r} \cdot_S x = r \cdot_R x = 0.$$

By definition of  $I$ , this implies  $r \in I$ . Thus  $\bar{r} = r + I = \bar{0}$  in  $S$ . Since  $\text{Ann}_S M = \{\bar{0}\}$ ,  $M$  is a faithful  $S$ -module. ■

### Problem 4

Let  $\varphi : M \rightarrow M$  be a homomorphism of  $R$ -modules such that  $\varphi\varphi = \varphi$ . Prove that

$$M = \ker \varphi \oplus \text{im } \varphi.$$

**Solution:** Let  $\psi = \text{id} - \varphi$ . First, we observe that:

$$\varphi \cdot \psi = \varphi \cdot (\text{id} - \varphi) = \varphi - \varphi^2 = \varphi - \varphi = 0,$$

$$\psi \cdot \varphi = (\text{id} - \varphi) \cdot \varphi = \varphi - \varphi^2 = 0.$$

We claim that  $M = \text{im } \psi \oplus \text{im } \varphi$  and  $\text{im } \psi = \ker \varphi$ .

1. For any  $x \in M$ , we can write:

$$x = \text{id}(x) = (\psi + \varphi)(x) = \psi(x) + \varphi(x).$$

Since  $\psi(x) \in \text{im } \psi$  and  $\varphi(x) \in \text{im } \varphi$ , it is clear that  $M = \text{im } \psi + \text{im } \varphi$ .

2. Since  $\varphi \cdot \psi = 0$ , for any  $y = \psi(z) \in \text{im } \psi$ , we have  $\varphi(y) = \varphi(\psi(z)) = 0$ . Thus  $\text{im } \psi \subseteq \ker \varphi$ . If  $x \in \ker \varphi$ , then  $x = \varphi(x) + \psi(x) = \psi(x)$ . This implies  $x \in \text{im } \psi$ , so  $\ker \varphi \subseteq \text{im } \psi$ . Hence,  $\text{im } \psi = \ker \varphi$ .

3. Suppose  $x \in \text{im } \psi \cap \text{im } \varphi$ . Using the identity from step 2:

$$x \in \text{im } \psi \implies x \in \ker \varphi \implies \varphi(x) = 0.$$

Meanwhile,  $x \in \text{im } \varphi \implies x = \varphi(y)$  for some  $y \in M$ . Applying  $\varphi$  to both sides:  $\varphi(x) = \varphi^2(y) = \varphi(y) = x$ . Since  $\varphi(x) = 0$  and  $\varphi(x) = x$ , we must have  $x = 0$ .

Therefore,  $M = \ker \varphi \oplus \text{im } \varphi$ . ■

## Problem 5

Show that any  $R$ -module is a homomorphic image of some free  $R$ -module.

**Solution:** Let  $M$  be an  $R$ -module. We construct a free  $R$ -module  $F$  using the elements of  $M$  as a basis. Specifically, let  $F = \bigoplus_{m \in M} R \cdot e_m$ , where  $\{e_m\}_{m \in M}$  is the set of free generators indexed by  $M$ .

By the universal property of free modules, there exists a unique  $R$ -module homomorphism  $\pi : F \rightarrow M$  such that  $\pi(e_m) = m$  for all  $m \in M$ . For any element in  $F$ , the map is defined as:

$$\pi \left( \sum_{i=1}^n r_i \cdot e_{m_i} \right) = \sum_{i=1}^n r_i \cdot m_i.$$

For any  $m \in M$ , there is a corresponding generator  $e_m \in F$  such that  $\pi(e_m) = m$ . This shows that  $\pi$  is a surjective homomorphism. Therefore,  $M = \text{im } \pi$  is a homomorphic image of the free  $R$ -module  $F$ . ■

## Problem 6

Let  $\mathbb{Q}$  be the field of rational numbers. Is the additive group  $(\mathbb{Q}, +)$  a free  $\mathbb{Z}$ -module? Explain your answer.

**Solution:** Suppose  $\mathbb{Q}$  is free over  $\mathbb{Z}$ . For any two non-zero elements  $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$  (where  $a, b, c, d \in \mathbb{Z} \setminus \{0\}$ ), they are  $\mathbb{Z}$ -linearly dependent since:

$$(b \cdot c) \cdot \frac{a}{b} - (a \cdot d) \cdot \frac{c}{d} = (a \cdot c) - (a \cdot c) = 0.$$

This implies that if  $\mathbb{Q}$  were free, its rank must be 1.

However, a free  $\mathbb{Z}$ -module of rank 1 is isomorphic to  $\mathbb{Z}$  itself, which is a cyclic group.  $\mathbb{Q}$  is not cyclic because for any  $q \in \mathbb{Q}$ , the element  $\frac{q}{2}$  cannot be expressed as  $n \cdot q$  for any  $n \in \mathbb{Z}$ . Thus,  $\mathbb{Q}$  cannot be a free  $\mathbb{Z}$ -module. ■

# 5 Week 3

## Problem 1

Let  $R$  be a commutative ring, and suppose that  $\varphi$  is an endomorphism of the free  $R$ -module  $R^m$ . Prove that if  $\varphi$  is surjective, then  $\varphi$  is injective. Consider that: if  $\varphi$  is injective, is  $\varphi$  surjective?

**Solution:** Suppose  $\{e_1, \dots, e_m\}$  is a basis of  $R^m$  and  $v_j = \varphi(e_j) = \sum_{i=1}^m a_{ij}e_i$ . Then we represent  $\varphi$  by matrix  $A = (a_{ij}) \in M_m(R)$ , we have

$$(v_1, v_2, \dots, v_m) = (e_1, e_2, \dots, e_m)A. \quad (\clubsuit)$$

Surjectivity implies that  $\{e_i\}$  can be linearly represented by  $\{v_i\}$ . Equivalently, there exists an  $m \times m$  matrix  $B = (b_{ij}) \in M_m(R)$  such that

$$(e_1, e_2, \dots, e_m) = (v_1, v_2, \dots, v_m)B. \quad (\spadesuit)$$

Combine  $\spadesuit$  and  $\clubsuit$ , we obtain that

$$(e_1, e_2, \dots, e_m) = (e_1, e_2, \dots, e_m)AB.$$

$\{e_i\}$  is a basis implies that  $AB = I$  and hence  $\det(A)\det(B) = 1$ ,  $\det(A)$  is a unit in  $R$ . Let  $A^*$  be adjoint matrix of  $A$ , then  $A^*A = \det(A)I$ . Let  $c = (c_1, \dots, c_m)^T \in R^m$  such that  $\sum c_i v_i = 0$ , then

$$(e_1, e_2, \dots, e_m)Ac = (v_1, v_2, \dots, v_m)c = 0$$

implies  $Ac = 0$ , whence  $c = \det(A)^{-1}A^*Ac = 0$ . Therefore,  $\{v_i\}$ , as linearly independent generators of  $R^m$ , is a basis. If  $w = \sum c_i e_i \in \ker(\varphi)$ , then  $\sum c_i v_i = 0$  implies  $c_i = 0$ . The injectivity of  $\varphi$  holds.

The converse is not true. Take  $R = \mathbb{Z}$  and consider an injective endomorphism

$$\begin{aligned} \varphi : \mathbb{Z}^m &\rightarrow \mathbb{Z}^m \\ (n_1, \dots, n_m) &\mapsto (2n_1, \dots, 2n_m), \end{aligned}$$

which is not surjective. ■

## Problem 2

Let  $\varphi : M \rightarrow N$  be a homomorphism of  $R$ -modules. Prove that:

1.  $\varphi$  is a monomorphism if and only if for any  $R$ -module  $T$  and any two  $R$ -module homomorphisms  $\xi, \eta : T \rightarrow M$ ,  $\varphi\xi = \varphi\eta$  implies  $\xi = \eta$ ;
2.  $\varphi$  is an epimorphism if and only if for any  $R$ -module  $S$  and any two  $R$ -module homomorphisms  $\rho, \theta : N \rightarrow S$ ,  $\rho\varphi = \theta\varphi$  implies  $\rho = \theta$ .

**Solution:** (1) Let  $\varphi$  be a monomorphism and  $\xi, \eta : T \rightarrow M$  be arbitrary module homomorphisms. Let  $f = \xi - \eta$ , then  $\varphi f = 0$  implies  $\text{Im}(f) \subseteq \ker \varphi = \{0\}$ . Therefore, It follows that  $f = 0$  and  $\xi = \eta$ . Conversely, setting  $T = \ker(\varphi)$  and  $\xi = \text{id}, \eta = 0$ , then  $\varphi\xi = \varphi\eta$  implies  $\xi = \eta$ . Consequently,  $\text{id} = 0$  yields  $\ker(\varphi) = T = 0$ ,  $\varphi$  is injective.

(2) Let  $\varphi$  be an epimorphism and  $\rho, \theta : N \rightarrow S$  be arbitrary module homomorphisms. Let  $g = \rho - \theta$ , then  $g\varphi = 0$  implies  $N = \text{Im}(\varphi) \subseteq \ker(g) \subseteq N$ , it follows  $\ker(g) = N$  and  $g = 0, \rho = \theta$ . Conversely, setting  $S = \text{coker}(\varphi) = N/\text{Im}(\varphi)$ , let  $\rho$  be the canonical module homomorphism and  $\theta$  be a null homomorphism.  $\rho\varphi = \theta\varphi$  implies  $\rho = \theta$  is a null homomorphism, which yields  $\text{Im}(\varphi) = N$ . Therefore, the surjectivity of  $\varphi$  holds. ■



### Problem 3

Let  $M$  and  $M_i (1 \leq i \leq n)$  be  $R$ -modules. Show that  $M \simeq \bigoplus_{i=1}^n M_i$  if and only if for every  $i \in \{1, \dots, n\}$ , there exists an  $\varphi_i \in \text{End}_R(M)$  such that  $\text{im } \varphi_i \simeq M_i$ ,  $\varphi_i \varphi_j = 0$  (for  $i \neq j$ ), and  $\varphi_1 + \varphi_2 + \dots + \varphi_n = \text{id}_M$ .

**Solution:** Assume  $M \simeq \bigoplus_{i=1}^n M_i$ . Without loss of generality, identify  $M$  with  $\bigoplus_{i=1}^n M_i$ . Let  $\pi_i : M \rightarrow M_i$  be the natural projection and  $\iota_i : M_i \rightarrow M$  be the natural inclusion. Define  $\varphi_i = \iota_i \circ \pi_i \in \text{End}_R(M)$ . Then  $\text{im}(\varphi_i) = \iota_i(M_i) \simeq M_i$  and for  $i \neq j$ ,  $\varphi_i \varphi_j = (\iota_i \pi_i)(\iota_j \pi_j) = \iota_i \circ (\pi_i \iota_j) \circ \pi_j = 0$ . For any  $x = (m_1, \dots, m_n) \in M$ , we have  $\sum_{i=1}^n \varphi_i(x) = \sum_{i=1}^n (0, \dots, m_i, \dots, 0) = x$ . Thus,  $\sum_{i=1}^n \varphi_i = \text{id}_M$ .

Assume there exist  $\varphi_1, \dots, \varphi_n \in \text{End}_R(M)$  satisfying the given conditions. Let  $N_i = \text{im } \varphi_i$ . By assumption,  $N_i \simeq M_i$ . Firstly, note that  $\varphi_i$  is idempotent. Multiplying  $\sum_{j=1}^n \varphi_j = \text{id}_M$  by  $\varphi_i$  gives  $\sum_{j=1}^n \varphi_i \varphi_j = \varphi_i$ . Since  $\varphi_i \varphi_j = 0$  for  $i \neq j$ , this reduces to  $\varphi_i^2 = \varphi_i$ . Consequently, for any  $x \in N_i$ ,  $\varphi_i(x) = x$ .  $M$  is the sum of  $N_i$  because for any  $x \in M$ ,  $x = \text{id}_M(x) = \sum_{i=1}^n \varphi_i(x)$ . Since  $\varphi_i(x) \in N_i$ , we have  $M = \sum_{i=1}^n N_i$ . Suppose  $\sum_{j=1}^n x_j = 0$  where  $x_j \in N_j$ . Apply  $\varphi_i$  to both sides:

$$\varphi_i \left( \sum_{j=1}^n x_j \right) = \varphi_i(0) = 0.$$

Because  $x_j \in \text{im } \varphi_j$ , we can write  $x_j = \varphi_j(y_j)$  for some  $y_j \in M$ . For  $j \neq i$ ,  $\varphi_i(x_j) = \varphi_i \varphi_j(y_j) = 0$ . Thus, the sum collapses to  $\varphi_i(x_i) = 0$ . Since  $x_i \in N_i$ ,  $x_i = \varphi_i(x_i) = 0$ . This holds for all  $i$ , proving the sum is direct. Therefore,  $M = \bigoplus_{i=1}^n N_i \simeq \bigoplus_{i=1}^n M_i$ . ■

### Problem 4

1. Prove  $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m, n)\mathbb{Z}$ , where  $m, n \in \mathbb{Z}$ .
2. Let  $A$  be an abelian group, prove  $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$ .
3. Let  $A, B$  be two finitely generated abelian groups. What is  $A \otimes_{\mathbb{Z}} B$ ?

**Solution:** (1) Let  $d = (m, n)$ . First, consider the map

$$\begin{aligned} f : \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} &\rightarrow \mathbb{Z}/d\mathbb{Z} \\ (a \bmod m, b \bmod n) &\mapsto ab \bmod d. \end{aligned}$$

$f$  is well-defined because if we suppose  $a' = a + km$  and  $b' = b + ln$  for some integers  $k, l$  then  $a'b' = (a + km)(b + ln) \equiv ab \bmod d$ . Furthermore, multiplication in  $\mathbb{Z}/d\mathbb{Z}$  naturally distributes over addition, making  $f$  a  $\mathbb{Z}$ -bilinear map. By the universal property of the tensor product, this bilinear map induces a unique  $\mathbb{Z}$ -module homomorphism

$$\begin{aligned} \phi : \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} &\rightarrow \mathbb{Z}/d\mathbb{Z} \\ a \otimes b &\mapsto ab \bmod d. \end{aligned}$$

Next, we construct the inverse homomorphism. Consider the  $\mathbb{Z}$ -module homomorphism

$$\begin{aligned} \psi : \mathbb{Z}/d\mathbb{Z} &\rightarrow \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \\ c \bmod d &\mapsto c(1 \otimes 1), \end{aligned}$$

which is well-defined because there exist integers  $x, y$  such that  $xm + yn = d$  by Bézout's identity and thus

$$d(1 \otimes 1) = (xm + yn)(1 \otimes 1) = xm(1 \otimes 1) + yn(1 \otimes 1) = x(m \otimes 1) + y(1 \otimes n) = 0,$$

then  $\psi(c + kd) = \psi(c)$ . Finally, we verify that  $\phi$  and  $\psi$  are mutually inverse. For any  $c \bmod d \in \mathbb{Z}/d\mathbb{Z}$ , we have  $\phi(\psi(c)) = \phi(c(1 \otimes 1)) = c \cdot \phi(1 \otimes 1) = c \cdot 1 = c \bmod d$ , so  $\phi \circ \psi = \text{id}_{\mathbb{Z}/d\mathbb{Z}}$ . Conversely, for any pure tensor  $a \otimes b \in \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$ , we have  $\psi(\phi(a \otimes b)) = \psi(ab) = ab(1 \otimes 1) = a \otimes b$ . Since the pure tensors generate the entire tensor product,  $\psi \circ \phi = \text{id}_{\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}}$ . Therefore,  $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m, n)\mathbb{Z}$ .

(2) First, consider the map

$$\begin{aligned} f : A \times \mathbb{Z}/n\mathbb{Z} &\rightarrow A/nA \\ (a, \bar{k}) &\mapsto ka \bmod nA, \end{aligned}$$

where  $\bar{k} = k \bmod n\mathbb{Z}$ . To verify this is well-defined, suppose  $k' = k + cn$  for some integer  $c$ . Then  $f(a, \bar{k}') = k'a \bmod nA = ka + cna \bmod nA = ka \bmod nA$ , since  $A$  is a  $\mathbb{Z}$ -module and  $cna \in nA$ . Furthermore,  $f$  naturally preserves addition in both components, making it a  $\mathbb{Z}$ -bilinear map. By the universal property of the tensor product, this induces a unique  $\mathbb{Z}$ -module homomorphism

$$\begin{aligned} \phi : A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} &\rightarrow A/nA \\ a \otimes \bar{k} &\mapsto ka \bmod nA. \end{aligned}$$

Next, we construct the inverse homomorphism. Consider the module homomorphism

$$\begin{aligned} \psi : A/nA &\rightarrow A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \\ a \bmod nA &\mapsto a \otimes \bar{1}, \end{aligned}$$

which is well-defined because if  $a' = a + n\alpha$  where  $\alpha \in A$  then  $\psi(a') = a \otimes \bar{1} + \alpha \otimes \bar{n} = a \otimes \bar{1}$ .

Finally, we verify that  $\phi$  and  $\psi$  are mutually inverse. For any class  $a + nA \in A/nA$ , we have  $\phi(\psi(a + nA)) = \phi(a \otimes \bar{1}) = 1a \bmod nA = a \bmod nA$ , meaning  $\phi \circ \psi = \text{id}_{A/nA}$ . Conversely, for any pure tensor  $a \otimes \bar{k} \in A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$ , we can rewrite it using scalar multiplication as  $a \otimes \bar{k} = ka \otimes \bar{1}$ . Applying the composition yields  $\psi(\phi(ka \otimes \bar{1})) = \psi(ka \bmod nA) = ka \otimes \bar{1} = a \otimes \bar{k}$ . Since pure tensors generate the entire tensor product,  $\psi \circ \phi = \text{id}_{A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}}$ . Therefore,  $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$ .

(3) By the Structure Theorem for Finitely Generated Abelian Groups, suppose

$$A \simeq \mathbb{Z}^r \oplus \bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z}, \quad B \simeq \mathbb{Z}^s \oplus \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z}$$

where  $r, s \geq 0$ ,  $\alpha_i, \beta_j > 0$  and  $p_i, q_j$  are prime numbers. Since the tensor product functor distributes over arbitrary direct sums, we can expand  $A \otimes_{\mathbb{Z}} B$  into four distinct blocks:

$$A \otimes_{\mathbb{Z}} B \simeq (\mathbb{Z}^r \otimes_{\mathbb{Z}} \mathbb{Z}^s) \oplus \left( \mathbb{Z}^r \otimes_{\mathbb{Z}} \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \right) \oplus \left( \bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}^s \right) \oplus \left( \bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \otimes_{\mathbb{Z}} \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \right).$$

By (2) and distributive law, we obtain that

$$\mathbb{Z}^r \otimes_{\mathbb{Z}} \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \simeq \bigoplus_{j=1}^l \left( \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \right)^r \quad \text{and} \quad \bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}^s \simeq \bigoplus_{i=1}^k \left( \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \right)^s.$$

Hence, (1) implies

$$\bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \otimes_{\mathbb{Z}} \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \simeq \bigoplus_{p_i=q_j=p} \mathbb{Z}/p^{\min(\alpha_i, \beta_j)} \mathbb{Z}.$$

Therefore we obtain that

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rs} \oplus \bigoplus_{j=1}^l \left( \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \right)^r \oplus \bigoplus_{i=1}^k \left( \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \right)^s \oplus \bigoplus_{p_i=q_j=p} \mathbb{Z}/p^{\min(\alpha_i, \beta_j)} \mathbb{Z}.$$

Alternatively, if we let  $T(A)$  and  $T(B)$  denote the torsion subgroups of  $A$  and  $B$  respectively, this entire result can be represented as

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rs} \oplus T(B)^r \oplus T(A)^s \oplus (T(A) \otimes_{\mathbb{Z}} T(B)).$$

■

### Problem 5

Let  $\varphi : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$  be the unique monomorphism. Prove that

$$\text{id} \otimes \varphi : \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$$

is a zero homomorphism.

**Solution:** The unique monomorphism  $\varphi : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$  must map the generator  $\bar{1} \in \mathbb{Z}/2\mathbb{Z}$  to the unique element of order 2 in  $\mathbb{Z}/4\mathbb{Z}$ , which is  $\bar{2}$ . Therefore,  $\varphi(\bar{1}) = \bar{2}$ . Thus, we obtain that

$$(\text{id} \otimes \varphi)(\bar{1} \otimes \bar{1}) = \text{id}(\bar{1}) \otimes \varphi(\bar{1}) = \bar{1} \otimes \bar{2} = 2(\bar{1}) \otimes \bar{1} = 0.$$

Therefore the only non-zero element  $\bar{1} \otimes \bar{1}$  in  $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$  is mapped to 0 meaning that  $\text{id} \otimes \varphi$  is a null homomorphism. ■

### Problem 6

Let  $V_1$  and  $V_2$  be two vector spaces over a field  $F$  with dimension  $n$  and  $m$  respectively. Let  $\{\epsilon_1, \dots, \epsilon_n\}$  and  $\{\eta_1, \dots, \eta_m\}$  be bases of  $V_1$  and  $V_2$  respectively. Suppose that  $\mathcal{A}$  (resp.  $\mathcal{B}$ ) is an  $F$ -linear map of  $V_1$  (resp.  $V_2$ ), and it is represented by the matrix  $A = (a_{ij})_{n \times n}$  (resp.  $B = (b_{ij})_{m \times m}$ ) with respect to the above bases. Prove that

$$\{\epsilon_i \otimes \eta_j \mid 1 \leq i \leq n, 1 \leq j \leq m\}$$

is an  $F$ -basis of  $V_1 \otimes_F V_2$ . Determine the matrix representing the  $F$ -linear map  $\mathcal{A} \otimes \mathcal{B}$  under this basis.

**Solution:** We first prove that the set  $\{\epsilon_i \otimes \eta_j \mid 1 \leq i \leq n, 1 \leq j \leq m\}$  forms an  $F$ -basis for  $V_1 \otimes_F V_2$ . For arbitrary  $v = \sum_{i=1}^n x_i \epsilon_i \in V_1$  and  $w = \sum_{j=1}^m y_j \eta_j \in V_2$ , we obtain that

$$v \otimes w = \left( \sum_{i=1}^n x_i \epsilon_i \right) \otimes \left( \sum_{j=1}^m y_j \eta_j \right) = \sum_{i=1}^n \sum_{j=1}^m (x_i y_j) (\epsilon_i \otimes \eta_j).$$

Since pure tensors of the form  $v \otimes w$  generate the space  $V_1 \otimes_F V_2$ , and any such pure tensor can be expressed as a linear combination of  $\{\epsilon_i \otimes \eta_j\}$ , this set spans  $V_1 \otimes_F V_2$ . To establish linear independence, suppose there exist scalars  $c_{ij} \in F$  such that  $\sum_{i=1}^n \sum_{j=1}^m c_{ij}(\epsilon_i \otimes \eta_j) = 0$ . There are natural isomorphisms

$$\text{Hom}_F(V_1 \otimes_F V_2, F) \simeq \text{Hom}_F(V_1, V_2^*) \simeq V_1^* \otimes_F V_2^*.$$

The first isomorphism arises directly from the Tensor-Hom adjunction and the second holds because of finite dimension of  $V_1$ . Therefore for any  $1 \leq k \leq n, 1 \leq l \leq m$ , we obtain that

$$c_{kl} = (\epsilon_k^* \otimes \eta_l^*) \left( \sum_{i=1}^n \sum_{j=1}^m c_{ij}(\epsilon_i \otimes \eta_j) \right) = 0$$

meaning that  $\{\epsilon_i \otimes \eta_j\}$  is a basis of  $V_1 \otimes V_2$ .

Next, we determine the matrix representing the  $F$ -linear map  $\mathcal{A} \otimes \mathcal{B}$  under this basis. By definition, the given matrices  $A = (a_{ij})_{n \times n}$  and  $B = (b_{ij})_{m \times m}$  describe the action of the linear maps on their respective bases as  $\mathcal{A}(\epsilon_j) = \sum_{i=1}^n a_{ij} \epsilon_i$  and  $\mathcal{B}(\eta_l) = \sum_{k=1}^m b_{kl} \eta_k$ . Evaluating the action of  $\mathcal{A} \otimes \mathcal{B}$  on a basis element  $\epsilon_j \otimes \eta_l$  yields:

$$(\mathcal{A} \otimes \mathcal{B})(\epsilon_j \otimes \eta_l) = \mathcal{A}(\epsilon_j) \otimes \mathcal{B}(\eta_l) = \left( \sum_{i=1}^n a_{ij} \epsilon_i \right) \otimes \left( \sum_{k=1}^m b_{kl} \eta_k \right) = \sum_{i=1}^n \sum_{k=1}^m (a_{ij} b_{kl}) (\epsilon_i \otimes \eta_k).$$

If we order the  $nm$  basis vectors lexicographically (i.e.,  $(1, 1), (1, 2), \dots, (1, m), (2, 1), \dots$ ), the resulting  $nm \times nm$  matrix representation is the Kronecker product of the matrices

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{pmatrix}$$

■

## 6 Week 4

### Problem 1

Let  $M, N, M_i (i \in I)$  and  $N_j (j \in J)$  be  $R$ -modules. Prove the following isomorphisms of abelian groups.

- (1)  $\text{Hom}_R(\bigoplus_{i \in I} M_i, N) \simeq \prod_{i \in I} \text{Hom}_R(M_i, N);$
- (2)  $\text{Hom}_R\left(M, \prod_{j \in J} N_j\right) \simeq \prod_{j \in J} \text{Hom}_R(M, N_j).$

**Solution:** (1) Let  $X = \text{Hom}_R(\bigoplus_{i \in I} M_i, N)$  and  $Y_i = \text{Hom}_R(M_i, N)$ . Let  $G$  be an arbitrary abelian group, and suppose we are given a family of group homomorphisms  $\varphi_i : G \rightarrow Y_i$  for each  $i \in I$ . For fixed  $g \in G$ , the family  $\{\varphi_i(g)\}_{i \in I}$  is a collection of  $R$ -module homomorphisms from  $M_i$  to  $N$ . Then by the universal property of the direct sum  $\bigoplus_{i \in I} M_i$  in  $R\text{-Mod}$ , there exists a unique  $R$ -module homomorphism  $f_g$  such that

$$f_g \circ \iota_i = \varphi_i(g) \quad \text{for all } i \in I.$$

where  $\iota_i$  are the canonical injections. Define  $\Phi : G \rightarrow X$  by setting  $\Phi(g) = f_g$ . I claim that  $\Phi$  is the unique group homomorphism such that the diagram

$$\begin{array}{ccc} G & & \\ \varphi_i \downarrow & \searrow \Phi & \\ Y_i & \xleftarrow{\pi_i} & X \end{array}$$

commutes where  $\pi_i = \text{Hom}_R(\iota_i, N)$  is the canonical projection induced by contravariant functor  $\text{Hom}_R(-, N)$ . Let  $g, h \in G$ . Since  $\varphi_i(g + h) = \varphi_i(g) + \varphi_i(h)$  we obtain that

$$f_{g+h} \circ \iota_i = \varphi_i(g) + \varphi_i(h) = (f_g + f_h) \circ \iota_i.$$

Then  $f_{g+h} = f_g + f_h$ , or equivalently  $\Phi(g + h) = \Phi(g) + \Phi(h)$ , by the uniqueness condition of the direct sum's universal property. Therefore,  $\Phi$  is a group homomorphism. To prove the uniqueness of  $\Phi$ , we suppose there is another group homomorphism  $\Psi : G \rightarrow X$  satisfying  $(\Psi(g)) \circ \iota_i = \varphi_i(g)$  for all  $i \in I$  and all  $g \in G$ . For any fixed  $g$ , both  $R$ -module homomorphisms  $\Psi(g)$  and  $\Phi(g)$  pull back along  $\iota_i$  to the same maps  $\varphi_i(g)$ . By the universal property of the direct sum, we must have  $\Psi(g) = \Phi(g)$  for all  $g$ , so  $\Psi = \Phi$ . The uniqueness and existence of  $\Phi$  yields  $\text{Hom}_R(\bigoplus_{i \in I} M_i, N) \simeq \prod_{i \in I} \text{Hom}_R(M_i, N)$ .

(2) Let  $X = \text{Hom}_R(M, \prod_{j \in J} N_j)$  and  $Y_j = \text{Hom}_R(M, N_j)$ . Let  $G$  be an arbitrary abelian group, and suppose we are given a family of group homomorphisms  $\varphi_j : G \rightarrow Y_j$  for each  $j \in J$ . Then for fixed  $g \in G$  the family of  $R$ -module homomorphisms  $\{\varphi_j(g) : M \rightarrow N_j\}_{j \in J}$  uniquely induced an  $R$ -module homomorphism  $f_g : M \rightarrow \prod_{j \in J} N_j$  such that  $\varphi_j(g) = p_j \circ f_g$ . Define the map  $\Phi : G \rightarrow \text{Hom}_R(M, \prod_{j \in J} N_j)$  by setting  $\Phi(g) = f_g$ . I claim  $\Phi$  is the unique group homomorphism such that the diagram

$$\begin{array}{ccc} G & & \\ \varphi_j \downarrow & \searrow \Phi & \\ Y_j & \xleftarrow{\pi_j} & X \end{array}$$

commutes where  $\pi_j = \text{Hom}_R(M, p_j)$  is the canonical projection induced by covariant functor  $\text{Hom}_R(M, -)$ . Let  $g, h \in G$ . Since  $\varphi_j(g + h) = \varphi_j(g) + \varphi_j(h)$  we obtain that

$$p_j \circ f_{g+h} = \varphi_j(g) + \varphi_j(h) = p_j \circ (f_g + f_h).$$

It implies  $\Phi(g + h) = \Phi(g) + \Phi(h)$  because of the uniqueness condition of the direct product's universal property. Therefore,  $\Phi$  is a group homomorphism. To prove the uniqueness of  $\Phi$ , we suppose there is another group homomorphism  $\Psi : G \rightarrow X$  satisfying  $p_j \circ (\Psi(g)) = \varphi_j(g)$  for all  $j \in J$  and all  $g \in G$ . For any fixed  $g$ , both  $R$ -module homomorphisms  $\Psi(g)$  and  $\Phi(g)$  push forward along  $p_j$  to the same maps  $\varphi_j(g)$ . By the universal property of the direct product, we must have  $\Psi(g) = \Phi(g)$  for all  $g$ , so  $\Psi = \Phi$ . The uniqueness and existence of  $\Phi$  yields  $\text{Hom}_R(M, \prod_{j \in J} N_j) \simeq \prod_{j \in J} \text{Hom}_R(M, N_j)$ .  $\blacksquare$

*Remark 6.1.* The isomorphism established in (1) is the direct consequence of the **Hom-Hom adjunction** and the fact that right adjoints preserve limits. Specifically, the contravariant functor  $\text{Hom}_R(-, N)$  is the right adjoint of contravariant functor  $\text{Hom}_{\mathbf{Ab}}(-, N)$ . More explicitly, there is a natural isomorphism

$$\text{Hom}_R(M, \text{Hom}_{\mathbf{Ab}}(A, N)) \cong \text{Hom}_{\mathbf{Ab}}(A, \text{Hom}_R(M, N)),$$

which holds for both left and right  $R$ -module categories. Similarly, **(2)** is the direct consequence of the **Tensor-Hom adjunction**. Specifically, the covariant functor  $\text{Hom}_R(M, -)$  is the right adjoint of the covariant functor  $- \otimes_{\mathbb{Z}} M$  if we consider right  $R$ -module category and  $M \otimes_{\mathbb{Z}} -$  for the left.

## Problem 2

(*Five Lemma*) Suppose that the following diagram of homomorphisms of  $R$ -modules

$$\begin{array}{ccccccccc} M_1 & \xrightarrow{f_1} & M_2 & \xrightarrow{f_2} & M_3 & \xrightarrow{f_3} & M_4 & \xrightarrow{f_4} & M_5 \\ \downarrow \varphi_1 & & \downarrow \varphi_2 & & \downarrow \varphi_3 & & \downarrow \varphi_4 & & \downarrow \varphi_5 \\ N_1 & \xrightarrow{g_1} & N_2 & \xrightarrow{g_2} & N_3 & \xrightarrow{g_3} & N_4 & \xrightarrow{g_4} & N_5 \end{array}$$

is commutative and has exact rows. Prove that

- (1) if  $\varphi_1$  is surjective, and  $\varphi_2$  and  $\varphi_4$  are injective, then  $\varphi_3$  is injective;
- (2) if  $\varphi_5$  is injective, and  $\varphi_2$  and  $\varphi_4$  are surjective, then  $\varphi_3$  is surjective.

**Solution:** **(1)** Suppose  $a_3 \in M_3$  such that  $\varphi_3(a_3) = 0$ , then  $\varphi_4 \circ f_3(a_3) = g_3 \circ \varphi_3(a_3) = 0$  implies  $f_3(a_3) = 0$  since  $\varphi_4$  is injective. Therefore,  $\text{Im}(f_2) = \ker(f_3)$  implies that there exists an element  $a_2 \in M_2$  such that  $f_2(a_2) = a_3$ . Let  $b_2 = \varphi_2(a_2)$ , then  $g_2(b_2) = \varphi_3 \circ f_2(a_2) = \varphi_3(a_3) = 0$  implies there exists  $b_1 \in N_1$  such that  $g_1(b_1) = b_2$ . Because  $\varphi_1$  is surjective then there exists  $a_1$  such that  $\varphi_1(a_1) = b_1$ . I claim that  $a_2 = f_1(a_1)$  and therefore  $a_2 \in \text{Im}(f_1) = \ker(f_2)$  implies  $a_3 = f_2(a_2) = 0$  and thus  $\varphi_3$  is injective. To prove it, we can consider  $\varphi_2(a_2 - f_1(a_1)) = \varphi_2(a_2) - g_1 \circ \varphi_1(a_1) = b_2 - b_2 = 0$  which implies  $a_2 - f_1(a_1) = 0$  because of injectivity of  $\varphi_2$ .

**(2)** Suppose  $b_3$  be arbitrary elements in  $N_3$  respectively. Let  $b_4 = g_3(b_3)$ . Because  $\varphi_4$  is surjective then we can let  $a_4 \in M_4$  such that  $\varphi_4(a_4) = b_4$ . Then  $\varphi_5 \circ f_4(a_4) = g_4 \circ \varphi_4(a_4) = g_4 \circ g_3(b_3) = 0$  implies  $f_4(a_4) = 0$  because of injectivity of  $\varphi_5$ . Then we can suppose  $a_3 \in M_3$  such that  $f_3(a_3) = a_4$ . Then  $g_3(b_3 - \varphi_3(a_3)) = b_4 - \varphi_4(a_4) = 0$  implies there exists  $b_2 \in N_2$  such that  $g_2(b_2) = b_3 - \varphi_3(a_3)$ . Then the surjectivity of  $\varphi_2$  implies there exists  $a_2$  such that  $\varphi_2(a_2) = b_2$ . Since diagram commutes, we obtain that  $\varphi_3 \circ f_2(a_2) = b_3 - \varphi_3(a_3)$ , or equivalently,  $b_3 = \varphi_3(f_2(a_2) + a_3) \in \text{Im}(\varphi_3)$  which implies  $\varphi_3$  is surjective. ■

## Problem 3

(*Snake Lemma*) Suppose that the following diagram of homomorphisms of  $R$ -modules

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\ & & \downarrow \varphi' & & \downarrow \varphi & & \downarrow \varphi'' & & \\ 0 & \longrightarrow & N' & \xrightarrow{\mu} & N & \xrightarrow{\nu} & N'' & \longrightarrow & 0 \end{array}$$

is commutative and has exact rows. Prove that

$$0 \rightarrow \ker \varphi' \xrightarrow{\alpha'} \ker \varphi \xrightarrow{\beta'} \ker \varphi'' \xrightarrow{\delta} \text{coker } \varphi' \xrightarrow{\mu'} \text{coker } \varphi \xrightarrow{\nu'} \text{coker } \varphi'' \rightarrow 0$$

is a long exact sequence, where  $\alpha', \beta', \mu', \nu'$  are homomorphisms induced by  $\alpha, \beta, \mu, \nu$ , respectively. (Hint: for any  $x \in \ker \varphi''$ , take  $y \in M$  such that  $\beta(y) = x$ , and take  $z \in N'$  such that  $\mu(z) = \varphi(y)$ . Define  $\delta(x) = \bar{z} = z + \text{im } \varphi'$ )

**Solution:**  $\alpha'$  is well-defined and injective. Suppose  $m' \in \ker(\varphi')$ , since diagram commutes, we obtain that  $\varphi \circ \alpha(m') = \mu \circ \varphi'(m') = 0$  which implies  $\alpha(m') \in \ker(\varphi)$ . Therefore,  $\alpha' = \alpha|_{\ker(\varphi')} : \ker(\varphi') \rightarrow \ker(\varphi)$  is well-defined. Hence,  $\alpha'$  is injective since  $\alpha$  is injective.  $\beta'$  is well-defined and  $\text{Im}(\alpha') = \ker(\beta')$ . Let  $m \in \ker(\varphi)$ , then  $\varphi'' \circ \beta(m) = \nu \circ \varphi(m) = 0$  yields well-definedness. Suppose  $m' \in \ker(\varphi')$  then  $\beta' \circ \alpha'(m') = \beta \circ \alpha(m') = 0$  where  $\beta \circ \alpha = 0$  because of the exactness of the first row. Therefore, we obtain that  $\text{Im}(\alpha') \subseteq \ker(\beta')$ . Conversely, if  $m \in \ker(\beta')$ , then  $\beta(m) = \beta'(m) = 0$ . By exactness, there exists  $m'$  such that  $\alpha(m') = m$ . Hence,  $m \in \ker(\varphi)$  implies that  $\mu \circ \varphi'(m') = \varphi \circ \alpha(m') = \varphi(m) = 0$ . Because  $\mu$  is injective, then we obtain that  $m' \in \ker(\varphi')$ . Therefore, for an arbitrary element  $m \in \ker(\beta')$  there exists  $m' \in \ker(\varphi')$  such that  $m = \alpha'(m')$  meaning that  $\ker(\beta') \subseteq \text{Im}(\alpha')$ .  $\delta$  is well-defined. Because  $\beta$  is surjective then for any  $x \in \ker(\varphi'')$  there exists an element  $y \in M$  such that  $\beta(y) = x$ . Since  $x \in \ker(\varphi'')$ , we obtain that  $\nu \circ \varphi(y) = \varphi''(x) = 0$ , which implies  $\varphi(y) \in \ker(\nu)$ . By exactness, there exists an element  $z \in N'$  such that  $\mu(z) = \varphi(y)$ . Let  $y_0, y_1 \in M$  be arbitrary two elements such that  $\beta(y_0) = \beta(y_1) = x$ . Let  $z_0, z_1 \in N'$  be arbitrary two elements such that  $\mu(z_0) = \varphi(y_0)$  and  $\mu(z_1) = \varphi(y_1)$ . Because  $\beta(y_1 - y_0) = 0$ , then there exists an element  $m' \in M'$  such that  $\alpha(m') = y_1 - y_0$ . Therefore,

$$\begin{aligned} \mu(\varphi'(m') - z_1 + z_0) &= \mu \circ \varphi'(m') - \mu(z_1 - z_0) \\ &= \varphi \circ \alpha(m') - \mu(z_1 - z_0) \\ &= \varphi(y_1 - y_0) - \mu(z_1 - z_0) = 0 \end{aligned}$$

implies  $\varphi'(m') = z_1 - z_0$  since  $\mu$  is injective. Equivalently, we obtain that  $\overline{z_1 - z_0} = 0$  which yields the well-definedness of  $\delta$ .  $\text{Im}(\beta') = \ker(\delta)$ . Suppose  $m \in \ker(\varphi)$ , then we can select  $z = 0 \in N'$  because  $\mu(z) = \varphi(m) = 0$  which implies  $\delta \circ \beta'(m) = 0$ . Therefore,  $\text{Im}(\beta') \subseteq \ker(\delta)$ . Suppose  $m'' \in \ker(\delta)$ , then there exists  $m \in M$  and  $n' \in N'$  such that  $\beta(m) = m''$  and  $\varphi(m) = \mu(n')$ . Hence  $\delta(m'') = 0$  implies there exists an element  $m' \in M'$  such that  $\varphi'(m') = n'$ . I claim that  $m - \alpha(m') \in \ker(\varphi)$  and thus  $\beta'(m - \alpha(m')) = \beta(m) - \beta \circ \alpha(m') = m''$  implies that  $\ker(\delta) \subseteq \text{Im}(\beta')$ . To see that, we consider  $\varphi(m - \alpha(m')) = \varphi(m) - \mu \circ \varphi'(m') = \varphi(m) - \mu(n') = 0$ .  $\mu'$  is well-defined and  $\text{Im}(\delta) = \ker(\mu')$ . Since the diagram commutes,  $\mu \circ \varphi' = \varphi \circ \alpha$  implies  $\mu(\text{Im}(\varphi')) \subseteq \text{Im}(\varphi)$  which yields well-definedness. Let  $m''$  be an arbitrary element in  $\ker(\varphi'')$  then there exists  $m \in M$  and  $n' \in N'$  such that  $\beta(m) = m''$  and  $\varphi(m) = \mu(n')$ . Then we have  $\mu' \circ \delta(m'') = \mu'(n' + \text{Im}(\varphi')) = \mu(n') + \text{Im}(\varphi) = 0 \in \text{coker}(\varphi)$ , which implies  $\text{Im}(\delta) \subseteq \ker(\mu')$ . For an arbitrary element  $\bar{n}' \in \ker(\mu')$ , then we have  $\mu(n') \in \text{Im}(\varphi)$ , which implies there exists an element  $m \in M$  such that  $\varphi(m) = \mu(n')$ . Let  $m'' = \beta(m)$  and I claim that  $m'' \in \ker(\varphi'')$  and thus  $\delta(m'') = \bar{n}'$  implies  $\ker(\mu') \subseteq \text{Im}(\delta)$ . To see that, just consider  $\varphi''(m'') = \nu \circ \varphi(m) = \nu \circ \mu(n') = 0$ .  $\nu'$  is well-defined and  $\ker(\nu') = \text{Im}(\mu')$ . Similarly, because diagram commutes, then  $\nu \circ \varphi = \varphi'' \circ \beta$  implies that  $\nu(\text{Im}(\varphi)) \subseteq \text{Im}(\varphi'')$  meaning that  $\nu'$  is well-defined. Let  $\bar{n}'$  be an arbitrary element in  $\text{coker}(\varphi')$ , then  $\nu' \circ \mu'(\bar{n}') = \nu'(\mu(n') + \text{Im}(\varphi)) = \nu \circ \mu(n') + \text{Im}(\varphi'') = 0 \in \text{coker}(\varphi'')$ . Therefore,  $\text{Im}(\mu') \subseteq \ker(\nu')$ . Let  $\bar{n}$  be an arbitrary element such that  $\nu'(\bar{n}) = 0$ , equivalently  $\nu(n) \in \text{Im}(\varphi'')$ . Then there exists an element  $m'' \in M''$  such that  $\varphi''(m'') = \nu(n)$ . Since  $\beta$  is surjective, then there exists an element  $m \in M$  such that  $\beta(m) = m''$ . I claim that  $n - \varphi(m) \in \ker(\nu)$  and thus there exists an element  $n' \in N'$  such that  $\mu(n') = n - \varphi(m)$ , which implies that  $\mu'(\bar{n}') = \bar{n}$  meaning that  $\ker(\nu') \subseteq \text{Im}(\mu')$ . To see that, we consider  $\nu(n - \varphi(m)) = \nu(n) - \varphi'' \circ \beta(m) = \nu(n) - \varphi''(m) = 0$ .  $\nu'$  is surjective. It is directly because  $\nu'$  is well-defined and induced by surjection  $\nu$ . Explicitly, for any arbitrary element  $\bar{n}'' \in \text{coker}(\varphi'')$ , there exists an element  $n \in N$  such that  $\nu(n) = \bar{n}''$ . Since  $\nu(\text{Im}(\varphi)) \subseteq \text{Im}(\varphi'')$  we have discussed, then we have  $\nu'(\bar{n}) = \bar{n}'' + \text{Im}(\varphi'') = \bar{n}''$ . ■

### Lemmas for Problem 3

**Lemma 6.2.** *A Let  $f : M \rightarrow M''$  and  $g : K \rightarrow M''$  be  $R$ -module homomorphisms. Let  $P = M \times_{M''} K$  be the pullback of  $f$  along  $g$ , defined explicitly by the subset of the direct sum:*

$$P = \{(m, k) \in M \times K \mid f(m) = g(k)\}$$

*equipped with the canonical projections  $\pi_M : P \rightarrow M, (m, k) \mapsto m$  and  $\pi_K : P \rightarrow K, (m, k) \mapsto k$ . Then the following properties hold:*

1. *If  $f$  is surjective, then the projection  $\pi_K$  is surjective.*
2. *If  $f$  is injective, then the projection  $\pi_K$  is injective.*

*Proof.* (1) Suppose  $f$  is surjective. Let  $k \in K$  be an arbitrary element. Because  $f$  is surjective, then there exists an element  $m \in M$  such that  $f(m) = g(k)$ , which implies  $(m, k) \in P$  then  $k = \pi_K(m, k)$  implies surjectivity.

(2) Suppose  $f$  is injective. Suppose  $(m, k) \in P$  such that  $\pi_K(m, k) = 0$ . Then we obtain that  $f \circ \pi_M(m, k) = f(m) = g(k) = g \circ \pi_K(m, k) = 0$ . The injectivity of  $f$  implies  $m = 0$ . Hence,  $k = \pi_K(m, k) = 0$  implies the injectivity of  $\pi_K$ .  $\square$

**Lemma 6.3.** *B Let  $\mu : N' \rightarrow N$  and  $h : N' \rightarrow L$  be  $R$ -module homomorphisms. Let  $Q = N \oplus_{N'} L$  be the pushout of  $\mu$  along  $h$ , defined explicitly as the quotient module of the direct sum:*

$$Q = (N \oplus L)/W$$

*where  $W = \{(\mu(n'), -h(n')) \mid n' \in N'\}$ . The induced maps are given by  $\bar{h} : N \rightarrow Q, n \mapsto \overline{(n, 0)}$  and  $i_L : L \rightarrow Q, l \mapsto \overline{(0, l)}$ . Then the following properties hold:*

1. *If  $\mu$  is surjective, then the induced map  $i_L$  is surjective.*
2. *If  $\mu$  is injective, then the induced map  $i_L$  is injective.*

*Proof.* (1) Suppose  $\mu$  is surjective. It suffices to prove that any pair with form  $\overline{(n, 0)}$  has preimage because  $\overline{(n, l)} = \overline{(n, 0)} + \overline{(0, l)}$  and  $\overline{(0, l)}$  has preimage. Because  $\mu$  is surjective, then there exists  $n' \in N'$  such that  $\mu(n') = n$ . I claim that  $\overline{(n, 0)} = \overline{(0, h(n'))} = i_L(h(n'))$  and thus surjectivity of  $i_L$  holds. To see that, we consider  $\overline{(n, 0)} - \overline{(0, h(n'))} = \overline{(n, -h(n'))} = 0$  since  $n = \mu(n')$ .

(2) Suppose  $\mu$  is injective. Let  $l \in L$  be an arbitrary element such that  $i_L(l) = \overline{(0, l)} = 0$ , or equivalently there exists  $n' \in N'$  such that  $\mu(n') = 0$  and  $-h(n') = l$ . The injectivity of  $\mu$  implies  $n' = 0$  and hence  $l = -h(n') = 0$ . Therefore,  $i_L$  is injective.  $\square$

## 7 Week 5

### Lemmas for Problem

**Lemma 7.1.** *A Let  $\mu : N' \rightarrow N$  and  $h : N' \rightarrow L$  be  $R$ -module homomorphisms. Let  $Q = N \oplus_{N'} L$  be the pushout of  $\mu$  along  $h$ , defined explicitly as the quotient module of the direct sum:*

$$Q = (N \oplus L)/W$$

*where  $W = \{(\mu(n'), -h(n')) \mid n' \in N'\}$ . The induced maps are given by  $\bar{h} : N \rightarrow Q, n \mapsto \overline{(n, 0)}$  and  $i_L : L \rightarrow Q, l \mapsto \overline{(0, l)}$ . Then the following properties hold:*

1. *If  $\mu$  is surjective, then the induced map  $i_L$  is surjective.*



2. If  $\mu$  is injective, then the induced map  $i_L$  is injective.

*Proof.* (1) Suppose  $\mu$  is surjective. It suffices to prove that any pair with form  $\overline{(n, 0)}$  has preimage because  $\overline{(n, l)} = \overline{(n, 0)} + \overline{(0, l)}$  and  $\overline{(0, l)}$  has preimage. Because  $\mu$  is surjective, then there exists  $n' \in N'$  such that  $\mu(n') = n$ . I claim that  $\overline{(n, 0)} = \overline{(0, h(n'))} = i_L(h(n'))$  and thus surjectivity of  $i_L$  holds. To see that, we consider  $\overline{(n, 0)} - \overline{(0, h(n'))} = \overline{(n, -h(n'))} = 0$  since  $n = \mu(n')$ .

(2) Suppose  $\mu$  is injective. Let  $l \in L$  be an arbitrary element such that  $i_L(l) = \overline{(0, l)} = 0$ , or equivalently there exists  $n' \in N'$  such that  $\mu(n') = 0$  and  $-h(n') = l$ . The injectivity of  $\mu$  implies  $n' = 0$  and hence  $l = -h(n') = 0$ . Therefore,  $i_L$  is injective.  $\square$

**Lemma 7.2.** *Let  $f : M \rightarrow M''$  and  $g : K \rightarrow M''$  be  $R$ -module homomorphisms. Let  $P = M \times_{M''} K$  be the pullback of  $f$  along  $g$ , defined explicitly by the subset of the direct sum:*

$$P = \{(m, k) \in M \times K \mid f(m) = g(k)\}$$

*equipped with the canonical projections  $\pi_M : P \rightarrow M, (m, k) \mapsto m$  and  $\pi_K : P \rightarrow K, (m, k) \mapsto k$ . Then the following properties hold:*

1. *If  $f$  is surjective, then the projection  $\pi_K$  is surjective.*
2. *If  $f$  is injective, then the projection  $\pi_K$  is injective.*

*Proof.* (1) Suppose  $f$  is surjective. Let  $k \in K$  be an arbitrary element. Because  $f$  is surjective, then there exists an element  $m \in M$  such that  $f(m) = g(k)$ , which implies  $(m, k) \in P$  then  $k = \pi_K(m, k)$  implies surjectivity.

(2) Suppose  $f$  is injective. Suppose  $(m, k) \in P$  such that  $\pi_K(m, k) = 0$ . Then we obtain that  $f \circ \pi_M(m, k) = f(m) = g(k) = g \circ \pi_K(m, k) = 0$ . The injectivity of  $f$  implies  $m = 0$ . Hence,  $k = \pi_K(m, k) = 0$  implies the injectivity of  $\pi_K$ .  $\square$

## Problem 1

Let  $J$  be an  $R$ -module. Show that the following are equivalent.

- (1)  $J$  is an injective module.
- (2) For any  $R$ -module monomorphism  $\varphi : J \rightarrow M$ , there exists an  $R$ -module homomorphism  $\psi : M \rightarrow J$  such that  $\psi\varphi = \text{id}_J$ .
- (3) If  $J$  is a submodule of some  $R$ -module  $M$ , then  $J$  is a direct summand of  $M$ .

**Solution:** We first prove that (1) implies (2). According to the definition of injective module, for morphism  $\text{id}_J$  and monomorphism  $\varphi : J \rightarrow M$ , there exists  $\psi : M \rightarrow J$  such that diagram

$$\begin{array}{ccc} J & \xhookrightarrow{\varphi} & M \\ & \searrow \text{id}_J & \downarrow \exists \psi \\ & & J \end{array}$$

commutes. In other words,  $\psi \circ \varphi = \text{id}_J$ .

Next, we prove that (2) implies (3). Because  $J$  is the submodule of  $M$ , then we obtain the canonical injection  $\varphi : J \hookrightarrow M$ , therefore (2) implies that there exists a morphism  $\psi : M \rightarrow J$  such that  $\psi \circ \varphi = \text{id}_J$ . I claim that  $M \cong J \oplus \ker(\psi)$ . To see that, we define maps

$$\begin{aligned} f : J \oplus \ker(\psi) &\rightarrow M \\ (j, h) &\mapsto \varphi(j) + h \end{aligned}$$

and

$$\begin{aligned} g : M &\rightarrow J \oplus \ker(\psi) \\ m &\mapsto (\psi(m), m - \varphi \circ \psi(m)). \end{aligned}$$

$f$  is a  $R$ -module morphism is quite clear because we have

$$\text{Hom}_R(J \oplus \ker(\varphi), M) \cong \text{Hom}_R(J, M) \oplus \text{Hom}_R(\ker(\varphi), M).$$

We need to verify  $g$  is a  $R$ -module morphism. The well-definedness of  $g$  is given by the fact that

$$\psi(m - \varphi \circ \psi(m)) = \psi(m) - \psi \circ \varphi \circ \psi(m) = 0,$$

and  $g$  is a  $R$ -module homomorphism since  $\psi$  and  $\varphi$  are  $R$ -module homomorphisms. I claim that  $f, g$  are bijection and  $f^{-1} = g$ . To see that, we just need to consider the composition of two maps

$$f \circ g(m) = \varphi \circ \psi(m) + (m - \varphi \circ \psi(m)) = m$$

and

$$\begin{aligned} g \circ f(j, h) &= (\psi(\varphi(j) + h), \varphi(j) + h - \varphi \circ \psi(\varphi(j) + h)) \\ &= (j + \psi(h), \varphi(j) + h - \varphi(j) + \varphi \circ \psi(h)) \\ &= (j, h). \end{aligned}$$

Therefore,  $M \cong J \oplus \ker(\psi)$  and then  $J$  is a direct summand of  $M$ .

Now we prove (3) implies (1). Suppose  $\varphi : M \rightarrow N$  is an arbitrary monomorphism and  $f : M \rightarrow J$  is an arbitrary homomorphism. Use the concepts already defined in **Lemma A**. We consider the pushout  $f$  along  $\varphi$

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & N \\ f \downarrow & & \downarrow j \\ J & \xrightarrow{i} & J \oplus_M N \end{array}$$

Since  $\varphi$  is injective,  $i$  is injective as well. Therefore,  $J$  can be interpreted as the submodule of  $J \oplus_M N$  by canonical inclusion. Hence, (3) implies that  $J$  is the direct summand of  $J \oplus_M N$ , then we can assume that  $J \oplus_M N \cong J \oplus K$ . Define map  $\psi = p \circ j : N \rightarrow J$  where  $p : J \oplus K \rightarrow J$  is the canonical projection. Then  $f = \psi \circ \varphi$  since  $f = p \circ i \circ f = p \circ j \circ \varphi = \psi \circ \varphi$ . ■

## Problem 2

Suppose that  $M_i$  ( $i \in I$ ) are  $R$ -modules. Prove that  $\bigoplus_{i \in I} M_i$  is a flat  $R$ -module if and only if each  $M_i$ ,  $i \in I$  is a flat  $R$ -module.

**Solution:** Let  $M = \bigoplus_{i \in I} M_i$ . Let  $f : A \hookrightarrow B$  be an arbitrary monomorphism. Assume that  $M_i$  are all flat, then by definition, the  $R$ -module morphisms  $f \otimes_R \text{id}_{M_i} : A \otimes_R M_i \hookrightarrow B \otimes_R M_i$  are all monomorphic. Therefore

$$\bigoplus_{i \in I} (f \otimes_R \text{id}_{M_i}) : \bigoplus_{i \in I} (A \otimes_R M_i) \hookrightarrow \bigoplus_{i \in I} (B \otimes_R M_i)$$

is monomorphism by definition of direct sum. Hence, the commutativity of tensor product and direct sum implies that

$$f \otimes_R \text{id}_M : A \otimes_R M \hookrightarrow B \otimes_R M$$

is monomorphism meaning that  $M$  is flat.

Conversely, we use concepts defined in **Lemma B**. For arbitrary index  $i$ , consider canonical injection  $j : B \otimes_R M_i \rightarrow B \otimes_R M$  pullback along  $f \otimes_R \text{id}_M$ . Let  $P = (B \otimes_R M_i) \times_{B \otimes_R M} (A \otimes_R M)$ , and according **Lemma B**, the pullback homomorphism  $g : P \rightarrow B \otimes_R M_i$  is monomorphism. Let  $\iota : A \otimes_R M_i \rightarrow P$  by sending pure tensor  $a \otimes m_i$  to  $(f(a) \otimes m_i, a \otimes m_i)$  and extending it linearly. It is clear that  $\iota$  is well-defined because  $f \otimes_R \text{id}_M(a \otimes m_i) = f(a) \otimes m_i$  and for arbitrary  $r \in R$ , we have

$$r \cdot \iota(a \otimes m_i) = (f(r \cdot a) \otimes m_i, r \cdot a \otimes m_i) = \iota(r \cdot a \otimes m_i)$$

and other requirements of  $R$ -module is easy to prove. Therefore  $\iota$  is a  $R$ -module morphism, hence it is injective since  $a \otimes m_i$  equals zero if and only if  $(f(a) \otimes m_i, a \otimes m_i)$  equals zero. Therefore  $f \otimes_R \text{id}_{M_i} = g \circ \iota$  is monomorphic meaning that  $M_i$  is flat. ■

### Problem 3

Prove: the following three conditions on an  $R$ -module  $V$  are equivalent.

- (1) (Ascending chain condition.) For every infinite chain of  $R$ -submodules of  $V$ ,

$$V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n \subseteq \cdots,$$

there is a number  $m$  such that  $V_m = V_{m+1} = \cdots$ .

- (2) Any non-empty set of submodules of  $V$  contains a maximal element with respect to inclusion.
- (3) Every  $R$ -submodule of  $V$  is finitely generated.

**Solution:** First we show that (1) implies (2). If there exists a non-empty set of submodules of  $V$ , denoted by  $\mathcal{S}$ , such that no maximal element exists. Then for every submodule  $V_1 \in \mathcal{S}$ , there exists a submodule  $V_2 \in \mathcal{S}$  such that  $V_1 \subsetneq V_2$ . Continuing this process, we obtain an infinite ascending chain contradicting to (1).

Now we prove (2) implies (3). Let  $W$  be an arbitrary submodule of  $V$ . Let  $\mathcal{S}$  be the collection of all finitely generated submodules of  $W$ . By (2),  $\mathcal{S}$  contains a maximal element  $W'$ . Suppose  $W' \subsetneq W$  is a proper submodule, then select an element  $\alpha \in W \setminus W'$  and let  $W''$  be the submodule generated by  $W'$  and  $\alpha$ , which is finitely generated because  $W'$  is finitely generated. However,  $W'$  is the maximal element of  $\mathcal{S}$  meaning that  $W'' \notin \mathcal{S}$ , which is a contradiction. Therefore,  $W' = W$  implies that  $W$  is finitely generated and thus every submodule of  $V$  is finitely generated.

Finally, we claim that (3) implies (1). If there exists an infinite ascending chain

$$V_1 \subsetneq V_2 \subsetneq \cdots \subsetneq V_n \subsetneq \cdots,$$

By (3), the submodule  $W = \cup_{i=1}^{\infty} V_i$  must be finitely generated. For arbitrary two elements  $x, y \in W$ , there exists an integer  $n$  such that  $x, y \in V_1 \cup \cdots \cup V_n = V_n$  by definition. Then all requirements of  $R$ -module are satisfied since  $V_n$  is a submodule. Suppose  $W = \langle w_1, w_2, \dots, w_m \rangle$  is generated by finite elements  $w_i$ , then there exists an integer  $s$  such that  $w_i \in V_1 \cup \cdots \cup V_s = V_s$  for all  $i$  meaning that  $W \subseteq V_s$  and therefore for every  $k \geq s$  we obtain that  $V_s = V_k$ , which is a contradiction. ■

#### Problem 4

Let  $W$  be an  $R$ -submodule of  $V$ . Prove:  $V$  is Noetherian if and only if both  $W$  and  $V/W$  are Noetherian.

**Solution:** Suppose  $V$  is Noetherian. Every ascending chain of  $W$  is also the ascending chain of  $V$  meaning that chain is stationary. Hence, Correspondence Theorem implies that if

$$\overline{V}_1 \subseteq \overline{V}_2 \subseteq \cdots \quad (4)$$

is an ascending chain of  $V/W$ , there exists ascending chain of  $V$

$$V_1 \subseteq V_2 \subseteq \cdots \quad (\textcircled{2})$$

such that  $V_i$  is the lift of  $\overline{V}_i$ . The chain  $(\textcircled{2})$  is stationary implies chain (4) is stationary because  $\overline{V}_i = V_i + W$ .

Conversely, suppose  $W$  and  $V/W$  are Noetherian and let

$$V_1 \subseteq V_2 \subseteq \cdots \quad (\textcircled{I})$$

be an arbitrary ascending chain of  $V$ . Since  $V/W$  is Noetherian, the ascending chain

$$V_1 + W \subseteq V_2 + W \subseteq \cdots \quad (\textcircled{M})$$

is stationary and we suppose it stationary at  $n$ . If chain  $(\textcircled{I})$  is not stationary, then select elements  $\alpha_i \in V_i \setminus V_{i-1}$  for  $i > n$ . Chain  $(\textcircled{M})$  stationary implies that  $\alpha_i \in V_i + W = V_n + W$ . Then we can suppose  $\alpha_i = \beta_i + \gamma_i$  where  $\beta_i \in V_n$  and  $\gamma_i \in W$ . Hence,  $V_n \subseteq V_i$  implies that  $\gamma_i \in V_i$  and  $\gamma_i \notin V_{i-1}$ . If not,  $\alpha_i$  must be the element contained in  $V_{i-1}$  contradicting our assumption. Consider ascending chain of  $W$

$$\langle \gamma_{n+1} \rangle \subseteq \langle \gamma_{n+1}, \gamma_{n+2} \rangle \subseteq \cdots$$

which is stationary since  $W$  is Noetherian. Suppose it is stationary at  $m$  meaning that  $\gamma_{m+1}$  can be linearly represented by  $\gamma_{n+1}, \dots, \gamma_m$  and therefore  $\gamma_{m+1} \in V_m$ , which is a contradiction, chain  $(\textcircled{I})$  must be stationary. ■

#### Problem 5

Let  $V$  be a Noetherian  $R$ -module and  $f : V \rightarrow V$  an epimorphism. Show that  $f$  is an  $R$ -isomorphism.

**Solution:** Let  $W_i = \ker(f^i)$  and consider an ascending chain

$$W_1 \subseteq W_2 \subseteq \dots \quad (\text{II})$$

because  $V$  is Noetherian then this chain is stationary. Suppose it is stationary at  $n$ . Then  $W_n$  is the submodule of  $V$  consisting exactly of all nilpotent elements associated to  $f$ . I claim that  $f(W_n) \subseteq W_n$  and for every element  $x \in V \setminus W_n$  we have  $f(x) \notin W_n$ . To see that, suppose  $w \in W_n$  is nilpotent, then  $f(w)$  is nilpotent as well, meaning that  $f(w) \in W_n$ . Hence, for arbitrary element  $x \in V \setminus W_n$ , if  $f(x) \in W_n$  then it implies that  $f(x)$  is nilpotent and thus  $x$  is nilpotent, which is contradiction. Suppose that  $n > 1$ , ascending chain (II) stationary at  $n$  means that  $W_{n-1} \subsetneq W_n$ , then  $W_n \setminus W_{n-1}$  has no preimage of  $f$  because  $f(W_n) = W_{n-1}$ . It is contradicts the assumption that  $f$  is epimorphism. If  $n = 1$  and  $W_1 \neq \{0\}$  then  $W_1 \setminus \{0\}$  has no preimage. Suppose  $x \in V$  such that  $f(x) \in W_1 \setminus \{0\}$ , then  $f^2(x) = 0$  implies that  $x \in W_2$ . However,  $W_2 = W_1$  implies that  $f(x) = 0$ , which is a contradiction. Therefore,  $f$  must be a monomorphism, hence an isomorphism. ■

## 8 Week 6

### Problem 1

Are the following statements correct?

- (1) Any submodule of a semisimple module is also semisimple.
- (2) Any semisimple Noetherian module is Artinian.

**Solution:** Both statements are correct. (1) Suppose  $N$  is an arbitrary submodule of semisimple module  $M$ . Then by discussion in our class, there exists simple submodules  $\{M_i\}_{i \in I}$  such that

$$M = \bigoplus_{i \in I} M_i$$

and subindex set  $J$  of  $I$  such that

$$M = N \oplus \left( \bigoplus_{j \in J} M_j \right)$$

Consider the canonical projection  $M \rightarrow N$  then by isomorphism theorem we obtain that

$$N \cong M / \left( \bigoplus_{j \in J} M_j \right) \cong \bigoplus_{i \in I \setminus J} M_i$$

meaning that  $N$  is semisimple.

- (2) Suppose  $M$  is an arbitrary semisimple Noetherian module with direct sum decomposition

$$M = \bigoplus_{i \in I} M_i.$$

Noetherian implies that  $M$  is finite-generated. Without lose of generality, we can suppose that  $x_1, \dots, x_n$  are generators and  $x_j \in M_j$ . Then  $M$  admits a finite decomposition

$$M = \bigoplus_{j \in J} M_j.$$

where  $J = \{1, \dots, n\}$  is finite. Therefore  $M$  has a finite composition series meaning that  $M$  is Artinian. ■

## Problem 2

Prove that: the following conditions on an  $R$ -module  $V$  with composition series are equivalent.

- (1)  $V$  is semisimple.
- (2) Every irreducible submodule of  $V$  is a direct summand of  $V$ .
- (3) Every maximal submodule of  $V$  is a direct summand of  $V$ .

**Solution:** (1) implies (2) and (3) just by equivalent definition of semisimple module we have discussed in our class.

Next, we prove that (2) implies (1). Suppose  $V \neq \langle 0 \rangle$ , otherwise it is trivially semisimple. Since  $V$  has a composition series, it must contain an irreducible submodule. let us denote it by  $V_1$ . Then by (2), we can let  $V = V_1 \oplus V_1'$ . We claim that for every irreducible submodule of  $V_1'$  also is a direct summand of  $V_1'$ . To see that, we suppose  $V_2$  is an irreducible submodule of  $V_1$  and hence it is also an irreducible submodule of  $V$ , then we can suppose that  $V = V_2 \oplus V_2'$ . Then we claim that

$$V_1' = V_2 \oplus (V_2' \cap V_1').$$

It is obvious that  $V_2 \cap (V_2' \cap V_1') = 0$  and  $V_2 \oplus (V_2' \cap V_1') \subseteq V_1'$ . For  $x \in V_1'$ , we can suppose that  $x = y + z$  where  $y \in V_2, z \in V_2'$ . Since  $x, y \in V_1'$  then  $z = x - y \in V_1'$  which implies that  $V_1' \subseteq V_2 \oplus (V_2' \cap V_1')$ . Continuing this process we obtain a decomposition

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_k \oplus W_k$$

where  $V_1, \dots, V_k$  are irreducible. Suppose that the length of composition series of  $V$  is  $n$ , then we obtain that  $W_n = 0$ , because in this case we have a strict ascending chain

$$\langle 0 \rangle \subsetneq V_1 \subsetneq \cdots \subsetneq V_1 \oplus V_2 \oplus \cdots \oplus V_n$$

which implies that

$$V = \bigoplus_{i=1}^n V_i$$

and  $W_n = \langle 0 \rangle$ . Then  $V$  is semisimple.

We now show that (3) implies (1). Similarly, we suppose that  $V_1'$  is an arbitrary maximal submodule of  $V$ . Then we suppose that  $V = V_1 \oplus V_1'$  and  $V_1 \cong V/V_1'$  implies that  $V_1$  is irreducible. We claim that every maximal submodule of  $V_1'$ , denoted by  $V_2'$ , is a direct summand of  $V_1'$ . To see that, we note that  $V_1 \oplus V_2'$  is a maximal submodule of  $V$  since

$$V/(V_1 \oplus V_2') \cong (V_1 \oplus V_1')/(V_1 \oplus V_2') \cong V_1'/V_2'.$$

Let  $V_1' = V_2 \oplus V_2'$  and continuing this process, then we obtain a decomposition

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_n \oplus W_n$$

and by same reason we have discussed  $W_n = \langle 0 \rangle$  and  $V$  is semisimple. ■

## Problem 3

Prove:  $\text{Soc}(U \oplus V) = \text{Soc}(U) \oplus \text{Soc}(V)$ .

**Solution:** It is obvious that

$$\text{Soc}(U \oplus V) \supseteq \text{Soc}(U) \oplus \text{Soc}(V).$$

Because if  $M$  is an irreducible submodule of  $U$  or  $V$ , it is also an irreducible submodule of  $U \oplus V$ . Conversely, suppose that  $M$  is a nonzero irreducible submodule of  $U \oplus V$ . Then we obtain that

$$M \subseteq M_1 \oplus M_2$$

where  $M_i = p_i(M)$  and  $p_1 : U \oplus V \rightarrow U$  and  $p_2 : U \oplus V \rightarrow V$  are canonical projections. I claim that  $M_i$  is also irreducible since by isomorphism theorem  $M_i \cong M / \ker(p_i|_M)$  and  $\ker(p_i|_M)$  as submodule of  $M$  must equal to zero or  $M$ , then  $M_i$  is isomorphic to zero or  $M$  meaning  $M_i$  is either the zero module or irreducible. Then we obtain that

$$\text{Soc}(U \oplus V) \subseteq \text{Soc}(U) \oplus \text{Soc}(V).$$

Therefore, the statement is true. ■

#### Problem 4

Let  $R$  be an arbitrary ring. Then every 2-sided ideal of the matrix ring  $M_n(R)$  can be expressed in the form  $M_n(I)$ , where  $I$  is a 2-sided ideal of  $R$ . In particular,  $R$  is simple if and only if  $M_n(R)$  is simple. [Hint: For an ideal  $L$  of  $M_n(R)$ , let  $I$  be the set of  $(1,1)$ -entries of  $L$ . Show that  $I$  is an ideal of  $R$  and  $L = M_n(I)$ .]

**Solution:** Use notations above. First we need to show  $I$  is an ideal. For an arbitrary element  $r \in R$ , let  $\text{diag}(r)$  denote diagonal matrix setting diagonal elements to be  $r$ . By our definition for any  $s \in I$ , there exists  $A_s \in L$  such that the  $(1,1)$ -entry of  $A_s$  is  $s$ . Since  $L$  is an ideal of  $M_n(R)$ , we obtain that

$$\text{diag}(r) \cdot A_s \in L \text{ and } A_s \cdot \text{diag}(r) \in L,$$

which implies that  $r \cdot s \in I$  and  $s \cdot r \in I$ . Furthermore, for any  $s_1, s_2 \in I$ , there exist  $A_{s_1}, A_{s_2} \in L$  whose  $(1,1)$ -entries are  $s_1$  and  $s_2$ , respectively. Since  $L$  is an additive subgroup,  $A_{s_1} - A_{s_2} \in L$ , and its  $(1,1)$ -entry is  $s_1 - s_2$ , meaning  $s_1 - s_2 \in I$ . Thus,  $I$  is an ideal. Next, we need to prove that  $L = M_n(I)$ . Let  $E_{ij}$  denote the matrix whose  $(i,j)$ -entry is 1 and other entries are 0.  $\{E_{ij}\}$  form a basis of  $M_n(R)$ . Suppose that  $B$  is an arbitrary matrix in  $L$ . Then we can suppose that

$$B = \sum_{i,j} b_{ij} E_{ij}$$

Since  $L$  is an ideal then by

$$E_{1k} \cdot B \cdot E_{l1} = b_{kl} E_{11} \in L$$

we know that arbitrary entry  $b_{kl}$  is contained in  $I$ . Therefore, we obtain that  $L \subseteq M_n(I)$ . Conversely, suppose  $B$  is an arbitrary matrix in  $M_n(I)$ . Then for  $(i,j)$ -entry of  $B$ , denoted by  $b_{ij}$ , there exists a matrix  $A_{ij} \in L$  such that the  $(1,1)$ -entry of  $A_{ij}$  is  $b_{ij}$ . Then by

$$B = \sum_{i,j} E_{i,1} \cdot A_{ij} \cdot E_{1,j}$$

we obtain that  $M_n(I) \subseteq L$ . Therefore,  $M_n(R)$  has no proper ideals if and only if  $R$  has no proper ideals. Equivalently,  $M_n(R)$  is simple if and only if  $R$  is simple. ■

### Problem 5

If  $W$  is a submodule of a semisimple module  $V$ , then  $W$  and  $V/W$  are also semisimple.

**Solution:** We first show that for an arbitrary submodule of  $W$ , it is a direct summand of  $W$  and thus  $W$  is semisimple. Let  $U$  be an arbitrary submodule of  $W$ . Since  $W \subseteq V$ ,  $U$  is also a submodule of  $V$ . Because  $V$  is semisimple,  $U$  is a direct summand of  $V$ . Thus, there exists a submodule  $U'$  of  $V$  such that

$$V = U \oplus U'.$$

Intersecting both sides with  $W$ , we get

$$W = V \cap W = (U \oplus U') \cap W.$$

Since  $U \subseteq W$ , we can apply Dedekind's modular law to obtain

$$W = U \oplus (U' \cap W).$$

Note that  $U' \cap W$  is a submodule of  $W$ . This shows that any submodule  $U$  of  $W$  is a direct summand of  $W$ .

Next, we show that  $V/W$  is semisimple. Since  $W$  is a submodule of the semisimple module  $V$ , it is a direct summand of  $V$ . Therefore, there exists a submodule  $W'$  of  $V$  such that

$$V = W \oplus W'.$$

By the First Isomorphism Theorem, we have

$$V/W = (W \oplus W')/W \cong W'.$$

Since  $W'$  is a submodule of the semisimple module  $V$ , by the conclusion from above,  $W'$  is also semisimple. Because  $V/W$  is isomorphic to a semisimple module  $W'$ , it follows that  $V/W$  itself is semisimple. ■

## 9 Week 7

### Problem 1

Let  $D$  be a division ring and  $n$  a positive integer.

- (a) Show that the natural left  $M_n(D)$ -module, consisting of column vectors of length  $n$  with entries in  $D$ , is a simple module.
- (b) Show that  $M_n(D)$  is semisimple and has up to isomorphism only one simple module.
- (c) Show that every ring of the form

$$M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

is semisimple, where  $D_i$  are division rings.

- (d) Show that  $M_n(D)$  is a simple ring, namely, one in which the only 2-sided ideals are the zero ideal and the whole ring.



**Solution:** (a) Let  $x$  be an arbitrary nonzero  $n$  dimensional column vector of  $D$ . Suppose that the  $i$ -entry of  $x$  is nonzero and denoted by  $x_i$ . Let  $E_{ji}$  denote the  $n \times n$  matrix such that the  $(j, i)$  entry is the identity in  $D$  and other entries are zero. Then we obtain that

$$(x_i^{-1}E_{ji})x = \varepsilon_j$$

where  $\varepsilon_j$  denotes the column vector whose  $j$  entry is the identity in  $D$  and other entries are zero. Since  $\{\varepsilon_j\}$  forms a basis, we obtain that if a submodule contains a nonzero element  $x$ , then we can use  $x$  to generate any vector. Therefore, the module of column vectors  $D^n$  is a simple  $M_n(D)$ -module.

(b) Let  $A_j$  denotes the submodule of  $M_n(D)$  consisting of matrices whose entries are zero except possibly in the  $i^{\text{th}}$  column. Then we obtain that

$$M_n(D) = A_1 \oplus \cdots \oplus A_n.$$

Since each  $A_j \simeq D^n$  and  $D^n$  is simple by (a). Therefore,  $M_n(D)$  is semisimple and has up to isomorphism only one simple module.

(c) Let  $R$  denote such a ring. I claim that each  $M_{n_i}(D_i)$  is semisimple as a  $R$ -module. Without loss of generality, we prove it for  $M_{n_1}(D_1)$ . By definition, we know

$$\text{Ann}_R(M_{n_1}(D_1)) = M_{n_2}(D_2) \oplus \cdots \oplus M_{n_r}(D_r).$$

Let  $I$  denote this annihilator. Since  $R/I \simeq M_{n_1}(D_1)$ ,  $M_{n_1}(D_1)$  is semisimple as  $R/I$ -module by (b). According the discussion in our class,  $M_{n_1}(D_1)$  is semisimple as  $R/I$ -module if and only if as  $R$ -module. Then each  $M_{n_i}(D_i)$  is semisimple as  $R$ -module, hence  $R$  is semisimple.

(d) We use a conclusion from Homework 6: Every 2-sided ideal of the matrix ring  $M_n(D)$  can be expressed in the form  $M_n(I)$  where  $I$  is a 2-sided ideal of  $D$ . Since  $D$  is a division ring, the 2-sided ideals of  $D$  are trivial. Therefore,  $M_n(D)$  is a simple ring. ■

## Problem 2

Let  $U$  be an  $A$ -module for a semisimple finite-dimensional algebra  $A$  (over a field). Show that if  $\text{End}_A(U)$  is a division ring then  $U$  is simple.

**Solution:** Since  $A$  is a semisimple finite-dimensional algebra, every  $A$ -module is semisimple. Therefore, we can suppose that

$$U \simeq S_1 \oplus \cdots \oplus S_k.$$

The endomorphism ring is then given by  $\text{End}_A(U) \cong \bigoplus_{p,q} \text{Hom}_A(S_p, S_q)$ . If  $U$  contained two non-isomorphic simple modules  $S_i$  and  $S_j$ ,  $\text{End}_A(U)$  would have a non-trivial idempotent corresponding to the projection onto one of the components, contradicting the fact that  $\text{End}_A(U)$  is a division ring.

Thus, all simple components must be isomorphic to some simple module  $S$ , meaning  $U \cong S^k$ . According to the properties of endomorphism rings of semisimple modules:

$$\text{End}_A(U) \cong M_k(\text{End}_A(S))$$

By Schur's Lemma,  $D' = \text{End}_A(S)$  is a division ring. Therefore,  $\text{End}_A(U) \cong M_k(D')$ . A matrix ring  $M_k(D')$  over a division ring is itself a division ring if and only if  $k = 1$ . Since if dimension greater than one, there exists non-invertible matrices. Consequently, we have

$$U \simeq S$$

showing that  $U$  is simple. ■

### Problem 3

Let  $A$  be a ring. Then  $A$  is a semisimple Artinian ring if and only if every  $A$ -module is projective.

**Solution:** Suppose that  $A$  is a semisimple Artinian ring. We want to show that every  $A$ -module is projective. For any arbitrary left  $A$ -module  $N$ , there exists a free  $A$ -module  $F$  and an epimorphism  $f : F \rightarrow N$ . Since the free module  $F$  is a direct sum of copies of semisimple module  ${}_A A$ , it follows that  $F$  is also a semisimple module. Therefore, the kernel of the epimorphism  $\ker(f)$  is a direct summand of  $F$ . This means the short exact sequence

$$0 \rightarrow \ker(f) \rightarrow F \xrightarrow{f} N \rightarrow 0$$

splits. Consequently,  $F \cong \ker(f) \oplus N$ , which implies that  $N$  is isomorphic to a direct summand of the free module  $F$  meaning that  $N$  is projective.

Conversely, suppose that every left  $A$ -module is projective. We want to show that  $A$  is a semisimple Artinian ring. Let  $I$  be an arbitrary left ideal of  $A$ . The quotient  $A/I$  is naturally a left  $A$ -module. Consider the canonical short exact sequence:

$$0 \rightarrow I \rightarrow A \xrightarrow{\pi} A/I \rightarrow 0$$

where  $\pi$  is the natural projection. By our assumption, every  $A$ -module is projective, so  $A/I$  is a projective module and thus the short exact sequence splits. We obtain that

$$A \simeq I \oplus A/I.$$

Therefore every submodule of  $A$  splits  $A$ , we obtain that  $A$  is semisimple. Since  $A$  is semisimple ring, it is also Artinian. To see that, We may suppose that

$${}_A A = \bigoplus_{i \in I} S_i$$

where  $S_i$  is simple left ideal. Suppose that  $1 = e_1 + \cdots + e_n$  with  $e_i \in S_i$  and therefore

$${}_A A = S_1 \oplus S_2 \oplus \cdots \oplus S_n$$

implying its Artinian. ■

### Problem 4

Let  $k$  be a field. Suppose the group algebra  $kG$  is semisimple, and the simple  $kG$ -modules are  $S_1, \dots, S_r$  with degrees  $d_i = \dim_k S_i$ . Show that

$$\sum_{i=1}^r d_i^2 \geq |G|$$

with equality if and only if  $\text{End}_{kG}(S_i) = k$  for all  $i$ .

**Solution:** According to Wedderburn-Artin theorem, we can suppose that

$$kG \cong M_{n_1}(D_1) \oplus M_{n_2}(D_2) \oplus \cdots \oplus M_{n_r}(D_r).$$

Hence, we can suppose that  $M_{n_i}(D_i) \simeq S_i^{n_i}$ . Since  $M_{n_i}(D_i)$  only consist of one simple module up to isomorphism and each simple module can be interpreted as the submodule of  $kG$ . Then

for each simple module  $S_j$ , it can only be contained in one component in our decomposition meaning that our assumption is valid. Therefore, we may assume that

$$kG \cong S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}.$$

Hence we obtain that

$$|G| = \dim_k(kG) = \sum_{i=1}^r n_i d_i.$$

Since  $S_i$  as a simple submodule of  $M_{n_i}(D_i)$ , then we can say  $S_i \simeq D_i^{n_i}$  as  $kG$ -module isomorphism. According to Schur's lemma,  $\text{End}_{kG}(S_i)$  must be a division ring. It means that  $\text{End}_{kG}(S_i)$  only consists of the matrices of the form  $dI_{n_i}$  where  $d \in D_i$ . Hence we can write

$$\text{End}_{M_{n_i}(D_i)}(S_i) \simeq D_i^{\text{op}}.$$

We can rewrite our formula of dimension by

$$n_i = \dim_k(S_i) / \dim_k(D_i).$$

We obtain that

$$|G| = \sum_{i=1}^r d_i^2 / \dim_k(D_i).$$

Therefore, the equality holds if and only if  $\dim_k(\text{End}_{M_{n_i}(D_i)}(S_i)) = \dim_k(D_i) = 1$ . It also equivalently say  $\text{End}_{M_{n_i}(D_i)}(S_i) \simeq k$ . ■

## Problem 5

Suppose that we have  $R$ -module homomorphisms  $U \xrightarrow{f} V \xrightarrow{g} W$ . Show that: if  $gf$  is an essential epimorphism and  $f$  is an epimorphism, then both  $f$  and  $g$  are essential epimorphisms.

**Solution:** Since  $gf$  is an epimorphism,  $g$  must be an epimorphism. Suppose that  $f$  is not essential. Equivalently, there exists a proper submodule  $N$  of  $U$  such that  $f|_N$  is an epimorphism. Since  $g$  is an epimorphism, we now obtain that

$$g \circ f|_N : N \rightarrow W$$

is also an epimorphism. However, since  $gf$  is essential, we obtain that  $N = U$ , which is a contradiction. Therefore,  $f$  is essential.

Suppose that  $g$  is not essential. Then we can find a proper submodule  $M$  of  $V$  such that  $g(M) = W$ . Then we consider the submodule  $Q = f^{-1}(M)$  of  $U$ . It is proper, since  $f(Q) = M \subsetneq V$ . Then we obtain that  $gf(Q) = W$ . Because  $gf$  is essential, we find that  $Q = U$ , which is a contradiction. Therefore  $g$  must be essential. Our claim holds. ■

## 10 Week 8

### Problem 1

Let  $V$  be an  $A$ -module.

- (a) Show that an endomorphism  $e : V \rightarrow V$  is a projection onto a  $A$ -submodule  $W$  if and only if  $e$  is idempotent as an element of  $\text{End}_A(V)$ .
- (b) Show that direct sum decompositions  $V = W_1 \oplus W_2$  as  $A$ -modules are in bijection with idempotent decompositions  $1 = e + f$  in  $\text{End}_A(V)$ .
- (c) Show that  $e \in \text{End}_A(V)$  is primitive if and only if  $e(V)$  is indecomposable.
- (d) Suppose that  $V$  is semisimple with finitely many simple summands and let  $e_1, e_2 \in \text{End}_A(V)$  be idempotents. Show that  $e_1(V) \cong e_2(V)$  as  $A$ -modules if and only if  $e_1$  and  $e_2$  are conjugate by an invertible element of  $\text{End}_A(V)$ .
- (e) Let  $k$  be a field. Show that all primitive idempotent elements in  $M_n(k)$  are conjugate under the action of the unit group  $GL_n(k)$ , which consists of all the invertible matrices in  $M_n(k)$ .

**Solution:** (a) If  $e$  is a projection, by definition, we can suppose that  $V = W \oplus W'$  such that  $e(W') = 0$  and for an arbitrary element  $w \in W$  we have  $e(w) = w$ . Therefore, let  $v$  be an arbitrary element in  $V$  with unique decomposition  $v = w + w'$  where  $w \in W$  and  $w' \in W'$ , then we obtain that  $e^2(v) = e^2(w + w') = e(w) = w$ , which implies  $e^2(v) = e(v)$  and thus  $e$  is idempotent. Conversely, let  $f = 1 - e$ , I claim that  $V = \text{im}(e) \oplus \text{im}(f)$ . Then for an arbitrary element  $v$  we obtain that  $v = e(v) + f(v)$  and thus  $V = \text{im}(e) + \text{im}(f)$ . Let  $v \in \text{im}(e) \cap \text{im}(f)$ , it means that there exists an element  $v'$  such that  $v = f(v')$ . Since  $e$  is idempotent, we then have  $v = e(v) = ef(v') = e(1 - e)(v') = 0$ . Therefore we obtain that  $V = \text{im}(e) \oplus \text{im}(f)$  and  $e$  act as an identity on  $\text{im}(e)$  and as null morphism on  $\text{im}(f)$ , which implies that  $e$  is a projection.

(b) Let  $\text{Dirsum}$  and  $\text{Idmdec}$  denotes the set of direct sum decompositions and the set of idempotent decompositions respectively. The process of (a) have shown that  $\text{Idmdec} \rightarrow \text{Dirsum}$ . Conversely, suppose that we have a direct sum decomposition  $V = W_1 \oplus W_2$ . Then we let  $e$  and  $f$  be the canonical projection to  $W_1$  and  $W_2$  respectively. Clearly,  $e + f = 1$  satisfies our requirement. To show bijectivity, given that an idempotent morphism  $e$ . Let  $f = 1 - e$ , we obtain that  $V = \text{im}(e) \oplus \text{im}(f)$  according to (a). Then  $e$  and  $f$  are identified as the canonical projection to  $\text{im}(e)$  and  $\text{im}(f)$  respectively. It means that  $\text{Idmdec} \rightarrow \text{Dirsum} \rightarrow \text{Idmdec}$  is an identity. Hence, given that direct sum decomposition  $V = W_1 \oplus W_2$ , let  $e$  and  $f$  denotes the canonical projection to  $W_1$  and  $W_2$  respectively. According to the definition of canonical projection, we know that  $e$  is idempotent and  $e + f = 1$ . Hence we have  $W_1 = \text{im}(e)$  and  $W_2 = \text{im}(f)$ . It means that  $\text{Dirsum} \rightarrow \text{Idmdec} \rightarrow \text{Dirsum}$  is an identity. Therefore  $\text{Dirsum} \rightarrow \text{Idmdec}$  is a bijection.

(c) Let  $W$  denote  $e(V)$ . If  $W$  is decomposable, according to (b) then there exists an idempotent decomposition  $e|_W = \text{id}|_W = \varphi + \psi$ . Let  $i : W \rightarrow V$  be the canonical inclusion by considering the direct sum decomposition induced by idempotent decomposition  $\text{id}|_V = e + (1 - e)$ . Let  $\tilde{\varphi} = i \circ \varphi \circ e$  and  $\tilde{\psi} = i \circ \psi \circ e$ . Both  $\tilde{\varphi}$  and  $\tilde{\psi}$  are an idempotent morphism on  $V$ . Clearly both of them act as null morphism on  $\text{im}(1 - e) = \ker(e)$  and for an arbitrary element  $v \in \text{im}(e)$ , we have  $\tilde{\varphi}^2(v) = i \circ \varphi \circ e \circ i \circ \varphi \circ e(v) = i \circ \varphi^2 \circ e(v) = \tilde{\varphi}(v)$ . Simlilar for  $\tilde{\psi}$ . Therefore, we obtain idempotent decomposition  $e = \tilde{\varphi} + \tilde{\psi}$ . Conversely, suppose that  $e$  is not primitive, then there exists an idempotent decomposition  $e = \varphi + \psi$ . Both  $\varphi$  and  $\psi$  act idempotently on  $W$ . Therefore we obtain an idempotent decomposition on  $W$  which is  $e|_W = \text{id}|_W = \varphi|_W + \psi|_W$ . According to (b), it corresponds to a direct sum decomposition of  $W$ . Therefore, our claim holds.

(d) Suppose that  $e_1$  and  $e_2$  are conjugate by  $a$ . Namely, we assume that  $e_1 = ae_2a^{-1}$  it gives an isomorphism  $e_2(V) \rightarrow e_1(V)$  by  $v \mapsto a(v)$ . We just need to verify its well-definedness. Let  $v$  be an arbitrary element of  $e_2(V)$ , it means that there exists an element  $v' \in V$  such that  $v = e_2(v')$ . Since  $a$  is invertible, then we obtain that  $v = e_2 \circ a^{-1} \circ a(v')$ . By our assumption,

then we have  $a(v) = e_1 \circ a(v') \in e_1(V)$ . Since  $a$  is invertible on  $V$ , the morphism  $e_2(V) \rightarrow e_1(V)$  is an isomorphism. Conversely, suppose there exists an isomorphism  $a : e_2(V) \rightarrow e_1(V)$ . I claim that there exists a morphism  $\tilde{a}$  such that  $\tilde{a}|_{e_2(V)} = a$  and  $e_1 = \tilde{a}e_2\tilde{a}^{-1}$ . Since  $V$  is semisimple, then we can assume that

$$V \simeq e_1(V) \oplus W_1 \simeq e_2(V) \oplus W_2.$$

where  $W_1$  and  $W_2$  are the complement of  $V_1$  and  $V_2$  induced by idempotent morphisms  $e_1$  and  $e_2$  respectively. Because  $e_1(V) \simeq e_2(V)$ , we have  $W_1 \simeq W_2$  by considering the simple module decomposition  $V \cong \bigoplus_{i=1}^k S_i^{n_i}$ . Therefore, we let  $b : W_2 \rightarrow W_1$  be an arbitrary isomorphism and let  $\tilde{a} = a \oplus b$ , we find that  $\tilde{a}|_{e_2(V)} = a$ . To see the equation  $e_1 = \tilde{a}e_2\tilde{a}^{-1}$  holds, by given an arbitrary element  $v \in V$ , there exists decomposition  $v = v_2 + w_2$  and  $\tilde{a}(v) = a(v_2) + b(w_2)$ . Then we obtain that

$$(e_1 \circ \tilde{a} - \tilde{a} \circ e_2)(v) = e_1(a(v_2) + b(w_2)) - \tilde{a}(v_2) = a(v_2) - a(v_2) = 0.$$

Our claim holds.

(e) Let  $V = k^n$  be a  $k$ -module. According to Artin–Wedderburn theorem, we obtain that  $\text{End}_k(V) = M_n(D)$  where  $D = \text{End}_k(k) = k$ . Given that two arbitrary primitive idempotent elements  $e_1$  and  $e_2$  in  $M_n(k)$ , we must have  $e_1(V) \simeq e_2(V) \simeq k$  according to (c). Hence, by (d), we obtain that  $e_1$  and  $e_2$  are conjugate by an invertible element of  $M_n(k)$ . Therefore, all primitive idempotent elements are conjugate under the action of unit group  $\text{GL}_n(k)$ . ■

## Problem 2

Prove: No non-zero idempotent  $e$  of  $A$  is contained in  $J(A)$ . [Hint:  $e(1 - e) = 0$ .]

**Solution:** Suppose that there exists a non-zero idempotent element  $e$  in  $J(A)$ . Then we know that  $1 - e$  is invertible. Therefore, we can reduce  $1 - e$  on each sides of equation  $e(1 - e) = 0$ , we know obtain that  $e = 0$  is trivial contradicting our assumption. ■

## Problem 3

Prove: Let  $A$  be an Artinian ring with Artinian center  $Z(A)$ . Then  $J(Z(A)) = J(A) \cap Z(A)$ .

**Solution:** Suppose that  $x \in J(A) \cap Z(A)$ . Then for arbitrary element  $y \in Z(A)$ , we obtain that  $1 - yx$  is invertible in  $A$ . Since  $1 - yx$  is commutative and therefore its inverse is included in  $Z(A)$ . It means that  $x \in J(Z(A))$ . Conversely, since  $A$  and  $Z(A)$  are Artinian, their Jacobson radicals are nilpotent ideals. Therefore, let  $x$  be a nilpotent element in  $Z(A)$ , it is also nilpotent in  $A$ . Of course, it is commutative. Therefore we obtain  $J(Z(A)) \subseteq J(A) \cap Z(A)$ . ■

## Problem 4

Prove: Let  $A = A_1 \oplus \cdots \oplus A_n$  be a direct sum of 2-sided ideals. Then

$$(i) \quad Z(A) = Z(A_1) \oplus Z(A_2) \oplus \cdots \oplus Z(A_n).$$

$$(ii) \quad J(A) = J(A_1) \oplus J(A_2) \oplus \cdots \oplus J(A_n).$$

**Solution:** (i) Let  $x \in Z(A)$ . It admits a decomposition  $x = \sum_i x_i$  where  $x_i \in A_i$ . Since  $A_i$  is double sided ideals, then we obtain that  $A_i A_j \subseteq A_i \cap A_j = 0$  and  $A_j A_i = 0$ . Therefore, for arbitrary  $y_i \in A_i$ , the equation  $xy_i = y_i x$  implies that  $x_i y_i = y_i x_i$ . It means that  $x_i \in Z(A_i)$  and thus  $Z(A) \subseteq \bigoplus_i Z(A_i)$ . Conversely, suppose that  $x_i \in Z(A_i)$  and  $x = \sum_i x_i$  is clearly commutative. Then  $Z(A) \supseteq \bigoplus_i Z(A_i)$  and thus the claim holds.

(ii) Given that  $x \in J(A)$  with decomposition  $x = \sum_i x_i$  and  $y_i \in A_i$ . We find that  $1 - y_i x = 1 - y_i x_i$  is invertible in  $A$ . Suppose that  $u(1 - y_i x) = 1$  and  $u$  admits a decomposition  $u = \sum_i u_i$ . Suppose 1 admits a decomposition  $1 = \sum_i e_i$ , then we obtain that  $u_i(e_i - y_i x_i) = e_i$ . Therefore,  $e_i - y_i x_i$  is invertible in  $A_i$  and thus  $J(A) \subseteq \bigoplus_i J(A_i)$ . Conversely, given that  $x_i \in J(A_i)$  and an arbitrary element  $y = \sum_i y_i \in A$ . We consider that  $1 - yx = \sum_i (e_i - y_i x_i)$ . Since  $x_i \in J(A_i)$ , we can suppose there exists  $u_i \in A_i$  such that  $u_i(e_i - y_i x_i) = e_i$ . Therefore, we obtain that  $u(1 - yx) = \sum_i u_i(1 - y_i x_i) = \sum_i e_i = 1$  and thus  $J(A) \supseteq \bigoplus_i J(A_i)$ . ■

## Problem 5

Let  $\phi : U \rightarrow V$  be a homomorphism of  $A$ -modules. Show that  $\phi(\text{Soc } U) \subseteq \text{Soc } V$ , and that if  $\phi$  is an isomorphism then  $\phi$  restricts to an isomorphism  $\text{Soc } U \rightarrow \text{Soc } V$ .

**Solution:** I claim that  $\phi(\text{Soc}(U))$  is semisimple. Since  $\text{Soc}(U)$  is semisimple, we suppose that  $\text{Soc}(U) = \bigoplus_{i \in I} U_i$ . Since  $U_i$  is simple, then  $\phi(U_i)$  must be zero module or isomorphic to  $U_i$ . In any case,  $\phi(U_i)$  must be either zero or simple.  $\phi(\text{Soc}(U)) \subseteq \sum_i \phi(U_i)$  implies that  $\phi(\text{Soc}(U))$  as a submodule of semisimple module  $\sum_i \phi(U_i)$  is also semisimple. It means that  $\phi(\text{Soc}(U)) \subseteq \text{Soc}(V)$ . Since  $\phi$  is an isomorphism, for any non zero simple submodule  $U_i$ , then  $\phi(U_i)$  is not zero submodule. For arbitrary two simple module  $U_i$  and  $U_j$ , we then have  $\phi(U_i) \cap \phi(U_j) = \{0\}$ . Therefore we can suppose that

$$\text{Soc}(U) = \bigoplus_{i \in I} U_i \text{ and } \phi(\text{Soc}(U)) = \bigoplus_{i \in I} \phi(U_i).$$

Let  $\text{Soc}(V) = \phi(\text{Soc}(U)) \oplus \bigoplus_{j \in J} V_j$ . By (i), we obtain that

$$\phi^{-1}(\text{Soc}(V)) = \text{Soc}(U) \oplus \bigoplus_{j \in J} \phi^{-1}(V_j) \subseteq \text{Soc}(U).$$

It means  $V_j \simeq \phi^{-1}(V_j) \simeq (0)$ . Equivalently, we obtain that  $\text{Soc } U \rightarrow \text{Soc } V$  is an isomorphism. ■

## Problem 6

If  $\varphi : A \rightarrow A'$  is an epimorphism of rings, then  $\varphi(J(A)) \subseteq J(A')$ .

**Solution:** By definition, we obtain that

$$\varphi(J(A)) = \varphi \left( \bigcap_{\mathfrak{m} \in \text{Spm}(A)} \mathfrak{m} \right) \subseteq \bigcap_{\mathfrak{m} \in \text{Spm}(A)} \varphi(\mathfrak{m}).$$

where  $\text{Spm}$  denotes the spectrum of maximal ideals. Since  $\varphi$  is surjective, it follows that every maximal ideal  $\mathfrak{m}'$  in  $A'$  corresponds to a maximal ideal  $\mathfrak{m}$  in  $A$  such that  $\varphi(\mathfrak{m}) = \mathfrak{m}'$ . Therefore,

we obtain that

$$\varphi(J(A)) \subseteq \bigcap_{\mathfrak{m}' \in \text{Spm}(A')} \mathfrak{m}' = J(A').$$

■

## 11 Week 9

### Problem 1

Let  $A$  be a finite-dimensional algebra over a field. Suppose that  $V$  is a f.g.  $A$ -module, and that a certain simple  $A$ -module  $S$  occurs as a composition factor of  $V$  with multiplicity 1. Suppose that there exist nonzero homomorphisms  $S \rightarrow V$  and  $V \rightarrow S$ . Prove that  $S$  is a direct summand of  $V$ .

**Solution:** Let  $\varphi$  and  $\psi$  denote nonzero homomorphism  $S \rightarrow V$  and  $V \rightarrow S$  respectively. Since  $S$  is simple, we obtain that  $\varphi$  must be injective. Otherwise, it should be a zero homomorphism which contradicts our conditions. For the same reason, we obtain that  $\psi$  must be surjective. Hence, I claim that  $\psi \circ \varphi \in \text{End}_A(S)$  is an isomorphism or a zero homomorphism according to Schur's Lemma. Suppose that  $\psi \circ \varphi = 0$ . We obtain that a descending chain of submodules of  $V$

$$V \supseteq \ker \psi \supseteq \text{im } \varphi \supseteq (0).$$

Since  $\varphi$  is injective, we obtain that  $S \simeq \text{im } \varphi$ . By the First Isomorphism Theorem for modules, we also have  $S \simeq V/\ker \psi$ . It implies that  $S$  occurs twice as a composition factor which is a contradiction. Therefore,  $\psi \circ \varphi$  must be an isomorphism. Let  $f = (\psi \circ \varphi)^{-1}$  then we obtain that  $\psi \circ (\varphi \circ f) = \text{id}_S$ . It follows that the short exact sequence

$$0 \rightarrow \ker \psi \rightarrow V \xrightarrow{\psi} S \rightarrow 0$$

is split. Equivalently, we say  $V \simeq S \oplus \ker \psi$ .

■

### Problem 2

Suppose that  $U$ ,  $V$  and  $W$  are modules for a finite-dimensional algebra over a field and  $P_W$  is the projective cover of  $W$ . Assume that these modules are finite-dimensional.

- Show that  $U \rightarrow W$  is an essential epimorphism if and only if there is a surjective homomorphism  $P_W \rightarrow U$  so that the composite  $P_W \rightarrow U \rightarrow W$  is a projective cover of  $W$ . In this situation, show that  $P_W \rightarrow U$  must be a projective cover of  $U$ .
- Prove the following “extension and converse” to Nakayama's Lemma: let  $V$  be any submodule of  $U$ . Then  $U \rightarrow U/V$  is an essential epimorphism if and only if  $V \subseteq \text{Rad } U$ .

**Solution:** (a) Let  $f$  denote the essential epimorphism  $U \rightarrow W$ . Let  $g$  denote  $P_W \rightarrow W$ . According to the universal property of projective module, there exists a morphism  $h : P_W \rightarrow U$  such that  $g = f \circ h$ . Then we can see  $h$  is the homomorphism that we want. According to discussion in our class, both  $f$  and  $g$  are essential epimorphisms implies that  $h$  is also an essential epimorphism. It follows that  $h : P_W \rightarrow U$  is a projective cover. Conversely, since  $g$

is an essential epimorphism and  $h$  is a surjection, according to our homework, we obtain that  $f : U \rightarrow W$  and  $h : P_W \rightarrow U$  are also essentially epimorphic. It follows that  $P_W \rightarrow U$  is a projective cover and  $U \rightarrow W$  is an essential epimorphism.

(b) Let  $f$  denote morphism  $U \rightarrow U/V$ . Suppose that  $V \not\subseteq \text{Rad } U$ , then there exists a maximal submodule  $M$  such that  $M + V = U$ . By isomorphism Theorem of module, we obtain that  $U/V \simeq M/(M \cap V)$ . It implies that

$$f|_M : M \rightarrow U/V \simeq M/(M \cap V)$$

is an epimorphism. Equivalently,  $f$  is not an essential epimorphism.

Suppose that  $V \subseteq \text{Rad } U$ . Suppose that  $f$  is not essentially epimorphic, equivalently, there exists a proper submodule  $M$  of  $U$  such that  $f|_M$  is epimorphic. For an arbitrary element  $x \in U$ ,  $f|_M$  is epimorphic implies that there exists  $m \in M$  and  $v \in V$  such that  $x = m + v$ . It means that  $M + V = U$ . However, by Nakayama's Lemma, since  $V \subseteq \text{Rad } U$ , it implies that  $M = U$ . It contradicts our assumption. Therefore,  $f$  must be essentially epimorphic. ■

*Remark 11.1.* There are some facts:

- Let  $\text{Idem}(R) = \{e \in R \mid e^2 = e\}$  be the set of idempotents of a ring  $R$ , and let  $\text{Summ}(M)$  be the set of all direct summands of  $M$ . There is a bijection:

$$\Phi : \text{Idem}(\text{End}_A(M)) \xrightarrow{\sim} \text{Summ}(M)$$

$$e \mapsto eM$$

- Let  $f : M \rightarrow N$  be an essential epimorphism ( $\text{Ker}(f) \subseteq \text{rad}(M)$ ). We define

$$\mathcal{L}_N = \{\bar{e} \in \text{Idem}(\text{End}_A(N)) \mid \exists e \in \text{Idem}(\text{End}_A(M)) \text{ s.t. } f \circ e = \bar{e} \circ f\}.$$

then there exists bijective correspondence  $\Psi$  maps these idempotents to the set of essential covers of summands:

$$\Psi : \mathcal{L}_N \xrightarrow{\sim} \{f|_{eM} : eM \rightarrow \bar{e}N \mid e \in \text{Idem}(\text{End}_A(M)) \text{ is a lift of } \bar{e}\}$$

$$\bar{e} \mapsto (f|_{eM} : eM \rightarrow \bar{e}N)$$

- A ring  $R$  is semiperfect if every finitely generated (left)  $R$ -module admits a projective cover. It is equivalent to the following conditions:
  - a The quotient  $R/J(R)$  is a semisimple Artinian ring;
  - b Idempotents lift modulo  $J(R)$ .
- Let  $\text{Top}(M) = M/\text{rad}(M)$  be its radical quotient. A module  $M$  admits a projective cover if and only if  $\text{Top}(M)$  admits a projective cover.

### Problem 3

The following holds for a valuation  $v$  of  $K$ :

- (i)  $v(\pm 1) = 0$
- (ii)  $v(a) = v(-a)$
- (iii)  $v(a^{-1}) = -v(a)$
- (iv)  $v(b) < v(a) \Rightarrow v(a + b) = v(b)$ .



**Solution:** (i) Since  $v(1) = v(1^n) = nv(1)$  for any integer  $n$ . For equation holds,  $v(1) = 0$ . Similarly,  $v(-1) = v((-1)^{2n+1}) = (2n+1)v(-1)$  for any integer  $n$ , then we obtain that  $v(-1) = 0$ . (ii) Since (i), we obtain that  $v(-a) = v(-1) + v(a) = v(a)$ . (iii) Consider that  $0 = v(1) = v(a \cdot a^{-1}) = v(a) + v(a^{-1})$ , then  $v(a^{-1}) = -v(a)$ . (iv) We note that by the strongly triangular inequality, we obtain that  $v(a+b) \geq \min\{v(a), v(b)\} = v(b)$ . By the same reason, we consider that

$$v(b) = v(a+b-a) \geq \min\{v(a+b), v(a)\}.$$

Since  $v(b) < v(a)$ , then it implies that  $v(b) \geq v(a+b)$  for the inequality holds. Therefore, we claim that  $v(a+b) = v(b)$  if  $v(b) < v(a)$ . ■

## 12 Week 10

### Problem 1

Let  $\varphi$  be a multiplicative valuation of a field  $K$ . Then the following holds:

- (i)  $(a_n)$  is a Cauchy sequence  $\Leftrightarrow \lim_{n \rightarrow \infty} \varphi(a_{n+1} - a_n) = 0$ .
- (ii) If  $\lim_{n \rightarrow \infty} a_n = a \neq 0$ , then  $\varphi(a_n) = \varphi(a)$  for a sufficiently large  $n$ .

**Solution:** (i) Suppose that  $(a_n)$  is a Cauchy sequence. Then for any  $\varepsilon > 0$ , there exists  $N \in \mathbb{Z}_{>0}$  such that

$$\varphi(a_m - a_n) < \varepsilon \text{ for any } m, n > N.$$

We take  $m = n + 1$ , then we obtain that  $\varphi(a_{n+1} - a_n) < \varepsilon$  for any  $n > N$ . Equivalently, we can say  $\lim_{n \rightarrow \infty} \varphi(a_{n+1} - a_n) = 0$ . Conversely, suppose that  $\varphi(a_{n+1} - a_n) = 0$ , it means that for any  $\varepsilon > 0$  there exists a positive integer  $N$  such that for any  $n > N$  we have

$$\varphi(a_{n+1} - a_n) < \varepsilon.$$

Consider the strongly triangular inequality, then we have

$$\varphi(a_{n+m} - a_n) = \varphi\left(\sum_{i=0}^{m-1} (a_{n+i+1} - a_{n+i})\right) = \max_i \{\varphi(a_{n+i+1} - a_{n+i})\} < \varepsilon.$$

Therefore, we obtain that  $(a_n)$  is a Cauchy sequence.

- (ii) Since  $a_n - a$  converge to zero, then there exists  $N > 0$  such that for any  $n > N$  we have

$$\varphi(a_n - a) < \varphi(a)/2.$$

Similarly, since  $a_n$  converge to  $a$ , then there exists  $M > 0$  such that for any  $n > M$  we have

$$\varphi(a_n) > \varphi(a)/2.$$

We can choose  $N \geq M$ , then we have  $\varphi(a_n) > \varphi(a_n - a)$ . It implies that

$$\varphi(a) = \varphi(a - a_n + a_n) = \max\{\varphi(a - a_n), \varphi(a_n)\} = \varphi(a_n).$$

Our claim holds. ■

## Problem 2

Let  $\varphi$  be a multiplicative valuation of a field  $K$ . Let  $V$  be a finite dimensional  $K$ -space with basis  $u_1, \dots, u_n$ . For  $v = a_1u_1 + \dots + a_nu_n \in V$ , define  $\|v\|_0$  by

$$\|v\|_0 = \max\{\varphi(a_i); 1 \leq i \leq n\}.$$

Then this is a norm on  $V$ . Moreover, the following holds concerning the metric induced by this norm.

- (i) Let  $v_r = a_{r1}u_1 + \dots + a_{rn}u_n \in V$  be given for  $r = 1, 2, \dots$ . Then the sequence  $(v_r)$  is a Cauchy sequence if and only if the sequence  $(a_{ri})_r$  in  $K$  is so for all  $i$ . Also  $\lim_{r \rightarrow \infty} v_r = \sum_{i=1}^n a_i u_i$  if and only if  $\lim_{r \rightarrow \infty} a_{ri} = a_i$  for all  $i$ .
- (ii) If  $K$  is complete, then so is  $V$ .

**Solution:** (i) Suppose that  $(v_r)_r$  is a Cauchy sequence. It means that for any  $\varepsilon > 0$ , there exists  $N > 0$  such that for any  $r > N$  and index  $i$  we have

$$0 \leq \varphi(a_{r+1,i} - a_{ri}) \leq \|v_{r+1} - v_r\|_0 < \varepsilon.$$

It implies that  $\lim_{r \rightarrow \infty} \varphi(a_{r+1,i} - a_{ri}) = 0$ . According to Problem 1.(i), we then have  $(a_{ri})_r$  is a Cauchy sequence for each  $i$ .

Conversely, suppose that  $(a_{ri})_r$  is a Cauchy sequence. Then for every  $\varepsilon > 0$ , there exists an integer  $N$  such that for any  $r > N$ , an integer  $m$  and index  $i$  satisfy

$$\varphi(a_{r+m,i} - a_{r,i}) < \varepsilon.$$

Hence, we have

$$\|v_{r+m} - v_r\|_0 = \max_i \varphi(a_{r+m,i} - a_{r,i}) < \varepsilon.$$

It shows  $(v_r)_r$  is a Cauchy sequence.

For next statement, it is suffice to show that  $v_r \rightarrow 0$  iff.  $a_{ri} \rightarrow 0$ . Because we just let  $w_r = v_r - \sum a_i u_i$ , and the statement  $w_r \rightarrow 0$  iff.  $(a_{ri} - a_i) \rightarrow 0$  is equivalent to the statement we need to prove. Suppose that  $v_r$  converge to zero. It means that for any  $\varepsilon > 0$ , we have  $N > 0$  and for any  $r > N$  and index  $i$  the inequality

$$0 \leq \varphi(a_{ri}) \leq \|v_r\|_0 < \varepsilon$$

Which implies that  $\varphi(a_{ri}) \rightarrow 0$  and  $a_{ri} \rightarrow 0$  follows. Conversely,  $a_{ri} \rightarrow 0$  implies  $\varphi(a_{ri}) \rightarrow 0$  and thus we have  $\|v_r\|_0 \rightarrow 0$ . And  $v_r \rightarrow 0$  follows. Therefore, our claim holds.

(ii) Let  $(v_r)_r$  be an arbitrary Cauchy sequence. According to (i), it is equivalent to  $(a_{ri})_r$  is a Cauchy sequence for each  $i$ . Since  $K$  is complete, it means that there exists  $a_i \in K$  such that  $(a_{ri} - a_i)_r$  converge to zero. According to our discussion above, it is equivalent to  $(v_r - \sum a_i u_i)_r$  converge to zero. Therefore,  $V$  is complete as well. ■

## 13 Midterm

### Lemmas for Problem

**Lemma 13.1.** *A  $\mathbb{Q}/\mathbb{Z}$  is an injective  $\mathbb{Z}$ -module.*

*Proof.* Given an arbitrary  $\mathbb{Z}$ -module  $G$  with its submodule  $H$  and a  $\mathbb{Z}$ -module homomorphism  $f : H \rightarrow \mathbb{Q}/\mathbb{Z}$ , let

$$\Omega = \{J \supseteq H \mid \text{there exists } f_J : J \rightarrow \mathbb{Q}/\mathbb{Z} \text{ such that } f_J|_H = f\}.$$

According to Zorn's lemma, the partially ordered set  $\Omega$  has a maximal element  $K$ . Suppose that  $K$  is a proper submodule of  $G$ . Choose an arbitrary element  $a \in G \setminus K$ , we let  $\tilde{K}$  denote the submodule generated by  $K$  and  $a$ .

If the intersection  $\langle a \rangle \cap K$  only contains zero, equivalently,  $\tilde{K} = K \oplus \mathbb{Z} \cdot a$ , then we construct a map  $\tilde{f}_K : \tilde{K} \rightarrow \mathbb{Q}/\mathbb{Z}$  by sending  $a$  to zero and identified with  $f_K$  on  $K$ . Clearly, it is a  $\mathbb{Z}$ -module homomorphism since  $\tilde{f}_K$  can be identified with the composite of  $\tilde{K} \rightarrow K \rightarrow \mathbb{Q}/\mathbb{Z}$  with the first homomorphism is the canonical projection. It is impossible because we already set  $K$  is the maximal element but we find a larger element  $\tilde{K}$  with homomorphism  $\tilde{f}_K$ .

Next, We let the integer  $n$  to be the minimal positive integer that makes  $na \in K$  holds. We construct a map  $\tilde{f}_K : \tilde{K} \rightarrow \mathbb{Q}/\mathbb{Z}$  by sending  $a$  to  $f_K(na)/n$  and  $\tilde{f}_K|_K = f_K$ . It is a well-defined module homomorphism since if we have  $k - ma = 0$  in  $\tilde{K}$  where  $k \in K$  then it implies that  $n \mid m$  otherwise there must exists a positive integer  $n' < n$  such that  $n \equiv n' \pmod{m}$  and then  $a^{n'} \in K$  which is a contradiction. Therefore, we have  $\tilde{f}_K(k) - \tilde{f}_K(ma) = f_K(k - ma) = 0$ .

The process that we extend  $f_K$  to  $\tilde{f}_K$  implies that  $K$  must not be proper, equivalently,  $K = G$ . Then we obtain that  $f$  can be extended to  $G \rightarrow \mathbb{Q}/\mathbb{Z}$ , it means that  $\mathbb{Q}/\mathbb{Z}$  is an injective  $\mathbb{Z}$ -module.  $\square$

**Lemma 13.2.** *Let  $A$ ,  $B$ , and  $C$  be  $R$ -modules such that  $A \subseteq B \subseteq C$ . If  $A$  is an essential submodule of  $B$  and  $B$  is an essential submodule of  $C$ , then  $A$  is an essential submodule of  $C$ .*

*Proof.* We let  $\subseteq_e$  denote essential extension. Let  $W$  be an arbitrary non-zero submodule of  $C$ . To prove that  $A \subseteq_e C$ , we need to show that  $W \cap A \neq 0$ . Since  $B \subseteq_e C$  and  $W$  is a non-zero submodule of  $C$ , the intersection of  $W$  and  $B$  must be non-trivial, i.e.,  $W \cap B \neq 0$ .

Since  $A \subseteq_e B$  and  $W \cap B$  is a submodule of  $B$ , we obtain that

$$(W \cap B) \cap A \neq 0$$

which implies that  $W \cap A \neq 0$ . Therefore, we obtain that  $A \subseteq_e C$ .  $\square$

**Lemma 13.3.** *Let  $A$  and  $B$  be  $R$ -modules such that  $A \subseteq_e B$ . Let  $f : B \rightarrow C$  be a  $R$ -module monomorphism, then  $f(A) \subseteq_e f(B)$ .*

*Proof.* Given that an arbitrary submodule  $W$  of  $f(B)$ . Since  $A \subseteq_e B$ , we have

$$f^{-1}(W) \cap A \neq 0.$$

Since  $f$  is monomorphic, we then have

$$W \cap f(A) \neq 0$$

which implies that  $A \subseteq_e B$ .  $\square$

## Problem 1

Let  $R$  be a ring with 1.

- (i) Let  $h : U \rightarrow V$  be a monomorphism of  $R$ -modules. Then the following two conditions are equivalent.

- (a) A homomorphism  $k : V \rightarrow Y$  must be a monomorphism whenever the composite map  $U \xrightarrow{h} V \xrightarrow{k} Y$  is a monomorphism.
- (b) If  $0 \neq W \subset V$ , then  $h(U) \cap W \neq 0$ .

The monomorphism  $h$  that satisfy the above conditions is said to be *essential*. If there is an injective  $R$ -module  $Q$  with an essential monomorphism  $h : V \rightarrow Q$ , then the diagram  $V \xrightarrow{h} Q$  is called an *injective hull* of  $V$ .

(ii) Prove that: any  $R$ -module  $V$  has an injective hull.

**Solution:** (i) We first show that (a) implies (b). Suppose that  $W \neq 0$  and  $h(U) \cap W = 0$ , then I claim that  $U \rightarrow V/W$  is injective. Since intersection  $h(U) \cap W = 0$ , we obtain that

$$h(U) + W \simeq h(U) \oplus W.$$

Since  $h$  is a monomorphism, there is a canonical isomorphism  $h(U) \simeq U$  and a canonical inclusion

$$U \hookrightarrow h(U) \oplus W.$$

In this sense, we find that the homomorphism  $U \rightarrow V/W$  can be decomposed by

$$U \hookrightarrow h(U) \oplus W \rightarrow (h(U) \oplus W)/W.$$

The last step is exactly the canonical projection, then we obtain that  $U \rightarrow V/W$  is an injection. However, the homomorphism  $V \rightarrow V/W$  is a monomorphism if and only if  $W = (0)$  which contradicts our assumption.

Conversely, we suppose that  $k \circ h$  is a monomorphism but  $k$  is not injective. Since  $k$  is not injective, it means that  $\ker k \neq 0$ . Hence, the injectivity of  $k \circ h$  implies that  $\ker(k \circ h) \supseteq (h(U) \cap \ker k)$  and thus  $h(U) \cap \ker k = 0$ . It is a contradiction because we require that for an arbitrary submodule  $W$  of  $V$  we must have  $h(U) \cap W \neq 0$  and  $\ker k$  does not satisfy this requirement.

(ii) For a fixed  $R$ -module  $V$ , we construct an injective hull for it as follow. According to **Lemma A.**, we know that  $\mathbb{Q}/\mathbb{Z}$  is an injective  $\mathbb{Z}$ -module, it equals the functor  $\text{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z})$  is exact. Let  $Q = \text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})$  be the Pontryagin dual module of  $R$ . Since the tensor-hom duality, we obtain that

$$\text{Hom}_R(M, Q) \simeq \text{Hom}_{\mathbb{Z}}(M \otimes_R R, \mathbb{Q}/\mathbb{Z}) \simeq \text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$$

For arbitrary  $R$ -module  $M$ . It means that the functor  $\text{Hom}_R(-, Q)$  is also exact, thus  $Q$  is an injective  $R$ -module. To construct an injective hull for  $R$ -module  $V$ , we first view  $V$  as a  $\mathbb{Z}$ -module and consider the  $\mathbb{Z}$ -module homomorphism

$$f : V \rightarrow \prod_{v \in V} \mathbb{Q}/\mathbb{Z}$$

determined by  $\psi_v : V \rightarrow \mathbb{Q}/\mathbb{Z}$  by extending the  $\mathbb{Z}$ -module homomorphism  $\langle v \rangle \rightarrow \mathbb{Q}/\mathbb{Z}$  by sending  $v$  to  $1/2 + \mathbb{Z}$  if  $\#\langle v \rangle = \infty$  or to  $1/n + \mathbb{Z}$  if  $\#\langle v \rangle = n$ . The extension exists because  $\mathbb{Q}/\mathbb{Z}$  is an injective module. We also note that  $f$  is injective since for an arbitrary element  $v \in \ker f$ , we have  $\psi_v(v) = \mathbb{Z}$ . It means that the order of  $\langle v \rangle$  is one, equivalently,  $1 \cdot v = 0$  and thus we have  $v = 0$ . The injectivity of  $f$  holds.

It induces an  $R$ -module homomorphism

$$\tilde{f} : \text{Hom}_{\mathbb{Z}}(R, V) \rightarrow \text{Hom}_{\mathbb{Z}}(R, \prod_{v \in V} \mathbb{Q}/\mathbb{Z}).$$

Since we have  $\text{Hom}_{\mathbb{Z}}(R, V) \simeq V$  and  $\text{Hom}_R(R, \prod_{v \in V} \mathbb{Q}/\mathbb{Z}) \simeq \prod_{v \in V} \mathbb{Q}$ . We can rewrite this homomorphism

$$\tilde{f} : V \rightarrow \prod_{v \in V} \mathbb{Q}.$$

First of all, the limit  $\prod_{v \in V} \mathbb{Q}$  is an injective  $R$ -module since given by an arbitrary  $R$ -module monomorphism  $A \rightarrow B$  with homomorphism  $A \rightarrow \prod_{v \in V} \mathbb{Q}$ , The second homomorphism induces homomorphisms  $A \rightarrow \mathbb{Q}$  for each index  $v \in V$  by composite the canonical projection. Because  $\mathbb{Q}$  is an injective  $R$ -module, there exists lift  $B \rightarrow \mathbb{Q}$  for each  $A \rightarrow \mathbb{Q}$ . Hence, by the universal property of product,  $B \rightarrow \mathbb{Q}$  for each index induces a  $R$ -module homomorphism  $B \rightarrow \prod_{v \in V} \mathbb{Q}$  such that  $A \rightarrow \prod_{v \in V} \mathbb{Q}$  can be decomposed to  $A \rightarrow B \rightarrow \prod_{v \in V} \mathbb{Q}$ . It means that  $\prod_{v \in V} \mathbb{Q}$  is an injective  $R$ -module. Hence,  $\tilde{f}$  is injective since  $f$  is injective and functor  $\text{Hom}_{\mathbb{Z}}(R, -)$  is left exact.

Now, we consider the partially ordered set

$$\Omega = \left\{ E \mid E \subseteq \prod_{v \in V} \mathbb{Q} \text{ and } E \cap \tilde{f}(V) \neq 0 \text{ for any nonzero submodule } F \text{ of } E \right\}.$$

According to Zorn's lemma,  $\Omega$  admits a maximal element we denote it by  $E$ . Hence, we consider the another partially ordered set

$$\Gamma = \left\{ K \mid K \subseteq \prod_{v \in V} \mathbb{Q} \text{ and } K \cap E = 0 \right\}.$$

Similarly, we let  $K$  denote a maximal element of  $\Gamma$ . I claim that  $\prod_{v \in V} \mathbb{Q} \simeq K \oplus E$ . To see that we consider a homomorphism  $E \rightarrow (\prod_{v \in V} \mathbb{Q})/K$ . Since  $E \cap K = 0$ , this homomorphism is injective. Therefore,  $\prod_{v \in V} \mathbb{Q}$  is an injective module implies that there exists a homomorphism

$$\Phi : \left( \prod_{v \in V} \mathbb{Q} \right) / K \rightarrow \prod_{v \in V} \mathbb{Q}$$

such that the diagram

$$\begin{array}{ccc} E & \xrightarrow{\iota} & (\prod_{v \in V} \mathbb{Q}) / K \\ & \searrow \psi & \downarrow \Phi \\ & & \prod_{v \in V} \mathbb{Q} \end{array}$$

commutes. We notice that  $\Phi$  is injective. Because  $\psi$  is injective, then  $\Phi|_{\iota(E)}$  is injective, therefore we have  $\ker \Phi \cap \iota(E) = 0$ . If  $\ker \Phi \neq (0)$ , we consider the canonical quotient

$$\pi : \prod_{v \in V} \mathbb{Q} \rightarrow \left( \prod_{v \in V} \mathbb{Q} \right) / K$$

and we obtain that  $\pi^{-1}(\ker \Phi) \cap E \subseteq K$ , since  $E \cap K = 0$ , then we have  $\pi^{-1}(\ker \Phi) \cap E = 0$ . Clearly,  $\pi^{-1}(\ker \Phi) \not\subseteq K$  then we find that  $(K + \pi^{-1}(\ker \Phi)) \cap E = 0$  and  $K \subsetneq (K + \pi^{-1}(\ker \Phi))$ .

It contradicts the maximality of  $K$ , therefore  $\Phi$  must be injective. The injectivity of  $\Phi$  implies that  $\iota(E)$  is an essential submodule of  $(\prod_{v \in V} Q)/K$  since if there exists a submodule  $M + K$  of  $(\prod_{v \in V} Q)/K$  such that  $(M + K) \cap \iota(E) = 0$ , then we have  $\Phi(M + K) \cap E = 0$ . It implies that  $\Phi(M + K) \subseteq K$  equivalently  $M + K \in \ker \Phi$ . Therefore, we obtain that  $\iota(E)$  is an essential submodule of  $(\prod_{v \in V} Q)/K$ .

According to **Lemma C.**, the essential extension  $\tilde{f}(V) \subseteq_e E$  implies that  $\iota \circ \tilde{f}(V) \subseteq_e \iota(E)$  since  $\iota$  is injective. Hence **Lemma B.** implies that

$$\iota \circ \tilde{f}(V) \subseteq_e (\prod_{v \in V} Q)/K$$

due to the fact that  $\iota \circ \tilde{f}(V) \subseteq_e \iota(E)$  and  $\iota(E) \subseteq_e (\prod_{v \in V} Q)/K$ . The injectivity of  $\Phi$  implies that

$$\tilde{f}(V) \subseteq_e \Phi \left( (\prod_{v \in V} Q)/K \right).$$

Since we also have  $E \subseteq \Phi((\prod_{v \in V} Q)/K)$ , the maximality of  $E$  implies that

$$E = \Phi \left( (\prod_{v \in V} Q)/K \right).$$

Equivalently we say  $E \simeq (\prod_{v \in V} Q)/K$  and therefore

$$\prod_{v \in V} Q = K \oplus E.$$

then  $E$  as a summand of injective  $R$ -module  $\prod_{v \in V} Q$  is also a injective  $R$ -module. The definition of  $\Omega$  yields  $E$  satisfies condition **(i.b)**, therefore  $E$  is an injective hull of  $V$ . Our discussion implies that every  $R$ -module has an injective hull. ■

## Problem 2

(i) The following holds for an idempotent  $e$  of  $R$ :

$$J(eRe) = eJ(R)e = eRe \cap J(R).$$

(ii) Prove:  $J(M_n(R)) = M_n(J(R))$ . Here  $M_n(R)$  means the matrix ring.

**Solution:** (i) Suppose that  $x = er_0e \in J(eRe)$ , then for arbitrary  $r \in R$ , we consider

$$1 - rx = (1 - e) + (e - rx).$$

Since  $x \in J(eRe)$ , then there exists  $t = er_1e \in eRe$  such that

$$t(e - ex) = e.$$

Hence we have

$$t(e - rx) = er_1e^2(e - rx) = t(e - ex) = e.$$

Now we consider

$$(1 - e + t)(1 - rx) = (1 - e + t)(1 - e + e - rx) = 1 - (1 - e)rx$$

Hence we have

$$((1-e)rx)^2 = (1-e)rx(1-e)rx = 0$$

since  $x \in eRe$  and  $x(1-e) = 0$ . Let  $\varphi = (1-e+t)$  and  $\psi = (1-rx)$  we then have

$$(2\varphi - \varphi\psi\varphi)(1-rx) = 1.$$

It means that  $x \in J(R)$ . Equivalently, we prove that  $J(eRe) \subseteq J(R)$ .

Since  $J(eRe) \subseteq eRe$  we obtain that

$$J(eRe) \subseteq eRe \cap J(R).$$

Let  $x \in eRe \cap J(R)$ , then we can suppose that  $x = er_0e$  and for any  $r \in R$ , we have

$$1 - rx = 1 - r(er_0e)$$

is invertible. Suppose that  $t(1-rx) = 1$  with  $r = er'e$  then we consider

$$(ete)(e-rx) = (ete)(e - (er'e)(er_0e)) = et(1-rx)e = e^2 = e.$$

it implies that

$$J(eRe) \supseteq eRe \cap J(R).$$

Then we obtain that  $J(eRe) = eRe \cap J(R)$ .

Next, we consider  $eJ(R)e$ . we consider that  $R$  as a left  $R$ -module  ${}_R R$ . Then we have idempotent decomposition

$${}_R R = eR \oplus (1-e)R$$

Then we have  $J(R) = eJ(R) + (1-e)J(R)$  and

$$J(R) \cap eR = eJ(R). \quad (\star)$$

Similarly, if we consider  $R$  as a right  $R$ -module  $R_R$ , we obtain that

$$J(R) \cap Re = J(R)e. \quad (\Delta)$$

I claim that  $eR \cap Re = eRe$ . To see that, let  $x \in eR \cap Re$  then suppose that  $x = er_1 = r_2e$ . Since  $e$  is idempotent, then we have

$$er_1e = xe = r_2e^2 = r_2e = x$$

it implies that  $eR \cap Re \subseteq eRe$ . And  $eRe \subseteq eR \cap Re$  is easy to see. By the same reason we have  $eJ(R) \cap J(R)e = eJ(R)e$ . Then we obtain that

$$(J(R) \cap eR) \cap (J(R) \cap Re) = J(R) \cap eRe$$

and therefore  $J(R) \cap eRe = eJ(R)e$ .

(ii) Let  $E_{ij}$  denote the matrix of  $(i, j)$ -entry is  $1_R$  and others entires is zero. Let  $M$  denote  $M_n(R)$ . Let  $A = (a_{ij})$  be an arbitrary element in  $J(M)$ . Since  $J(M)$  is a double sided ideal, then we have  $\tilde{A} = E_{1i}AE_{j1} \in J(M)$ . According to (i), we have

$$E_{11}J(M)E_{11} = J(E_{11}ME_{11}) \simeq J(R).$$

Then it implies that  $E_{11}\tilde{A}E_{11} \in J(E_{11}ME_{11})$ . Equivalently, we have  $a_{ij} \in J(R)$  and thus  $J(M) \subseteq M_n(J(R))$ . Conversely, we first show that  $J(R)E_{ii} \subseteq J(M)$  since

$$J(R)E_{ii} = J(E_{ii}ME_{ii}) = E_{ii}J(M)E_{ii} \subseteq J(M).$$

Therefore, we have

$$J(R)E_{ij} = J(R)E_{ii}E_{ij} \subseteq J(M)E_{ij} \subseteq J(M).$$

It implies that  $M_n(J(R)) \subseteq J(M)$  and thus

$$J(M_n(R)) = M_n(J(R)).$$

■

### Problem 3

**Prove:** A submodule of a finitely generated free module over a PID is free.

**Solution:** Let  $R$  be an arbitrary PID. We first to show that submodules of  ${}_R R$  is free. Let  $H$  be an arbitrary submodule of  $R$ , equivalently,  $H$  be an arbitrary ideal of  $R$ . Since  $R$  is PID, then there exists an element  $x \in R$  such that  $H = Rx$ . Since  $R$  is integral then there is not nontrivial zero divisor in  $R$ . Therefore, we find that

$$R \rightarrow Rx \quad r \mapsto rx$$

is an isomorphism. Equivalently,  $H$  is free  $R$ -module.

Now, we consider the induction on rank of finitely generated free modules. Suppose that our claim holds for  $R^{n-1}$ . We consider the case that rank of  $n$ . Let  $H$  be an arbitrary submodule of  $R^n$ . Let  $p_n$  be the canonical projection on  $n^{\text{th}}$  position of  $R^n$ . Then we obtain a short exact sequence

$$0 \rightarrow \ker p_n \cap H \rightarrow H \rightarrow p_n(H) \rightarrow 0.$$

Since  $p_n(H)$  is a submodule of  $R$ , it means that  $p_n(H)$  is a free  $R$ -module. Therefore, the exact sequence split. We can write

$$H \simeq (\ker p_n \cap H) \oplus p_n(H).$$

Since  $\ker p_n \cap H$  is the submodule of  $R^{n-1}$ , then by our assumption, it is also a free  $R$ -module. Therefore, our claim holds. ■

## 14 Week 11

### Problem 1

Let  $R$  be a ring and  $V$  be an  $R$ -module. Let  $I$  be an ideal of  $R$ . Suppose that  $d_I$  is defined on  $V$  and on  $R$ . Then

- (i) Addition is continuous from  $V \times V$  to  $V$ .
- (ii) Multiplication is continuous from  $R \times V$  to  $V$ .
- (iii) If  $d_I$  is defined on the  $R$ -module  $W$  and  $f \in \text{Hom}_R(V, W)$ , then  $f$  is continuous.



**Solution:** (1) Let the addition map be  $f : V \times V \rightarrow V$ , where  $f(x, y) = x + y$ . Let  $O$  be an arbitrary open set in the target space  $V$ . We need to prove that its preimage  $f^{-1}(O)$  is an open set in  $V \times V$ . If  $f^{-1}(O)$  is empty, it is trivially open. If  $f^{-1}(O)$  is non-empty, pick an arbitrary point  $(x, y) \in f^{-1}(O)$ . It means  $x + y \in O$ . Since  $O$  is an open set in  $V$ , there exists an integer  $n$  such that:

$$(x + y) + I^n V \subseteq O.$$

Let  $U_x = x + I^n V$  and  $U_y = y + I^n V$  be the open neighborhood of  $x, y$  respectively. Then  $U_x \times U_y$  is an open neighborhood of  $(x, y)$  in  $V \times V$ . Since

$$f(U_x \times U_y) = U_x + U_y = (x + I^n V) + (y + I^n V) = x + y + I^n V$$

we have  $f(U_x \times U_y) \subseteq O$  which implies

$$U_x \times U_y \subseteq f^{-1}(O).$$

Therefore, for arbitrary  $(x, y) \in f^{-1}(O)$ , we have an open neighborhood  $U_x \times U_y$  contained in  $f^{-1}(O)$ . It turns out that  $f^{-1}(O)$  is open and thus  $f$  is continuous.

(2) Let the multiplication map be  $g : R \times V \rightarrow V$ , where  $g(r, v) = rv$ . Let  $O$  be an arbitrary open set in the target space  $V$ . We need to prove that its preimage  $g^{-1}(O)$  is an open set in  $R \times V$ . If  $g^{-1}(O)$  is empty, it is trivially open. If  $g^{-1}(O)$  is non-empty, pick an arbitrary point  $(r, v) \in g^{-1}(O)$ . It means  $rv \in O$ . Since  $O$  is an open set in  $V$ , there exists an integer  $n$  such that:

$$rv + I^n V \subseteq O.$$

Let  $U_r = r + I^n R$  and  $U_v = v + I^n V$  be the open neighborhood of  $r, v$  respectively. Then  $U_r \times U_v$  is an open neighborhood of  $(r, v)$  in  $R \times V$ . Since

$$g(U_r \times U_v) = U_r U_v = (r + I^n R)(v + I^n V) \subseteq rv + r(I^n V) + (I^n R)v + I^{2n} V \subseteq rv + I^n V$$

we have  $g(U_r \times U_v) \subseteq O$  which implies

$$U_r \times U_v \subseteq g^{-1}(O).$$

Therefore, for arbitrary  $(r, v) \in g^{-1}(O)$ , we have an open neighborhood  $U_r \times U_v$  contained in  $g^{-1}(O)$ . It turns out that  $g^{-1}(O)$  is open and thus  $g$  is continuous.

(3) Let  $O$  be an arbitrary open set in the target space  $W$ . We need to prove that its preimage  $f^{-1}(O)$  is an open set in  $V$ . If  $f^{-1}(O)$  is empty, it is trivially open. If  $f^{-1}(O)$  is non-empty, pick an arbitrary point  $v \in f^{-1}(O)$ . It means  $f(v) \in O$ . Since  $O$  is an open set in  $W$ , there exists an integer  $n$  such that:

$$f(v) + I^n W \subseteq O.$$

Let  $U_v = v + I^n V$  be the open neighborhood of  $v$  in  $V$ . Since  $f$  is an  $R$ -module homomorphism, we have  $f(I^n V) = I^n f(V) \subseteq I^n W$ . Then

$$f(U_v) = f(v + I^n V) = f(v) + f(I^n V) \subseteq f(v) + I^n W$$

we have  $f(U_v) \subseteq O$  which implies

$$U_v \subseteq f^{-1}(O).$$

Therefore, for arbitrary  $v \in f^{-1}(O)$ , we have an open neighborhood  $U_v$  contained in  $f^{-1}(O)$ . It turns out that  $f^{-1}(O)$  is open and thus  $f$  is continuous. ■

## Problem 2

The following algebra isomorphisms hold for  $R$ -algebras  $A$  and  $B$ , where  $R$  is a commutative ring.

- (i)  $M_n(A) \otimes_R B \simeq M_n(A \otimes_R B)$ .
- (ii)  $M_n(A) \otimes_R M_m(B) \simeq M_{nm}(A \otimes_R B)$ .

**Solution:** (i) Let

$$\begin{aligned}\phi : M_n(A) \times B &\rightarrow M_n(A \otimes_R B) \\ ((a_{ij}), b) &\mapsto (a_{ij} \otimes b)\end{aligned}$$

Since  $\phi$  is  $R$ -bilinear, it induces a well-defined  $R$ -linear map

$$\begin{aligned}\Phi : M_n(A) \otimes_R B &\rightarrow M_n(A \otimes_R B) \\ (a_{ij}) \otimes b &\mapsto (a_{ij} \otimes b)\end{aligned}$$

For any  $(a_{ij}), (c_{ij}) \in M_n(A)$  and  $b, d \in B$ , we have

$$\Phi((a_{ij}) \otimes b)((c_{ij}) \otimes d) = \Phi((a_{ij})(c_{ij}) \otimes bd) = \left( \sum_k a_{ik} c_{kj} \otimes bd \right).$$

On the other hand,

$$\Phi((a_{ij}) \otimes b)\Phi((c_{ij}) \otimes d) = (a_{ij} \otimes b)(c_{ij} \otimes d) = \left( \sum_k (a_{ik} \otimes b)(c_{kj} \otimes d) \right) = \left( \sum_k a_{ik} c_{kj} \otimes bd \right).$$

Thus  $\Phi$  preserves multiplication and is an algebra homomorphism. The map

$$\begin{aligned}\Psi : M_n(A \otimes_R B) &\rightarrow M_n(A) \otimes_R B \\ (a \otimes b)E_{ij} &\mapsto aE_{ij} \otimes b\end{aligned}$$

provides a two-sided inverse to  $\Phi$ . Therefore, we obtain that

$$M_n(A) \otimes_R B \simeq M_n(A \otimes_R B).$$

- (ii) By applying the isomorphism from (i) and treating  $M_m(B)$  as an  $R$ -algebra, we have

$$M_n(A) \otimes_R M_m(B) \simeq M_n(A \otimes_R M_m(B)).$$

Since  $R$  is commutative, the tensor product is symmetric up to isomorphism, yielding

$$A \otimes_R M_m(B) \simeq M_m(B) \otimes_R A.$$

Applying (i) again to  $M_m(B) \otimes_R A$ , we get

$$M_m(B \otimes_R A) \simeq M_m(A \otimes_R B).$$

Substituting this sequence of isomorphisms back gives:

$$M_n(A \otimes_R M_m(B)) \simeq M_n(M_m(A \otimes_R B)).$$

For any ring  $S$ , there is a natural ring isomorphism  $M_n(M_m(S)) \simeq M_{nm}(S)$  given by viewing an  $n \times n$  block matrix of  $m \times m$  matrices as an  $nm \times nm$  matrix. Letting  $S = A \otimes_R B$ , we conclude

$$M_n(A) \otimes_R M_m(B) \simeq M_{nm}(A \otimes_R B).$$

■

## 15 Week 12

### Problem 1

- (i) Let  $(X, \rho)$  be a transitive **Set**-representation of  $G$ . For  $x \in X$ , let  $G_x$  denote the fixator of  $x$  in  $G$ . Then the map

$$G/G_x \rightarrow X, \quad gG_x \mapsto \rho(g)(x)$$

defines an isomorphism  $(G/G_x, \rho_{G_x}) \xrightarrow{\sim} (X, \rho)$ .

- (ii) For subgroups  $H$  and  $H'$  of  $G$ , the representations  $(G/H, \rho_H)$  and  $(G/H', \rho_{H'})$  are isomorphic if and only if  $H$  and  $H'$  are conjugate in  $G$ .

**Solution:** (i) Let  $f : G/G_x \rightarrow X$  denote the map we discuss. First we show that  $f$  is well-defined. Suppose

$$gG_x = hG_x.$$

Then  $h^{-1}g \in G_x$ , so

$$\rho_{h^{-1}g}(x) = x.$$

Hence

$$\rho_g(x) = \rho_h(\rho_{h^{-1}g}(x)) = \rho_h(x).$$

Therefore  $f(gG_x) = f(hG_x)$ , so  $f$  is well-defined. Next, We show that  $f$  is a bijection of sets. Let  $y$  be an arbitrary element of  $X$  that distinct to  $x$ . Since  $G$  act transitively on  $X$  then there exists  $g \in G$  such that  $\rho_g(x) = y$ . Therefore, we have

$$f(gG_x) = \rho_g(x) = y.$$

It implies that  $f$  is surjective. Let  $g, h \in G$  such that  $f(gG_x) = f(hG_x)$ . It implies that  $\rho_g(x) = \rho_h(x)$  and thus

$$x = \rho_{h^{-1}} \circ \rho_g(x) = \rho_{h^{-1}g}(x).$$

Then we have  $h^{-1} \cdot g \in G_x$ . Equivalently, we have  $gG_x = hG_x$ . Therefore,  $f$  is injective. For arbitrary  $g \in G$ , we then have

$$f(g \cdot hG_x) = f(ghG_x) = \rho_{gh}(x) = \rho_g \circ \rho_h(x) = \rho_g \cdot f(hG_x).$$

Therefore,  $f$  is isomorphism of representation.

- (ii) Suppose first that

$$f : G/H \rightarrow G/H'$$

is an isomorphism of  $G$ -sets. Write

$$f(H) = gH'$$

for some  $g \in G$ . Since  $f$  is  $G$ -equivariant, it preserves stabilizers. Hence

$$H = \text{Stab}_G(H) = \text{Stab}_G(f(H)) = \text{Stab}_G(gH') = gH'g^{-1}.$$

Thus  $H$  and  $H'$  are conjugate.

Conversely, suppose

$$H = gH'g^{-1}$$

for some  $g \in G$ . Define

$$\Phi : G/H \rightarrow G/H', \quad \Phi(aH) = agH'.$$

This is well-defined: if  $aH = bH$ , then  $a = bh$  for some  $h \in H$ . Since

$$g^{-1}hg \in H',$$

we get

$$agH' = bhgH' = bgH'.$$

Thus  $\Phi(aH) = \Phi(bH)$ .

The map  $\Phi$  is bijective, with inverse

$$\Psi : G/H' \rightarrow G/H, \quad \Psi(aH') = ag^{-1}H.$$

Moreover, for  $k \in G$ ,

$$\Phi(k \cdot aH) = \Phi(kaH) = kagH' = k \cdot \Phi(aH).$$

Therefore  $\Phi$  is  $G$ -equivariant. Hence

$$(G/H, \rho_H) \cong (G/H', \rho_{H'})$$

if and only if  $H$  and  $H'$  are conjugate in  $G$ . ■

## Problem 2

Assume that  $G$  is the cyclic (additive) group  $\mathbb{Z}/p\mathbb{Z}$  where  $p$  is a prime number, and let  $k$  be a field of characteristic  $p$ . Consider the morphism

$$G \rightarrow \mathrm{GL}_2(k), \quad \lambda \mapsto \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}.$$

Prove that the line generated by the first standard basis vector of  $k^2$  is  $G$ -stable but has no  $G$ -stable complement.

**Solution:** Let

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Since  $\mathrm{char}(k) = p$ , the map

$$\rho : \mathbb{Z}/p\mathbb{Z} \rightarrow \mathrm{GL}_2(k), \quad \lambda \mapsto \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}$$

is well-defined and is a group homomorphism.

Let

$$W = ke_1.$$

Then for every  $\lambda \in \mathbb{Z}/p\mathbb{Z}$ ,

$$\rho(\lambda)e_1 = e_1,$$

so  $W$  is  $G$ -stable.

We claim that  $W$  has no  $G$ -stable complement. Suppose, for contradiction, that

$$k^2 = W \oplus U$$

with  $U$  is  $G$ -stable. Then  $U$  is one-dimensional and  $U \neq W$ , so

$$U = k(ae_1 + e_2)$$

for some  $a \in k$ . Since  $U$  is  $G$ -stable, it is stable under the action of  $1 \in \mathbb{Z}/p\mathbb{Z}$ . Thus

$$\rho(1)(ae_1 + e_2) = (a+1)e_1 + e_2$$

must lie in  $U$ . Hence there exists  $c \in k$  such that

$$(a+1)e_1 + e_2 = c(ae_1 + e_2).$$

Comparing coefficients gives  $c = 1$ , and then

$$a+1 = a,$$

a contradiction. Therefore  $W$  has no  $G$ -stable complement. ■

### Problem 3

Let  $(M, \rho)$  and  $(N, \sigma)$  be two  $k$ -representations of  $G$ . Fix basis  $(e_i)_{1 \leq i \leq m}$  and  $(f_j)_{1 \leq j \leq n}$  of  $M$  and  $N$  respectively. Then  $(e_i \otimes f_j)_{1 \leq i \leq m, 1 \leq j \leq n}$  is a basis of  $M \otimes_k N$ . Denote by  $(m, \rho)$ ,  $(n, \sigma)$ ,  $(mn, \rho \otimes \sigma)$  be the matrix representations associated with  $(M, \rho)$ ,  $(N, \sigma)$ ,  $(M \otimes_k N, \rho \otimes \sigma)$  respectively. Determine the relation between  $(\rho \otimes \sigma)(g)$  and  $\rho(g)$ ,  $\sigma(g)$ .

**Solution:** Suppose

$$\rho_g(e_i) = \sum_{k=1}^m a_{ki} e_k, \quad \sigma_g(f_j) = \sum_{\ell=1}^n b_{\ell j} f_\ell.$$

Then

$$(\rho \otimes \sigma)_g(e_i \otimes f_j) = \rho_g(e_i) \otimes \sigma_g(f_j) = \sum_{k=1}^m \sum_{\ell=1}^n a_{ki} b_{\ell j} (e_k \otimes f_\ell).$$

With respect to the ordered basis

$$(e_1 \otimes f_1, \dots, e_1 \otimes f_n, e_2 \otimes f_1, \dots, e_m \otimes f_n),$$

the column corresponding to  $e_i \otimes f_j$  is

$$(a_{1i}b_{1j}, \dots, a_{1i}b_{nj}, a_{2i}b_{1j}, \dots, a_{2i}b_{nj}, \dots, a_{mi}b_{1j}, \dots, a_{mi}b_{nj})^T.$$

Therefore

$$[(\rho \otimes \sigma)_g] = [\rho_g] \otimes [\sigma_g],$$

where the tensor product on the right is the Kronecker product of matrices. ■

### Problem 4

Let  $(M, \rho)$  be a  $k$ -representations of  $G$ . Fix basis  $(e_i)_{1 \leq i \leq m}$  of  $M$ . Let  $(M^*, \rho^*)$  be the contragredient representation of  $(M, \rho)$ . Denote by  $(m, \rho)$ ,  $(m, \rho^*)$  be the matrix representations associated with  $(M, \rho)$ ,  $(M^*, \rho^*)$ . Determine the relation between  $\rho(g)$  and  $\rho^*(g)$ .

**Solution:** Let  $e_1, \dots, e_n$  be a basis of  $V$ , with dual basis  $e_1^*, \dots, e_n^*$ . Suppose

$$\rho_g(e_i) = \sum_{k=1}^n a_{ki} e_k.$$

Thus

$$[\rho_g] = (a_{ki}).$$

The contragredient representation  $\rho^*$  is defined by

$$(\rho_g^* \varphi)(v) = \varphi(\rho_{g^{-1}} v).$$

Write

$$\rho_{g^{-1}}(e_j) = \sum_{k=1}^n b_{kj} e_k.$$

Then  $[\rho_{g^{-1}}] = (b_{kj}) = [\rho_g]^{-1}$ . For each  $i, j$ ,

$$(\rho_g^* e_i^*)(e_j) = e_i^*(\rho_{g^{-1}} e_j) = e_i^* \left( \sum_{k=1}^n b_{kj} e_k \right) = b_{ij}.$$

Hence the coefficient of  $e_j^*$  in  $\rho_g^*(e_i^*)$  is  $b_{ij}$ . Therefore

$$[\rho_g^*] = (b_{ij}) = [\rho_{g^{-1}}]^T = ([\rho_g]^{-1})^T.$$

Thus the matrix of the contragredient representation is the inverse transpose of the original matrix. ■

### Problem 5

- (i) Let  $G$  be a finite group acting on a finite set  $\Omega$ . Let  $\text{Fix}^G(\Omega)$  denote the set of fixed points of  $\Omega$  under  $G$ , and let  $\Omega^\#$  denote the set of orbits of  $G$  on  $\Omega \setminus \text{Fix}^G(\Omega)$ . Prove that

$$|\Omega| = |\text{Fix}^G(\Omega)| + \sum_{\omega \in \Omega^\#} |\omega|.$$

From now on, we assume that  $p$  is a prime number and that  $G$  is a nontrivial  $p$ -group.

- (ii) Assume now that  $|\Omega|$  is a power of  $p$  different from 1. Prove that  $|\text{Fix}^G(\Omega)|$  is divisible by  $p$ .
- (iii) Let  $k$  be a (commutative) field of characteristic  $p$ . Let  $M$  be a  $kG$ -module. Prove that  $\text{Fix}^G(M) \neq 0$ . (Hint. (a) Let  $(e_i)_{i=1, \dots, r}$  be a  $k$ -basis of  $M$ . Prove that the abelian group generated by  $(ge_i)_{i=1, \dots, r}$  ( $g \in G$ ) is an  $\mathbb{F}_p G$ -module. (b) Apply question (2) above to conclude.)
- (iv) Prove that, up to isomorphism, the trivial representation of  $G$  is the unique irreducible  $kG$ -module.

**Solution:** (i) The set  $\Omega$  decomposes as a disjoint union

$$\Omega = \text{Fix}^G(\Omega) \sqcup (\Omega \setminus \text{Fix}^G(\Omega)).$$

Moreover,  $\Omega \setminus \text{Fix}^G(\Omega)$  is  $G$ -stable, hence it is a disjoint union of its  $G$ -orbits:

$$\Omega \setminus \text{Fix}^G(\Omega) = \bigsqcup_{\omega \in \Omega^\#} \omega.$$

Therefore

$$\Omega = \text{Fix}^G(\Omega) \sqcup \bigsqcup_{\omega \in \Omega^\#} \omega.$$

Taking cardinalities gives

$$|\Omega| = |\text{Fix}^G(\Omega)| + \sum_{\omega \in \Omega^\#} |\omega|.$$

(ii) We just prove that for any  $\omega \in \Omega^\#$  we have  $|\omega|$  is divisible by  $p$ . Consider  $\omega$  as transitive  $G$ -set, according to (1), we choose  $x \in \omega$  then it induces the bijection

$$G/G_x \xrightarrow{\sim} \omega.$$

Since the only prime divisor of  $|G|$  is  $p$ , we can suppose that  $|\omega| = |G/G_x| = p^s$ . If  $s = 0$ , it turns out that  $x \in \text{Fix}^G(\Omega)$ , which contradicts to our assumption. Therefore,  $|\Omega|$  and each  $|\omega|$  is divisible by  $p$  implies that

$$|\text{Fix}^G(\Omega)| = |\Omega| - \sum_{\omega \in \Omega^\#} |\omega|.$$

is also divisible by  $p$ .

(iii) Assume  $M \neq 0$ , and let  $e_1, \dots, e_r$  be a  $k$ -basis of  $M$ . Set

$$M' = \langle ge_i \mid g \in G, 1 \leq i \leq r \rangle_{\mathbb{F}_p} \subset M.$$

Since  $\text{char}(k) = p$ , this is a finite  $\mathbb{F}_p$ -subspace of  $M$ . It is  $G$ -stable, because for  $h \in G$ ,

$$h(ge_i) = (hg)e_i \in M'.$$

Thus  $G$  acts on the finite set  $M'$ .

Since  $M' \neq 0$ , we have  $|M'| = p^n$  for some  $n \geq 1$ . By part (ii),

$$|\text{Fix}^G(M')|$$

is divisible by  $p$ . But  $0 \in \text{Fix}^G(M')$ , so the fixed point set cannot consist only of 0. Hence there exists

$$0 \neq m \in \text{Fix}^G(M').$$

Since  $M' \subset M$ , we get

$$0 \neq m \in \text{Fix}^G(M).$$

Therefore

$$\text{Fix}^G(M) \neq 0.$$

(iv) Let  $M$  be an irreducible  $kG$ -module. By part (iii),

$$\text{Fix}^G(M) \neq 0.$$

Choose

$$0 \neq m \in \text{Fix}^G(M).$$

Then  $gm = m$  for all  $g \in G$ . Hence the one-dimensional subspace

$$km \subset M$$

is  $G$ -stable. Since  $M$  is irreducible, we must have

$$M = km.$$

Thus  $M$  is one-dimensional and  $G$  acts trivially on  $M$ . Therefore

$$M \cong k$$

as  $kG$ -modules, where  $k$  is the trivial representation.

Conversely, the trivial representation  $k$  is one-dimensional, hence irreducible. Therefore, up to isomorphism, the trivial representation is the unique irreducible  $kG$ -module. ■

## 16 Week 13

### Problem 1

Write out a proof of Maschke's Theorem in the case of representations over  $\mathbb{C}$  along the following lines.

1. Given a representation

$$\rho : G \rightarrow GL(V),$$

where  $V$  is a vector space over  $\mathbb{C}$ , let  $(\cdot, \cdot)$  be any positive definite Hermitian form on  $V$ . Define a new form  $(\cdot, \cdot)_1$  on  $V$  by

$$(v, w)_1 = \frac{1}{|G|} \sum_{g \in G} (gv, gw).$$

Show that  $(\cdot, \cdot)_1$  is a positive definite Hermitian form, preserved under the action of  $G$ ; that is,

$$(v, w)_1 = (gv, gw)_1$$

always.

2. If  $W$  is a subrepresentation of  $V$ , show that

$$V = W \oplus W^\perp$$

as representations.

**Solution:** (1) For  $a, b \in \mathbb{C}$ ,

$$(av_1 + bv_2, w)_1 = \frac{1}{|G|} \sum_{g \in G} (a gv_1 + b gv_2, gw) = a(v_1, w)_1 + b(v_2, w)_1,$$

and similarly the second variable is conjugate-linear. Moreover,

$$(w, v)_1 = \frac{1}{|G|} \sum_{g \in G} (gw, gv) = \overline{\frac{1}{|G|} \sum_{g \in G} (gv, gw)} = \overline{(v, w)_1}.$$



Thus  $(\cdot, \cdot)_1$  is Hermitian. Next, for  $v \neq 0$ , since each  $g \in G$  acts invertibly, we have  $gv \neq 0$ . Hence

$$(v, v)_1 = \frac{1}{|G|} \sum_{g \in G} (gv, gv) > 0.$$

Therefore  $(\cdot, \cdot)_1$  is positive definite. Finally, we show that this form is preserved by the action of  $G$ . Let  $h \in G$ . Then

$$(hv, hw)_1 = \frac{1}{|G|} \sum_{g \in G} (ghv, ghw).$$

As  $g$  runs over  $G$ , so does  $gh$ . Therefore

$$(hv, hw)_1 = \frac{1}{|G|} \sum_{u \in G} (uv, uw) = (v, w)_1.$$

Thus

$$(v, w)_1 = (hv, hw)_1$$

for all  $h \in G$ , so  $(\cdot, \cdot)_1$  is  $G$ -invariant.

(2) Choose a basis of  $V$ , so that  $V \simeq \mathbb{C}^n$ , and take the standard positive definite Hermitian form

$$(v, w) = \sum_i v_i \overline{w_i}.$$

Averaging this form over the  $G$ -action gives a new Hermitian form

$$(v, w)_1 = \frac{1}{|G|} \sum_{g \in G} (gv, gw).$$

By the previous part,  $(\cdot, \cdot)_1$  is positive definite and  $G$ -invariant.

Now let  $W \subset V$  be a subrepresentation, and define

$$W^\perp = \{v \in V \mid (v, w)_1 = 0 \text{ for all } w \in W\}.$$

Since  $(\cdot, \cdot)_1$  is positive definite, we have

$$V = W \oplus W^\perp$$

as vector spaces.

It remains to show that  $W^\perp$  is  $G$ -stable. Let  $u \in W^\perp$ ,  $h \in G$ , and  $w \in W$ . Since  $W$  is a subrepresentation,  $h^{-1}w \in W$ . Using  $G$ -invariance of  $(\cdot, \cdot)_1$ , we get

$$(hu, w)_1 = (u, h^{-1}w)_1 = 0.$$

Hence  $hu \in W^\perp$ , so  $W^\perp$  is a subrepresentation. Therefore

$$V = W \oplus W^\perp$$

as representations. ■

## Problem 2

Prove the following theorem of Burnside: let  $G$  be a finite group,  $k$  an algebraically closed field in which  $|G|$  is invertible, and

$$\rho : G \rightarrow GL(V)$$

be a representation over  $k$ . By taking a basis of  $V$ , write each endomorphism  $\rho(g)$  as a matrix. Let  $\dim V = n$ . Show that the representation is simple if and only if there exist  $n^2$  elements  $g_1, \dots, g_{n^2}$  of  $G$  so that the matrices

$$\rho(g_1), \dots, \rho(g_{n^2})$$

are linearly independent, and that this happens if and only if the algebra homomorphism

$$kG \rightarrow \text{End}_k(V)$$

is surjective. Note that  $\rho$  itself is generally not surjective.

**Solution:** Let

$$A = \text{span}_k\{\rho(g) : g \in G\} \subseteq \text{End}_k(V).$$

Then  $A$  is the image of the algebra homomorphism  $kG \rightarrow \text{End}_k(V)$ .

Suppose that there exist  $g_1, \dots, g_{n^2} \in G$  such that  $\rho(g_1), \dots, \rho(g_{n^2})$  are linearly independent. Since  $\dim_k \text{End}_k(V) = n^2$ , they form a basis of  $\text{End}_k(V)$ . Hence every elementary matrix  $E_{ij}$ , sending  $e_j$  to  $e_i$  and all other basis vectors to 0, can be written as a  $k$ -linear combination of the matrices  $\rho(g)$ . Let  $0 \neq W \subseteq V$  be a subrepresentation and choose  $0 \neq v = \sum_{\ell=1}^n c_\ell e_\ell \in W$ . Pick  $j$  with  $c_j \neq 0$ . Then for every  $i$ ,

$$\frac{1}{c_j} E_{ij} v = e_i.$$

Since  $W$  is stable under every  $\rho(g)$ , it is stable under their linear combinations. Thus  $e_i \in W$  for all  $i$ , so  $W = V$ . Therefore  $V$  is simple.

Conversely, suppose that  $V$  is simple. Let

$$A = \text{im}(kG \rightarrow \text{End}_k(V)).$$

Since  $|G|$  is invertible in  $k$ , Maschke's theorem implies that  $kG$  is semisimple. Hence its quotient  $A$  is also semisimple.

Since  $V$  is simple as a  $G$ -representation, it is simple as an  $A$ -module. By Schur's lemma, because  $k$  is algebraically closed,

$$\text{End}_A(V) = k.$$

By the Artin–Wedderburn theorem,  $A$  is a product of matrix algebras over  $k$ . Since  $V$  is simple, only one simple factor acts nontrivially on  $V$ . But  $A$  is defined as the image in  $\text{End}_k(V)$ , so its action on  $V$  is faithful. Therefore  $A$  has only this one factor, and its action on  $V$  identifies it with

$$\text{End}_k(V).$$

Thus

$$A = \text{End}_k(V).$$

Hence  $\dim_k A = n^2$ . Since  $A$  is spanned by the matrices  $\rho(g)$ , one can choose  $n^2$  linearly independent matrices among them. ■

### Problem 3

Let

$$G = \langle x \mid x^n = 1 \rangle$$

be a cyclic group of order  $n$ , and let  $\zeta_n \in \mathbb{C}$  be a primitive  $n$ -th root of unity. Then the simple complex characters of  $G$  are the  $n$  functions

$$\chi_r(x^s) = \zeta_n^{rs},$$

where  $0 \leq r \leq n-1$ .

**Solution:** Let  $V$  be a simple complex representation of  $G$ , and set  $T = \rho(x)$ . Since  $x^n = 1$ , we have  $T^n = I$ . Hence the minimal polynomial of  $T$  divides  $t^n - 1$ . Over  $\mathbb{C}$ , the polynomial  $t^n - 1 = \prod_{r=0}^{n-1} (t - \zeta_n^r)$  has distinct roots, so  $T$  is diagonalizable and its eigenvalues are among  $\zeta_n^0, \dots, \zeta_n^{n-1}$ .

For an eigenvalue  $\lambda$ , the eigenspace  $V_\lambda = \{v \in V \mid Tv = \lambda v\}$  is  $G$ -stable, because  $G = \langle x \rangle$ . Since  $V$  is simple, only one eigenspace can be nonzero. Thus  $V = V_\lambda$  for some  $\lambda$ , so  $\dim V = 1$ .

Therefore every simple complex representation of  $G$  is one-dimensional. Since  $\rho(x)^n = 1$ , we have  $\rho(x) = \zeta_n^r$  for some  $0 \leq r \leq n-1$ . Hence, for every  $s$ ,

$$\chi_r(x^s) = \rho(x^s) = \rho(x)^s = (\zeta_n^r)^s = \zeta_n^{rs}.$$

These  $n$  characters are distinct since  $\chi_r(x) = \zeta_n^r$ . Hence they are exactly all the simple complex characters of  $G$ . ■

## 17 Week 14

### Problem 1

Let  $G$  be a finite group and let  $p$  be a prime. Prove that each element  $x \in G$  can be uniquely written as

$$x = st,$$

where  $s$  is  $p$ -regular,  $t$  is  $p$ -singular, and  $st = ts$ . If  $x_1 = s_1 t_1$  is such a decomposition of an element  $x_1$  that is conjugate to  $x$ , then  $s$  is conjugate to  $s_1$ , and  $t$  is conjugate to  $t_1$ .

**Solution:** Let  $x \in G$ , and write  $|x| = n = p^a m$ , where  $(m, p) = 1$ . Choose integers  $A, B$  such that  $Ap^a + Bm = 1$ . Set  $s = x^{Ap^a}$  and  $t = x^{Bm}$ . Then  $s$  and  $t$  commute, since both are powers of  $x$ , and  $st = x^{Ap^a + Bm} = x$ . Moreover,  $s^m = x^{Ap^a m} = x^{An} = 1$ , so  $|s| \mid m$ ; hence  $s$  is  $p$ -regular. Similarly,  $t^{p^a} = x^{Bmp^a} = x^{Bn} = 1$ , so  $|t| \mid p^a$ ; hence  $t$  is a  $p$ -element. Thus the required decomposition exists.

Now suppose  $x = s_1 t_1$  is another such decomposition, with  $s_1$   $p$ -regular,  $t_1$  a  $p$ -element, and  $s_1 t_1 = t_1 s_1$ . Since  $s_1$  and  $t_1$  commute and have relatively prime orders,  $|x| = |s_1 t_1| = |s_1| |t_1|$ . Comparing the  $p$ -part and the prime-to- $p$  part of  $|x| = p^a m$ , we get  $|s_1| = m$  and  $|t_1| = p^a$ . Therefore

$$x^{Ap^a} = (s_1 t_1)^{Ap^a} = s_1^{Ap^a} t_1^{Ap^a} = s_1,$$

because  $Ap^a \equiv 1 \pmod{m}$  and  $t_1^{p^a} = 1$ . Similarly,  $x^{Bm} = (s_1 t_1)^{Bm} = t_1$ . Hence  $s_1 = s$  and  $t_1 = t$ , so the decomposition is unique.

Finally, if  $x_1 = g x g^{-1}$ , then  $x_1 = (g s g^{-1})(g t g^{-1})$ . The first factor is still  $p$ -regular, the second is still a  $p$ -element, and they commute. By uniqueness, if  $x_1 = s_1 t_1$  is the corresponding

decomposition, then  $s_1 = gsg^{-1}$  and  $t_1 = gtg^{-1}$ . Thus  $s$  is conjugate to  $s_1$ , and  $t$  is conjugate to  $t_1$ . ■

## 18 Extra Problems

### 18.1 Problem 1

**(Principle of idealization)** Let  $U$  be an  $(A, A)$ -bimodule and let  $B = A \oplus U$ . Then  $B$  becomes a ring with multiplication defined by

$$(a_1, u_1)(a_2, u_2) = (a_1a_2, a_1u_2 + u_1a_2),$$

and  $U$  is an ideal of  $B$ .

**Solution:** Verify it by definition of rings directly. ■

### 18.2 Problem 2

Let  $I_1, \dots, I_r$  be ideals of  $A$ , and suppose that  $I_i + I_j = A$  whenever  $i \neq j$ . Then the following hold.

(i)  $I_i + \bigcap_{j \neq i} I_j = A$ .

(ii) **(Chinese remainder theorem)** The natural homomorphism

$$\varphi : A \longrightarrow \bigoplus_{j=1}^r A/I_j, \quad a \longmapsto (a + I_j)_j,$$

is an epimorphism. Consequently, the following isomorphism holds:

$$A / \bigcap_{j=1}^r I_j \cong \bigoplus_{j=1}^r A/I_j.$$

**Solution:** (1) Consider  $\prod_{j \neq i} I_j \subseteq \bigcap_{j \neq i} I_j$  then let  $y_j \equiv 1 \pmod{I_1}$ . Then  $1 - \prod_{j \neq 1} y_j \in I_1$ . (2) Directly by (1). ■

### 18.3 Problem 3

Let  $V_A$  be a Noetherian module. Then any epimorphism  $\varphi : V \rightarrow V$  is necessarily an isomorphism.

### 18.4 Problem 4

Let  $U = V \oplus W$ , and let  $\pi_V$  and  $\pi_W$  be projections onto  $V$  and  $W$ , respectively. Then the following hold.

(i) For a submodule  $L$  of  $U$ ,

$$\pi_V(L)/(L \cap V) \cong \pi_W(L)/(L \cap W).$$

(ii) If  $U$  has a composition series, the following two conditions are equivalent:

(1) Every submodule  $L$  of  $U$  is expressed in the form

$$L = L_1 \oplus L_2, \quad L_1 \subset V, \quad L_2 \subset W.$$

(2)  $V$  and  $W$  have no irreducible constituent in common.

**Solution:** (i) Consider  $\pi_V(l) = \pi_W(l) + (L \cap W)$  with kernel  $L \cap V$ . (ii) If  $V$  and  $W$  has common  $E$  with

$$V \supseteq \cdots \supseteq V' \supseteq V'' \supseteq \cdots$$

and

$$W \supseteq \cdots \supseteq W' \supseteq W'' \supseteq \cdots$$

such that  $V'/V'' \simeq W'/W'' \simeq E$ . Consider pullback. Conversely, consider the piece of composite series

$$\pi_V(L) \supseteq \cdots \supseteq L \cap V \quad \pi_W(L) \supseteq \cdots \supseteq L \cap W$$

Then condition (2) implies that its length is zero. Equivalently,  $\pi_V(L) = L \cap V$  and  $\pi_W(L) = L \cap W$ . ■

### 18.5 Problem 5

If  $\varphi : A \rightarrow A'$  is an epimorphism of rings, then

$$\varphi(J(A)) \subset J(A').$$

**Solution:** Use maximal ideal. ■

### 18.6 Problem 6

Suppose that  $\bar{A} = A/J(A)$  is Artinian, and let  $V$  be a finitely generated  $A$ -module. Then  $VJ(A)$  coincides with the intersection of all maximal  $A$ -submodules of  $V$ .

**Solution:** Let  $M$  be an arbitrary maximal  $A$ -submodule of  $V$ . Consider that  $J(A)(V/M) = 0$  which implies that  $J(A)V \subseteq M$ . Then we have

$$VJ(A) \subseteq \bigcap_{M \text{ maximal}} M.$$

Conversely, consider  $\bar{V}$  be  $A/J(A)$  module. Since  $A/J(A)$  is artinian and hence

$$J(A/J(A)) = 0$$

since all maximal ideal of  $A$  has intersection  $J(A)$ . Then the intersection of maximal ideal of  $A/J(A)$  is zero.  $\bar{V}$  is f.g. semisimple artinian  $\bar{A}$ -module. Let  $\bar{v} \in \bar{V}$  be an arbitrary nonzero element. Then we have

$$\bar{V} = \langle \bar{v} \rangle \oplus \bar{V}'$$

which means that for any  $v \notin VJ(A)$  there exists an maximal  $A$ -submodule  $V'$  such that  $V'/V'J(A) \simeq \bar{V}'$  and  $v \notin V'$ . ■

### 18.7 Problem 7

Suppose that the identity of  $A$  is expressed as orthogonal sums of  $n$  idempotents in two ways:

$$1 = \sum_{i=1}^n e_i = \sum_{i=1}^n f_i.$$

If  $e_i A \cong f_i A$  for all  $i$ , then there exists a unit  $u$  of  $A$  such that

$$u^{-1} e_i u = f_i$$

for all  $i$ .

**Solution:**  $e_i$  and  $f_i$  gives

$$A = \bigoplus_i e_i A = \bigoplus_i f_i A$$

Then  $\psi$  gives isomorphism such that  $\psi(a) = ua$  where  $u \in A^\times$  since  $\text{End}_A(A_A) \simeq A$ . ■

### 18.8 Problem 8

Let  $I$  be an ideal of  $A$  contained in  $J(A)$ , and let  $\bar{A} = A/I$ . If two idempotents  $e, f$  of  $A$  satisfy  $\bar{e} = \bar{f}$ , then there exists  $c \in I$  such that

$$(1 + c)^{-1} e (1 + c) = f.$$

**Solution:** Consider  $u = 1 + c = (1 - e)(1 - f)$  and verify it. ■

### 18.9 Problem 9

The following three conditions on  $V_A$  with a composition series are equivalent.

- (1)  $V$  is completely reducible.
- (2) Every irreducible submodule of  $V$  is a direct summand of  $V$ .
- (3) Every maximal submodule of  $V$  is a direct summand of  $V$ .

### 18.10 Problem 10

$$\text{soc}(U \oplus V) = \text{soc}(U) \oplus \text{soc}(V).$$

### 18.11 Problem 11

Let  $A$  be a semisimple Artinian ring and let

$$A = A_1 \oplus \cdots \oplus A_n$$

be the decomposition into the direct sum of simple components  $A_i$ . Then the following hold.

- (i) Every ideal of  $A$  is a direct sum of some  $A_i$ 's.
- (ii) If  $A'$  is a homomorphic image of  $A$ , then  $A'$  is isomorphic to a direct sum of some of the  $A_i$  as rings.

**Solution:** Use idempotent decomposition on ideal and kernel. ■

### 18.12 Problem 12

Let  $A$  be an Artinian ring. Then the following hold.

- (i) There are only a finite number of maximal ideals, and the intersection of them coincides with  $J(A)$ .
- (ii) Let  $I$  be an ideal of  $A$ . If  $I$  contains no idempotent, then  $I \subset J(A)$ .

**Solution:** Consider  $A/J(A)$  is semisimple and its maximal ideals are 1-1 corresponding to  $A$ . ■

### 18.13 Problem 13

Let  $A$  be Noetherian and let  $x, y \in A$ . If  $xy = 1$ , then  $yx = 1$ .

**Solution:** Use Problem 3, and  $L_x$  is epimorphism then is isomorphism. ■

### 18.14 Problem 14

Let  $P, Q$  be finitely generated projective  $A$ -modules. Then

$$P/PJ(A) \cong Q/QJ(A) \iff P \cong Q.$$

**Solution:** The projectivity induces  $f : P \rightarrow Q$  such that  $f(P)$  1-1 correspondence to representation of  $Q/QJ(A)$ . Then we have

$$Q = f(P) + QJ(A).$$

By Nakayama lemma, we have  $Q = f(P)$ . To prove the injectivity, we consider

$$0 \rightarrow \ker(f) \rightarrow P \rightarrow Q \rightarrow 0$$

splits. ■

### 18.15 Problem 15

Let  $e$  be an idempotent of  $A$  such that  $AeA = A$ . Let  $\Gamma = eAe$ , and set

$$\mathfrak{F}(V) = Ve$$

for  $V_A$ , and

$$\mathfrak{G}(W) = W \otimes_{\Gamma} eA$$

for  $W_{\Gamma}$ . Note that  $\mathfrak{F}(V)$  and  $\mathfrak{G}(W)$  are a  $\Gamma$ - and  $A$ -module under the natural actions, respectively. Then the following hold.

- (i)  $\mathfrak{G}(\mathfrak{F}(V)) \cong V$ , and  $\mathfrak{F}(\mathfrak{G}(W)) \cong W$ .

(ii) The restriction map induces an isomorphism

$$\mathrm{Hom}_A(U, V') \cong \mathrm{Hom}_\Gamma(Ue, V'e).$$

In particular, there is a ring isomorphism

$$A \cong \mathrm{End}_\Gamma(Ae), \quad a \mapsto \varphi_a,$$

where  $\varphi_a(xe) = axe$ .

(iii)  $\mathfrak{F}$  and  $\mathfrak{G}$  preserve direct sums, inclusions, and projectivity.

**Solution:** (i) ■

### 18.16 Problem 16

Suppose that

$$1 = \sum_{i=1}^k \sum_{j=1}^{n_i} e_{ij}$$

is a primitive idempotent decomposition of 1 such that

$$e_{ij}A \cong e_{i'j'}A \iff i = i'.$$

If we set  $e_{i1} = e_i$  and  $e = e_1 + \cdots + e_k$ , then  $AeA = A$ . Moreover, if we let  $\Gamma = eAe$  as in the preceding problem, then

$$\Gamma = \bigoplus_{i=1}^k e_i \Gamma,$$

and a  $\Gamma$ -isomorphism  $e_i \Gamma \cong e_j \Gamma$  holds only if  $i = j$ . The ring  $\Gamma$  is called the basic ring of  $A$ .

### 18.17 Problem 17

Let  $P$  be a finitely generated projective  $A$ -module and let

$$E = \mathrm{End}_A(P).$$

Then the following hold.

(i)

$$J(E) = \{\varphi \in E \mid \mathrm{Im} \varphi \subset PJ(A)\}.$$

(ii)

$$E/J(E) \cong \mathrm{End}_A(P/PJ(A)).$$

### 18.18 Problem 18

The following condition on  $U_A$  is a necessary and sufficient condition for  $U$  to be a finitely generated projective  $A$ -module:

There exist  $u_1, \dots, u_n \in U$  and  $\varphi_1, \dots, \varphi_n \in \mathrm{Hom}_A(U, A)$  such that

$$u = \sum_i u_i \varphi_i(u)$$

for all  $u \in U$ .



### 18.19 Problem 19

Let  $I$  be an ideal of  $A$ . If  $Z(A)$  has no idempotent other than the identity, then the same is true for  $Z(A/I)$ , provided  $I$  satisfies one of the following conditions:

- (1)  $I = NA$ , where  $N$  is a nilpotent ideal of  $Z(A)$ .
- (2)  $I \subset J(A)^2$ , where  $J(A)$  is assumed to be nilpotent.

### 18.20 Problem 20

Given two exact sequences

$$V' \xrightarrow{\lambda} V \xrightarrow{\mu} V'' \rightarrow 0, \quad W' \xrightarrow{\lambda'} W \xrightarrow{\mu'} W'' \rightarrow 0,$$

the following sequence is also exact:

$$(V' \otimes_A W) \oplus (V \otimes_A W') \xrightarrow{f} V \otimes_A W \xrightarrow{\mu \otimes \mu'} V'' \otimes_A W'' \rightarrow 0,$$

where

$$f = \lambda \otimes \text{id}_W + \text{id}_V \otimes \lambda'.$$

### 18.21 Problem 21

Let  $V_A$  be a flat module and let  ${}_A W$  be a left  $A$ -module. The following equality holds in  $V \otimes_A W$  for two  $A$ -submodules  $W_1, W_2$  of  $W$ :

$$V \otimes_A W_1 \cap V \otimes_A W_2 = V \otimes_A (W_1 \cap W_2).$$

### 18.22 Problem 22

Let  $R$  be a commutative ring and let  $I, L$  be ideals of  $R$ . Then the following isomorphism holds:

$$R/I \otimes_R R/L \cong R/(I + L).$$